

---

# Senior Design 2 Final Report

## Smart and Grid-Interactive Electrical Panels

---



### Group 37

#### Authors:

|                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|
| Oscar Rodriguez    | Nicholas Sally     | Jack Dwyer         | Abraham Alcala     |
| <i>Computer</i>    | <i>Electrical</i>  | <i>Electrical</i>  | <i>Computer</i>    |
| <i>Engineering</i> | <i>Engineering</i> | <i>Engineering</i> | <i>Engineering</i> |

#### Reviewer Committee:

|                     |     |                             |
|---------------------|-----|-----------------------------|
| Dr. Saleem Sahawneh | ECE | Lecturer                    |
| Dr. Sonali Das      | ECE | Lecturer                    |
| Dr. Xun Gong        | ECE | Associate Professor         |
| Dr. Zhou Sun        | ECE | Sponsor/Engineering Support |

#### Mentor:

Dr. Lei Wei

# Table of Contents

|                                                                |    |
|----------------------------------------------------------------|----|
| Table of Contents .....                                        | ii |
| Table of Tables .....                                          | ix |
| Table of Figures .....                                         | x  |
| Chapter 1 – Executive Summary .....                            | 1  |
| Chapter 2 – Project Description .....                          | 2  |
| 2.1: Project Background.....                                   | 2  |
| 2.1.1: Previous Works .....                                    | 3  |
| 2.1.2: Existing Products .....                                 | 3  |
| 2.2: Project Motivation .....                                  | 3  |
| 2.3: Project Function .....                                    | 4  |
| 2.4: Project Objectives .....                                  | 5  |
| Basic Goals: .....                                             | 5  |
| Advanced Goals: .....                                          | 5  |
| Stretch Goals:.....                                            | 5  |
| 2.5: Requirement Specifications.....                           | 5  |
| 2.6: House of Quality .....                                    | 7  |
| 2.7: Hardware Block Diagram .....                              | 9  |
| 2.8: MCU Firmware Block Diagram .....                          | 9  |
| 2.9: Mobile Software Block Diagram .....                       | 10 |
| Chapter 3 – Research .....                                     | 12 |
| 3.1: Service Panel and Breaker.....                            | 12 |
| 3.1.1: Service Panel Specifications.....                       | 12 |
| 3.1.2: Service Panel Selection .....                           | 13 |
| 3.1.3: Circuit Breaker Specifications .....                    | 13 |
| 3.1.4: Service Panel Selection .....                           | 14 |
| 3.2: Power Supply .....                                        | 14 |
| 3.2.1: Power Supply Overview.....                              | 14 |
| 3.2.2: The Transformer .....                                   | 15 |
| 3.2.2.1: Transformer Overview.....                             | 15 |
| 3.2.2.2 Step Up vs. Step Down vs. Isolation Transformers.....  | 16 |
| 3.2.2.3: Step Up vs. Step Down vs. Isolation Transformers..... | 16 |

|                                                             |    |
|-------------------------------------------------------------|----|
| 3.2.2.4: Transformer Comparison Summary.....                | 18 |
| 3.2.2.5: Transformer Selection .....                        | 19 |
| 3.2.3: Rectification.....                                   | 19 |
| 3.2.3.1: Rectification Overview.....                        | 20 |
| 3.2.3.2: Half Wave vs. Full Wave Rectification.....         | 20 |
| 3.2.4: The Power Management Integrated Circuit.....         | 21 |
| 3.2.4.1: Power Management Integrated Circuit Overview ..... | 21 |
| 3.2.4.2: Linear vs Switch Mode Power Supplies .....         | 21 |
| 3.2.4.3: Switching PMIC Selection .....                     | 22 |
| 3.3: Frequency Detector.....                                | 24 |
| 3.3.1: Frequency Detector Overview .....                    | 24 |
| 3.3.2: Frequency Measuring Methods .....                    | 25 |
| 3.3.2.1: Cross-Zero Detector Overview .....                 | 25 |
| 3.3.2.2: Cross-Zero Detector Technologies .....             | 25 |
| 3.3.2.3: Frequency Detector Technology Selection .....      | 28 |
| 3.3.2.4: Cross-Zero Detector Part Selection .....           | 28 |
| 3.4: Mechanical Systems .....                               | 30 |
| 3.4.1: Mechanism of Breaker Automation .....                | 30 |
| 3.4.2: Switching mechanism .....                            | 31 |
| 3.4.3: Gantry System Overview .....                         | 31 |
| 3.4.4: Guide Selection .....                                | 32 |
| 3.4.5: Motor Load Calculations .....                        | 32 |
| 3.4.6: Motor and Control Schema Comparison .....            | 33 |
| 3.4.7: Comparison of Stepper Motors .....                   | 34 |
| 3.4.8: Comparison of Stepper Motor Drivers.....             | 35 |
| 3.4.10: Physical Mounting of Gantry Structure.....          | 38 |
| 3.4.11: Software Control of motors .....                    | 39 |
| 3.4.12: End effector .....                                  | 39 |
| 3.5: MCU .....                                              | 40 |
| 3.5.1: MCU vs FPGA Considerations.....                      | 40 |
| 3.5.2: Selecting Microcontrollers for the Project.....      | 41 |
| 3.5.2.1: Raspberry Pi Series .....                          | 41 |

|                                                              |    |
|--------------------------------------------------------------|----|
| 3.5.2.2: Comparison with ESP32, STM32, and Arduino Nano..... | 42 |
| 3.6: Wireless Communication Protocols.....                   | 44 |
| 3.6.1: Wi-Fi.....                                            | 44 |
| 3.6.2: Bluetooth .....                                       | 45 |
| 3.6.3: LoRaWAN.....                                          | 45 |
| 3.7: Hardware Programming Language Comparisons.....          | 46 |
| 3.7.1: Python .....                                          | 46 |
| 3.7.2: C.....                                                | 47 |
| 3.7.3: C++ .....                                             | 47 |
| 3.7.4: Rust.....                                             | 48 |
| 3.8: Web Stack Comparisons.....                              | 48 |
| 3.8.1: Database Comparisons .....                            | 49 |
| 3.8.1.1: PostgreSQL.....                                     | 49 |
| 3.8.1.2: MongoDB .....                                       | 49 |
| 3.8.1.3: MySQL .....                                         | 49 |
| 3.8.2: Frontend Comparisons .....                            | 50 |
| 3.8.2.1: React.....                                          | 50 |
| 3.8.2.2: Vue.js .....                                        | 50 |
| 3.8.2.3: Angular .....                                       | 51 |
| 3.8.2.4: Flutter .....                                       | 51 |
| 3.8.3: Backend Comparisons.....                              | 52 |
| 3.8.3.1: Golang.....                                         | 52 |
| 3.8.3.2: JavaScript (Node.js).....                           | 52 |
| 3.8.3.3: PHP.....                                            | 52 |
| 3.8.3.4: Python (Django/Flask).....                          | 53 |
| Chapter 4 – Standards and Design Constraints.....            | 53 |
| 4.1: Form Factor .....                                       | 53 |
| 4.2: Standards .....                                         | 54 |
| 4.3: National Electric Code (NEC).....                       | 54 |
| 4.3.1: Article 240.....                                      | 54 |
| 4.3.2: Article 312.....                                      | 54 |
| 4.3.3: Article 408.....                                      | 54 |

|                                                                 |    |
|-----------------------------------------------------------------|----|
| 4.4: National Fire Protection Association (NFPA) .....          | 55 |
| 4.4.1: NFPA 70A.....                                            | 55 |
| 4.5: OSHA Electrical Safety Standards .....                     | 55 |
| 4.5.1: Examination.....                                         | 55 |
| 4.5.2: Installation and Use .....                               | 56 |
| 4.5.3: Identification of Disconnecting Mains and Circuits ..... | 56 |
| 4.5.4: Guarding of Live Parts and Equipment .....               | 56 |
| 4.5.5: Grounding.....                                           | 57 |
| 4.6: Electromagnetic Compatibility (EMC) Standards.....         | 57 |
| 4.6.1: IEC 61000 and FCC Part 15.....                           | 57 |
| 4.7: Communication Standards .....                              | 58 |
| 4.7.1: SPI .....                                                | 58 |
| 4.8: User Interactivity and Data Constraints .....              | 60 |
| 4.9: Consumer Constraints .....                                 | 61 |
| 4.9.1: Utility Constraints.....                                 | 61 |
| 4.9.2: Homeowner Constraints .....                              | 62 |
| 4.10: Environmental Constraints .....                           | 62 |
| 4.11: Remaining Constraints .....                               | 63 |
| 4.11.1: Political Constraints .....                             | 64 |
| 4.11.2: Ethical Constraints .....                               | 64 |
| Chapter 5 - Comparison of AI Platforms.....                     | 65 |
| 5.1: Technical Assistance and Idea Generation .....             | 65 |
| 5.2: Code Refinement and Error Resolution.....                  | 65 |
| 5.3: Data Analysis Support.....                                 | 66 |
| 5.4: Limitations and Potential Pitfalls.....                    | 66 |
| 5.5: Creativity and Problem-Solving .....                       | 67 |
| 5.6: Test Case.....                                             | 67 |
| 5.6.1: Zero-Crossing Detection.....                             | 68 |
| 5.6.2: Phase-Locked Loop (PLL) .....                            | 68 |
| 5.6.3: Frequency Measurement Using a Microcontroller .....      | 68 |
| 5.7: Result.....                                                | 68 |
| Chapter 6: Hardware Design.....                                 | 70 |
| 6.1: Overall electrical Schematic .....                         | 70 |

|                                                     |    |
|-----------------------------------------------------|----|
| 6.2: 12V PSU Electrical Schematic .....             | 72 |
| 6.3: 3.3V PSU Electrical Schematic .....            | 73 |
| 6.4: Frequency Detector Electrical Schematic.....   | 74 |
| 6.5: MCU Subsystem .....                            | 75 |
| 6.6: Motor control .....                            | 77 |
| 6.7: Mechanical Systems .....                       | 78 |
| Chapter 7 – Software Design .....                   | 79 |
| 7.1: API Endpoint Flowchart.....                    | 79 |
| 7.2: Entity Relationship Diagram (ERD) .....        | 80 |
| 7.2.1: Users.....                                   | 80 |
| 7.2.2: Devices.....                                 | 81 |
| 7.2.3: Breakers .....                               | 81 |
| 7.2.4: Connections .....                            | 81 |
| 7.3: Case Diagram .....                             | 82 |
| 7.3.1: Account Creation and Authentication .....    | 83 |
| 7.3.2: Device Management.....                       | 83 |
| 7.3.2.1: Creating a New Device .....                | 83 |
| 7.3.2.2: Reading/Searching an Existing Device ..... | 84 |
| 7.3.2.3: Updating an Existing Device.....           | 85 |
| 7.3.2.4: Deleting an Existing Device.....           | 85 |
| 7.3.3: Breaker Management .....                     | 86 |
| 7.3.4: User Actions and Device Commands.....        | 86 |
| 7.3.5: Error Handling and User Experience .....     | 87 |
| 7.4: State Diagram .....                            | 87 |
| 7.5: Class Diagram .....                            | 88 |
| 7.6: API Unit Testing.....                          | 89 |
| 7.7: Mobile Unit Testing.....                       | 90 |
| 7.8: User Interface .....                           | 90 |
| 7.8.1: Login Screen .....                           | 90 |
| 7.8.2: Sign Up Screen .....                         | 90 |
| 7.8.3: Reset Password Screen .....                  | 90 |
| 7.8.4: Device Management Screen.....                | 90 |
| 7.8.5: Device Detail/Interaction Screen.....        | 91 |

|                                                                   |     |
|-------------------------------------------------------------------|-----|
| 7.8.6: User Management Screen .....                               | 91  |
| 7.9: Frequency Detection Software .....                           | 91  |
| 7.9.1: Software Overview .....                                    | 91  |
| 7.9.2: Frequency Detection and Feedback Flowchart .....           | 91  |
| 7.9.3: Implementation .....                                       | 93  |
| 7.9.3.1: Frequency Detection.....                                 | 93  |
| 7.10: Frequency Detection and Feedback Case Diagram.....          | 94  |
| 7.11: Frequency Detection and Feedback State Diagram .....        | 95  |
| 7.12: Frequency Detection and Feedback Data Transfer Diagram..... | 95  |
| 7.13: Mechanical Systems Software .....                           | 96  |
| 7.13.1: G-code.....                                               | 97  |
| Chapter 8: System Fabrication and Construction .....              | 99  |
| 8.1: PCB Design General Guidelines .....                          | 99  |
| 8.2: PCB Layout and Design .....                                  | 100 |
| 8.2.1: 12V PSU PCB .....                                          | 100 |
| 8.2.2: 3.3V PSU.....                                              | 101 |
| 8.2.3: Frequency Detector PCB .....                               | 102 |
| 8.2.4: MCU PCB.....                                               | 103 |
| 8.2.5: Motor Control PCB .....                                    | 106 |
| 8.3: Mechanical system fabrication and construction .....         | 107 |
| 8.3.1: Face Plate.....                                            | 107 |
| 8.3.2: Frame .....                                                | 108 |
| 8.3.3: Wire management.....                                       | 108 |
| 8.3.4: Linear movement considerations .....                       | 109 |
| Chapter 9 - System Testing and Evaluation .....                   | 110 |
| 9.1: Hardware Testing and Evaluation .....                        | 110 |
| 9.1.1: 12V PSU PCB Testing and Evaluation .....                   | 110 |
| 9.1.2: 3.3V PSU PCB Testing and Evaluation .....                  | 111 |
| 9.1.3: Frequency Detector.....                                    | 111 |
| 9.1.4: Motor Controller.....                                      | 112 |
| 9.1.5: MCU Board.....                                             | 112 |
| 9.2: Software Testing and Evaluation .....                        | 112 |
| 9.2.1: Frequency Detection and Feedback Software .....            | 113 |

|                                                                       |     |
|-----------------------------------------------------------------------|-----|
| 9.2.2: Mobile App Software.....                                       | 113 |
| 9.2.2: – TMCStepper library.....                                      | 114 |
| 9.2.3: – GRBL.....                                                    | 114 |
| 9.3: Overall Integration and Testing.....                             | 114 |
| 9.3.1: Overall System Integration.....                                | 114 |
| 9.3.2: Overall System Testing .....                                   | 115 |
| 9.4: Plan for SD2.....                                                | 116 |
| Chapter 10 – Administrative Content .....                             | 119 |
| 10.1: Budget .....                                                    | 119 |
| 10.2: Bill of Materials.....                                          | 119 |
| 10.3: Distribution of Work.....                                       | 121 |
| 10.4: Project Milestones.....                                         | 122 |
| Chapter 11 - Conclusion.....                                          | 124 |
| Appendix A: Reference.....                                            | 126 |
| Appendix B: Image Permissions .....                                   | 132 |
| Appendix C: Texas Instruments Webench Power Designer Schematics ..... | 137 |

## Table of Tables

|                                                               |     |
|---------------------------------------------------------------|-----|
| Table 1: Requirement Specifications.....                      | 6   |
| Table 2: Service Panel Selection Matrix.....                  | 13  |
| Table 3: Transformer Types Comparison Matrix.....             | 18  |
| Table 4:Transformer Selection Matrix .....                    | 19  |
| Table 5: PMIC Configuration Matrix .....                      | 22  |
| Table 6: 12V PMIC Selection Matrix .....                      | 23  |
| Table 7: 3.3V PMIC Selection Matrix .....                     | 23  |
| Table 8: Comparator vs. Operational Amplifier Comparison..... | 27  |
| Table 9: Cross-Zero Detector Technologies Comparison .....    | 28  |
| Table 10: Comparator Comparison Matrix .....                  | 29  |
| Table 11: Guide Selection Matrix [25].....                    | 32  |
| Table 12: Lead-Torque Matrix .....                            | 33  |
| Table 13: Stepper motor comparison matrix .....               | 34  |
| Table 14: Stepper Motor Comparison Matrix.....                | 35  |
| Table 15: MCU Decision Matrix .....                           | 44  |
| Table 16: Embedded Programming Languages .....                | 48  |
| Table 17: Database Technology Matrix.....                     | 50  |
| Table 18: Frontend Technology Matrix.....                     | 52  |
| Table 19: Backend Technology Matrix.....                      | 53  |
| Table 20: SPI Modes.....                                      | 59  |
| Table 21: G Code Command Matrix [47].....                     | 98  |
| Table 22: Budget Matrix .....                                 | 119 |
| Table 23: Itemized BOM.....                                   | 120 |
| Table 24: Distribution of Work Matrix.....                    | 122 |
| Table 25: Senior Design 1 Project Timeline.....               | 122 |
| Table 26: Senior Design 2 Project Timeline.....               | 123 |

## Table of Figures

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| Figure 1: House of Quality .....                                              | 8   |
| Figure 2: Hardware Block Diagram .....                                        | 9   |
| Figure 3: Embedded System BD .....                                            | 10  |
| Figure 4: Mobile Software BD .....                                            | 11  |
| Figure 5: Power Flow Block Diagram .....                                      | 15  |
| Figure 6: Example of an autotransformer set-up from Electrical .....          | 17  |
| Figure 7: Current sense transformer from Electronics-Tutorial .....           | 17  |
| Figure 8: Rectification Comparison from Rohm Semiconductor [15] .....         | 20  |
| Figure 9: Frequency Detector Block Diagram .....                              | 25  |
| Figure 10: Cross-Zero Detector Base Courtesy of Texas Instruments .....       | 29  |
| Figure 11: Bipolar stepper motor phases .....                                 | 35  |
| Figure 12: TMC 5160 STEP/DIR Mode BD [30] .....                               | 37  |
| Figure 13: Daisy-Chained SPI Interface .....                                  | 60  |
| Figure 14: Overall Electrical Schematic Part 1 .....                          | 70  |
| Figure 15: Overall Electrical Schematic Part 2 .....                          | 71  |
| Figure 16: Overall Electrical Schematic Part 3 .....                          | 71  |
| Figure 17: 12V PSU Electrical Schematic .....                                 | 72  |
| Figure 18: 3.3V PSU Electrical Schematic .....                                | 74  |
| Figure 19: Frequency Detector Electrical Schematic .....                      | 75  |
| Figure 20: MCU Electrical Schematic .....                                     | 76  |
| Figure 21: Motor Controllers Electrical Schematic .....                       | 77  |
| Figure 22: Mechanical Systems CAD Model .....                                 | 78  |
| Figure 23: SW w/API endpoint Flowchart .....                                  | 80  |
| Figure 24: Entity Relationship Diagram (ERD) .....                            | 80  |
| Figure 25: Use Case Diagram .....                                             | 82  |
| Figure 26: State Diagram .....                                                | 88  |
| Figure 27: Class Diagram .....                                                | 89  |
| Figure 28: API Unit Testing .....                                             | 89  |
| Figure 29: ESP32 Frequency Detection and Feedback Flowchart .....             | 93  |
| Figure 30: ESP32 Frequency Detection and Feedback Case Diagram .....          | 94  |
| Figure 31: ESP32 Frequency Detection and Feedback State Diagram .....         | 95  |
| Figure 32: ESP32 Frequency Detection and Feedback Data Transfer Diagram ..... | 96  |
| Figure 33: ESP32 Mechanical System Block Diagram .....                        | 97  |
| Figure 34: 12V PSU PCB Layout .....                                           | 100 |
| Figure 35: 3.3V PSU PCB Layout .....                                          | 102 |
| Figure 36: Frequency Detector PCB Layout .....                                | 103 |
| Figure 37: MCU PCB Layout .....                                               | 104 |
| Figure 38: Motor Controller PCB Layout .....                                  | 107 |
| Figure 39: Simulated Output of 12V PSU .....                                  | 110 |
| Figure 40: Simulated Output of the Frequency Detector .....                   | 111 |
| Figure 41: Daisy-Chained SPI Interface Permission (Requested) .....           | 132 |
| Figure 42: Autotransformer Permission (Request) .....                         | 132 |
| Figure 43: Current Transformer Permission (Granted) .....                     | 133 |

|                                                                             |     |
|-----------------------------------------------------------------------------|-----|
| Figure 44: Transformer Specifications and Diagram Permission (Granted)..... | 133 |
| Figure 45: Full Wave Rectifier Image Permission (Requested) .....           | 134 |
| Figure 46: Cross-Zero Detector Image Permission (Requested) .....           | 134 |
| Figure 47: TMC 5160 block diagram permission (Requested) .....              | 135 |
| Figure 48: Stepper Motor Diagram Permission (Requested).....                | 135 |
| Figure 49: 12V PSU Webench Schematic Courtesy of Texas Instruments.....     | 137 |
| Figure 50: 3.3V PSU Webench Schematic Courtesy of Texas Instruments.....    | 137 |

# Chapter 1 – Executive Summary

The goal of our project was to design a smart and grid-interactive electrical panel that can control multiple circuit breakers, allowing for dynamic load management and responsive interaction with grid frequency signals. This system provides users with the ability to control their electrical breakers remotely via a mobile app, facilitating easier access to load management and allowing users to reduce electricity usage as well as costs. Additionally, the system supports automated responses to grid signals, helping to integrate more renewable energy sources into the grid and reduce dependence on fossil fuels.

The electrical panel controller is the primary design focus of our project, and we have carefully engineered it for cost-effectiveness to encourage widespread adoption. Designed to be retrofitted onto existing breaker panels, this solution provides homeowners with the convenience of remote control over their electrical circuits, either manually or automatically based on real-time grid conditions. With compact and durable components, the system is intended to be easy to install and adaptable to a range of electrical configurations.

The smart panel communicates with a central server via Wi-Fi, making it responsive to commands sent from a mobile app. For this aspect of our project, we utilized an ESP32 microcontroller for Wi-Fi connectivity, providing a reliable and flexible means of communication. Users interact with the system through a Flutter-based mobile application, which connects to a cloud-hosted backend on Heroku. To facilitate user authentication and data storage, we have employed Golang for the server-side logic, PostgreSQL as our database to house all information that the application may need, and Heroku for hosting the backend services. The app allows users to view real-time data on breaker status and issue commands to the ESP32 device, giving users control over their energy usage and system response settings.

The back-end infrastructure has been designed to provide both manual and automated breaker control options. The mobile app interfaces with the backend via RESTful API endpoints, enabling users to manage their accounts and devices effectively. Core API functions include the ability to create, update, and delete users and devices, as well as to issue commands to the breakers. In response to the need for secure access and smooth user experience, we implemented JWT-based authentication in our backend, ensuring secure and efficient communication between the user app and the server.

Overall, this smart panel system allows homeowners to participate in demand-side management, reducing the need for additional power generation and supporting a greener, more sustainable grid. Through convenient and secure mobile app control, users can optimize their power consumption, enhance their contribution to renewable energy integration, and take proactive steps toward a more efficient energy future.

## Chapter 2 – Project Description

This chapter provides a detailed overview of the project's purpose, objectives, and expected outcomes. The project centers around designing a responsive electrical monitoring and control system capable of detecting variations in grid frequency, managing circuit breakers, and providing mobile accessibility for user control. The ultimate goal is to create a robust and user-friendly solution for consumers seeking automated electrical management, with added functionalities to ensure ease of installation, secure access, and minimal maintenance.

The design emphasizes reliability, fast response, and energy efficiency. By integrating IoT technology and advanced control features, the system can detect changes in grid frequency and automatically adjust circuit breakers, thereby ensuring stable and secure operation of connected devices. The system also allows users to monitor and manage the setup remotely through a mobile app, offering convenience and greater control. This chapter outlines the project requirements, functional specifications, and constraints to clarify the expected capabilities and design considerations needed to meet both consumer needs and technical demands.

### 2.1: Project Background

Electrical Breaker Panels serve a hub that houses any homes circuit subsystems. They are essential to every home as they are directly responsible for managing the distribution of electrical power to various parts of the building. The circuit breakers protect against electrical overloading by disconnecting the circuits when overloading is detected. However, traditional panels require physical interaction with it to allow for the managing the distribution of power at a moment's notice. As times change and the introduction of innovative smart home technology, smart home enabled electric panels has largely fallen behind. Other renewable sources being introduced as subsystems also means more dynamic and responsive panels are needed. The ability to switch from distance allows for high load circuits to be offloaded from the system when other power sources are being introduced.

This project is to design and implement a smart grid like-interactive electrical panel that can automatically control multiple circuit breakers based on grid frequency signals or manual commands from a smartphone app. This system would allow homeowners to easily manage their electrical load, potentially reducing electricity bills and contributing to a more stable and sustainable grid by responding to real-time grid conditions.

The project stems from the need to make electrical systems more adaptable and user-friendly, especially in the context of increasing renewable energy integration. By allowing for automatic load adjustments based on grid signals, the system can help balance supply and demand, reducing the need for fossil fuel-based power plants to meet peak demands. Additionally, providing homeowners with the ability to control their electrical loads remotely offers convenience and can encourage more energy-efficient behaviors.

### 2.1.1: Previous Works

This project aims to iterate upon existing work done by another team year(s) prior to our team. This team also worked alongside our sponsor Dr. Qun Zhou Sun to develop a system that reliably senses frequency variations within a given electrical grid. Their system was developed for the purpose of keeping power consumption to a minimum in power intensive subsystems found commonly in buildings including but not limited to Heating, Ventilation and Air Conditioning Units (HVAC) and Grid Interactive Water Heaters (GIWH).

This previous iteration was a hardware implementation to solve a common problem. The hardware was intricately designed and proved to be somewhat cumbersome to install and operate considering the I/O interface was limited to an on-device solution. This iteration used relays to control the power input to the water heater which meant it was more of a bespoke solution as it didn't have a generic solution for most electrical grids.

### 2.1.2: Existing Products

Little to no existing products seem to exist to solve this problem in the fashion that we aim to address. Very expensive smart-home-enabled breaker boxes exist that aim to solve this problem as we do. These products, while effective and do work very well, are clearly without a shroud of a doubt a bespoke solution and don't work as a catch all solution. The whole electrical grid panel must be replaced with the expensive COTS system. We aim to produce a system that can be retrofitted for most standard breaker panels.

## 2.2: Project Motivation

The motivation behind the Smart and Grid-Interactive Electrical Panels project is to create a system that enhances the adaptability and efficiency of residential electrical systems, particularly in the context of increasing reliance on renewable energy sources. As the global energy landscape shifts towards sustainability, there is a growing need for intelligent systems that can manage electrical loads in response to real-time grid conditions. This project aims to leverage modern technology to provide homeowners with the tools needed to actively participate in energy management, contributing to both cost savings and environmental benefits.

The development of this project is driven by the recognition that traditional electrical panels, while functional, lack the capability to interact with the grid in a dynamic and automated manner. With the rise of renewable energy sources like solar and wind, which are inherently variable, the stability of the grid can be challenged during periods of high demand or low supply. By designing a system that can automatically adjust electrical loads based on grid frequency signals, this project seeks to mitigate these challenges and support the integration of more renewable energy into the grid.

Additionally, the convenience offered to homeowners by enabling remote control of their electrical panels through a smartphone app is a significant motivation for this project. In today's connected world, consumers are increasingly seeking smart

home solutions that offer both ease of use and tangible benefits, such as reduced energy bills. By giving users, the ability to manage their electrical consumption more effectively, this system can promote more energy-conscious behaviors and help households play a role in the broader effort to reduce carbon emissions.

Existing technologies in smart grids and home automation systems have laid the groundwork for projects like this. For example, smart thermostats and energy monitoring systems have demonstrated the potential of integrating real-time data and user control to optimize energy use. However, there is a gap in the market for a system that directly interacts with the home's electrical panel to manage loads in response to grid conditions. By addressing this gap, the Smart and Grid-Interactive Electrical Panels project aims to advance the capabilities of smart home technology and contribute to a more resilient and sustainable energy future.

Our team is motivated to not only improve the functionality and convenience of home electrical systems but also to empower consumers to actively participate in the transition to a cleaner energy grid. By developing a system that is both innovative and user-friendly, we hope to make a meaningful impact on energy efficiency and sustainability at the residential level. The success of this project will be determined by the system's ability to respond to grid frequency signals effectively, manage multiple circuit breakers simultaneously, and offer an intuitive and reliable user interface via a smartphone app. The system should also be easy to install and integrate with existing electrical panels, minimizing the need for extensive modifications.

## 2.3: Project Function

The Smart and Grid-Interactive Electrical Panels system will enable homeowners to manage their home's electrical load more efficiently and responsively, either automatically in response to grid conditions or manually through a smartphone app. The system will be integrated into existing electrical breaker panels, allowing it to control multiple circuit breakers based on real-time grid frequency signals or user input.

When a grid frequency signal indicates a need for load adjustment, the system will automatically activate or deactivate specific circuit breakers to balance the electrical load. This functionality helps stabilize the grid by reducing demand during peak times or increasing it when excess renewable energy is available. Homeowners can also manually control the system using a smartphone app, giving them the ability to turn appliances on or off remotely, providing both convenience and potential cost savings on energy bills.

The system is designed with ease of installation and use in mind. It must be straightforward to retrofit into existing electrical panels without requiring significant modifications to the home's wiring. The mechanical components required to control the breakers will be compact and durable, ensuring reliable operation without the need for frequent maintenance.

Given that multiple breakers may need to be coordinated simultaneously, the system will include a control unit capable of managing these operations seamlessly. This coordination is crucial for responding effectively to grid frequency signals, where timing and synchronization across different circuits are essential for maintaining overall grid stability.

To address the challenge of integrating this system into existing infrastructure, the design will focus on creating a solution that is both user-friendly and robust. The system will be tested to ensure it can handle various scenarios, such as different grid conditions and user preferences, without compromising the safety or functionality of the electrical panel.

Overall, the Smart and Grid-Interactive Electrical Panels system will empower homeowners to contribute to grid stability and energy efficiency while offering the convenience of remote control and automation. By focusing on ease of installation, reliability, and seamless operation, the project aims to provide a practical solution that can be widely adopted in residential settings.

## 2.4: Project Objectives

The main objective of our smart grid project is to successfully and accurately switch electric breakers on a grid panel must have specific goals in mind to achieve such a task. The goals that are listed below vary in levels of complexity and abstraction that would have differing amounts of time allocated to fulfilling each goal.

### Basic Goals:

- Design and execute a mobile interface
- Automatic Routines for frequency sensing
- Easy installation

### Advanced Goals:

- Robust user security & verification for mobile app
- User defined scenes for breaker management
- Pathfinding for efficient switch flipping

### Stretch Goals:

- Current sensing
- Robust on-device MCU communication security

## 2.5: Requirement Specifications

The requirement specifications for this project outline key performance and design parameters essential for meeting functional, usability, and marketability goals. Table 1 below details these requirements, spanning critical areas such as frequency detection, installation ease, relay response time, and user interaction.

Each requirement has been assigned a specific value or benchmark to ensure the product is both functional and competitive in real-world applications.

The requirements cover technical performance, like frequency detection within  $60\pm 1$  Hz, which ensures the device can monitor and react to grid stability effectively. Physical constraints, such as the weight being under 30lbs and dimensions under 8 inches, are set to make the product feasible for residential or commercial installation. User experience factors are also addressed with specifications for mobile app latency, which aims to keep relay switching times under 2 seconds, ensuring responsive control via a mobile interface. Additional security features, such as login and verification systems within the app, contribute to user safety and data integrity. Together, these requirements provide a comprehensive framework for developing a reliable, user-friendly, and safe product.

*Table 1: Requirement Specifications*

| Requirement              | Specification                                                                                                                                   | Value        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Frequency Detection      | The project should be able to detect frequency variation in the electrical grid.                                                                | $60\pm 1$ Hz |
| Installation Time        | The project should be assembled with minimal tooling.                                                                                           | <3hrs        |
| Relay Switching Time     | Once frequency is out of bounds or a manual override is present, the circuit breaker is flipped quickly                                         | <10sec       |
| Product Weight           | Minimizing weight is always valuable for consumer products.                                                                                     | <30lbs       |
| Product Power            | Minimizing idle power further improves marketability, it is still inconclusive as to how to test this. Product will have a low quiescent power. | <10W         |
| Mobile Latency           | The time it takes for the product to start flipping switches after the mobile app command will be quick                                         | <2s          |
| Physical Size            | Height and length shall not exceed that of the breaker receptacle itself. Depth as measured from the mounting wall                              | <8"          |
| Product noise            | Product should be quiet. No louder than the switching of the switch itself.                                                                     | <70dB        |
| Product should be secure | App should have a login. Mobile app should have a verification system.                                                                          | 2 systems    |

## 2.6: House of Quality

This diagram is a House of Quality chart, commonly used in engineering design to map customer requirements to specific product requirements. Here's a breakdown of its main sections:

1. **Customer Requirements:** The left side lists customer-focused goals, such as easy installation, remote control via an app, automatic grid response, low cost, durability, energy efficiency, and quick response time. Each requirement is assigned an "importance" rating based on customer feedback.
2. **Product Requirements:** The top row lists technical specifications that address the customer requirements, including frequency detection, relay switching time, power consumption, mobile latency, physical size, product noise, and security.
3. **Interrelationship Matrix:** This matrix links customer requirements to product specifications, with symbols (X, 5, 3, 1) indicating the strength of the relationship. A strong relationship (5) means a product requirement strongly impacts a customer requirement, while a weak relationship (1) indicates minimal impact.
4. **Correlation Matrix:** At the top of the house, the correlation matrix shows how each product requirement interacts with others, with "P" indicating a positive relationship, "N" a negative tradeoff, and empty cells indicating no correlation.
5. **Competitor Analysis:** The right side includes a comparison of your company with competitors (Company A and Company B), showing how well each meets customer requirements.
6. **Organizational Difficulty:** This row indicates the difficulty of achieving each product requirement, rated from 1 (easiest) to 5 (most difficult).
7. **Target Benchmark:** At the bottom, this row shows the target specifications, providing concrete goals for each product requirement.

The House of Quality table offers some unique advantages during the initial phases of project planning, research, and development, namely, an objective ranking of goals. Having an objective ranking allows us to properly allocate time and resources to the systems that are most critical to the project's development. Without it, a team member or the whole group may become fixated on certain issues or try to perfect one of their subsystems while neglecting the other. If we are ever in doubt then we can always refer back to Figure 1 to ensure that we are focused on the most critical aspects of the project first, then dedicate the remainder time to nice-to-have aspects of the project.

Our rankings of the individual categories are set based on our sponsor's demands. These requirements that are not flexible include the mobile app, automatic

switching, switching speed, and ease of installation. If we do not meet these specifications, then we lose any sponsorship. The relationship between these requirements and the rest of goals is illustrated in the previous paragraphs. As for the remaining requirements, we had the luxury of assigning goals we would deem most critical for a mass-produced, off-the-shelf good and scale these goals relative to the sponsor's wishes.

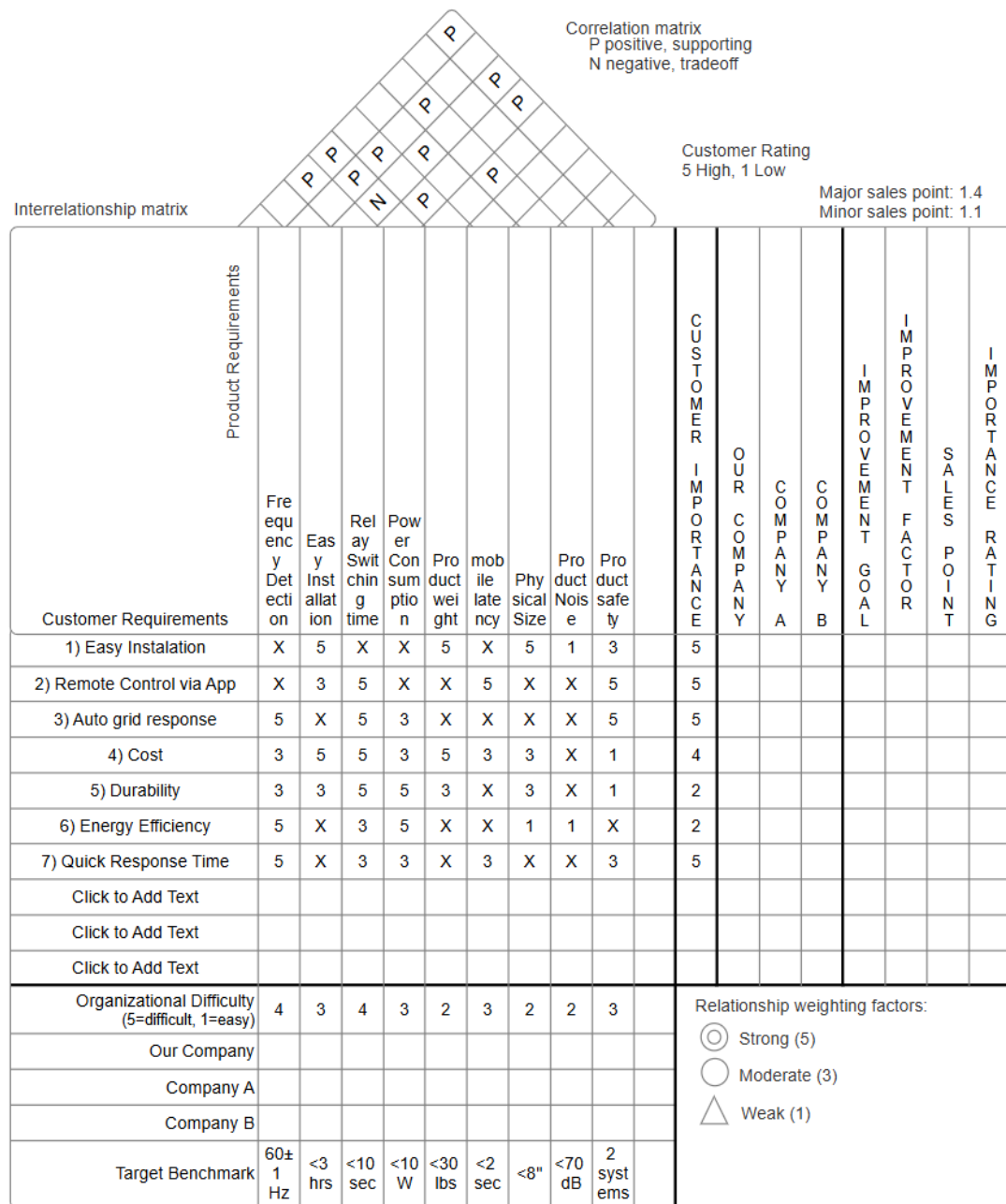


Figure 1: House of Quality

## 2.7: Hardware Block Diagram

The hardware block diagram shown in Figure 2 describes how the various physical components will interact with each other. At the core is the processor. The processor takes input from the frequency sensing and cloud software (mobile app would represent the hardware) and makes decisions. These decisions control the motors which flip or reset the circuit breakers.

The work allocations are also represented in the block diagram. Each person is responsible for researching, designing, integrating, and testing their subsection.

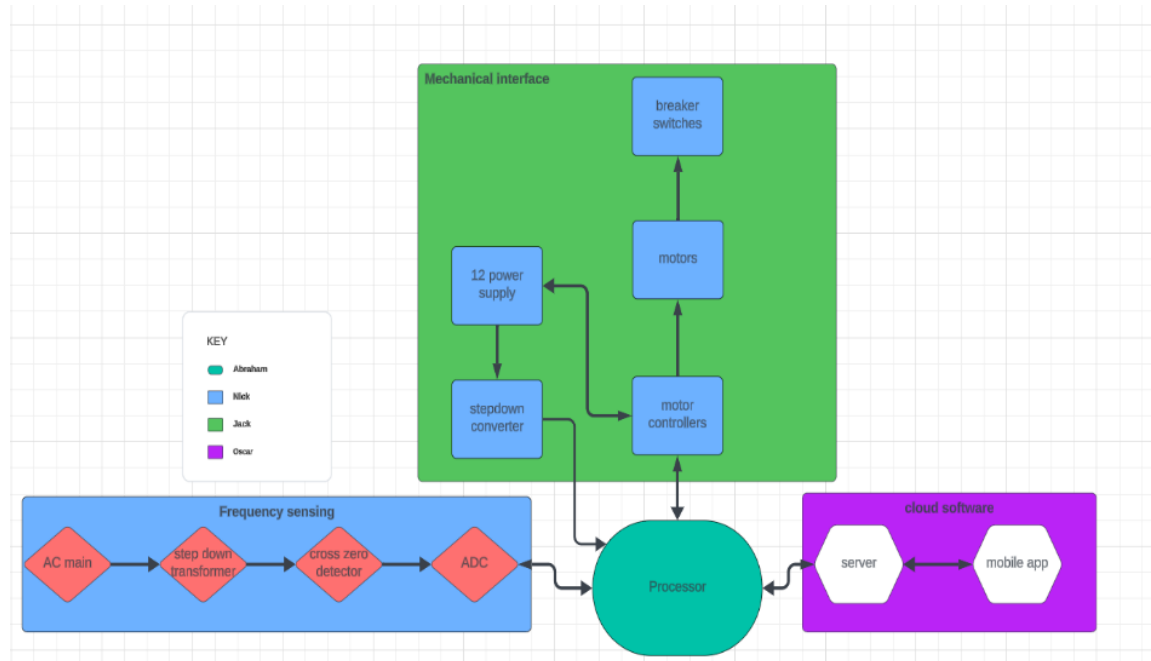


Figure 2: Hardware Block Diagram

## 2.8: MCU Firmware Block Diagram

Figure 3 represents the logic flowchart the MCU will follow to complete the sponsors requirements. The project must accept two inputs: the grid's frequency and the user's choice via the mobile application. The grid frequency response should be automatic. If a significant deviation from the nominal 60Hz is detected, regardless of user input, the project should flip some breakers to lighten the power grid's load. The user's input is another branch of the block diagram. It will bypass the frequency-changing logic and immediately activate the motors to flip the switches. Figure 3 only shows the highest level of operation that the MCU has to perform. Abstracted away in the arrows of the block diagram are the nuances of frequency detection, integration with the motor controllers, and connecting to Wi-Fi. As the paper progresses, a more in-depth look at the embedded and smartphone application programming will be explored.

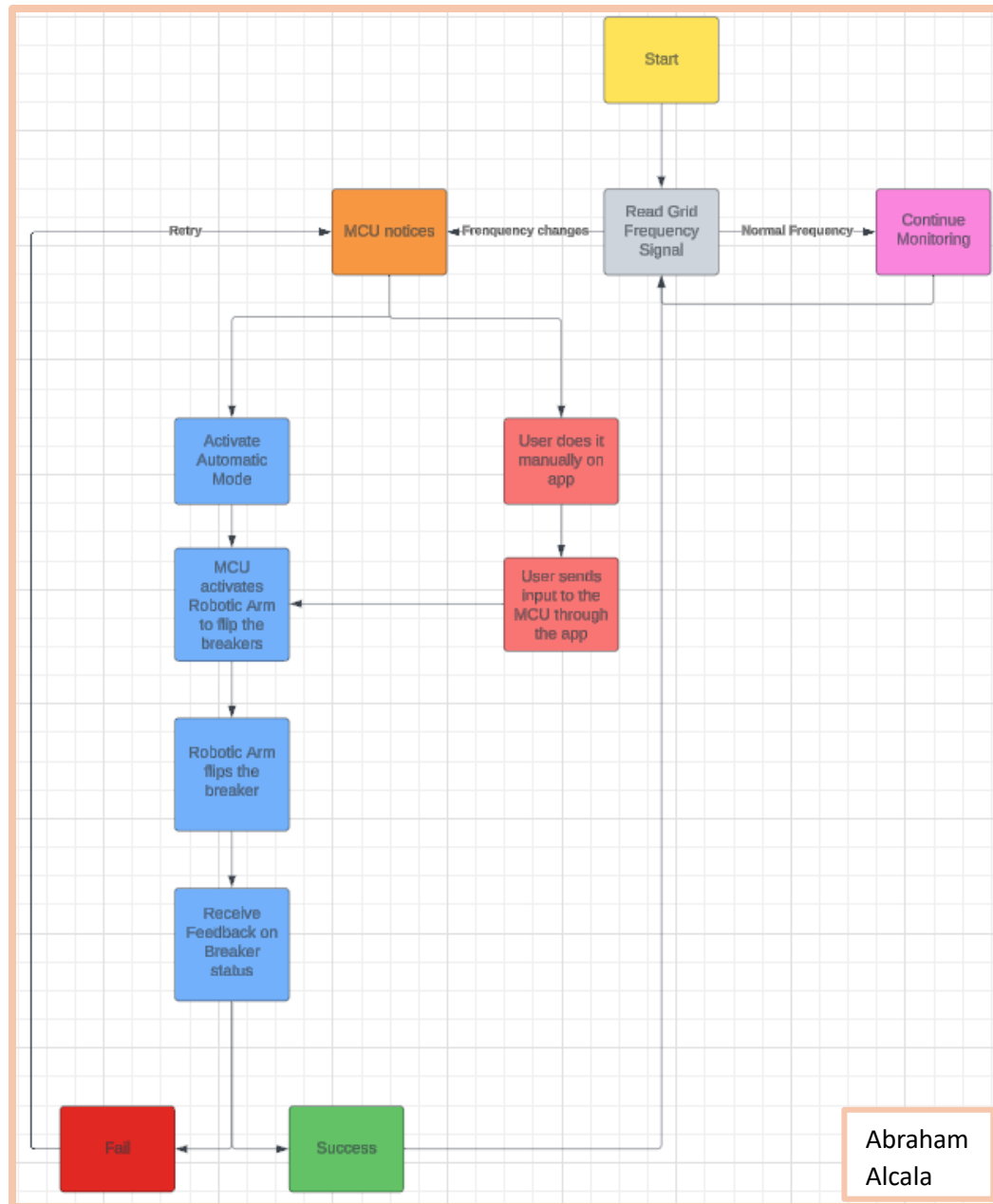


Figure 3: Embedded System BD

## 2.9: Mobile Software Block Diagram

Figure 4 represents the logic flowchart the software for the mobile application will roughly follow. The mobile application will have a variety of action buttons and text inputs that will be parsed to execute a variety of basic functions. The flowchart includes error checking where the path of execution flow will vary based on the presence of the error or lack thereof. The flowchart also delves into primary responsibility for division of labor per the legend shown below.

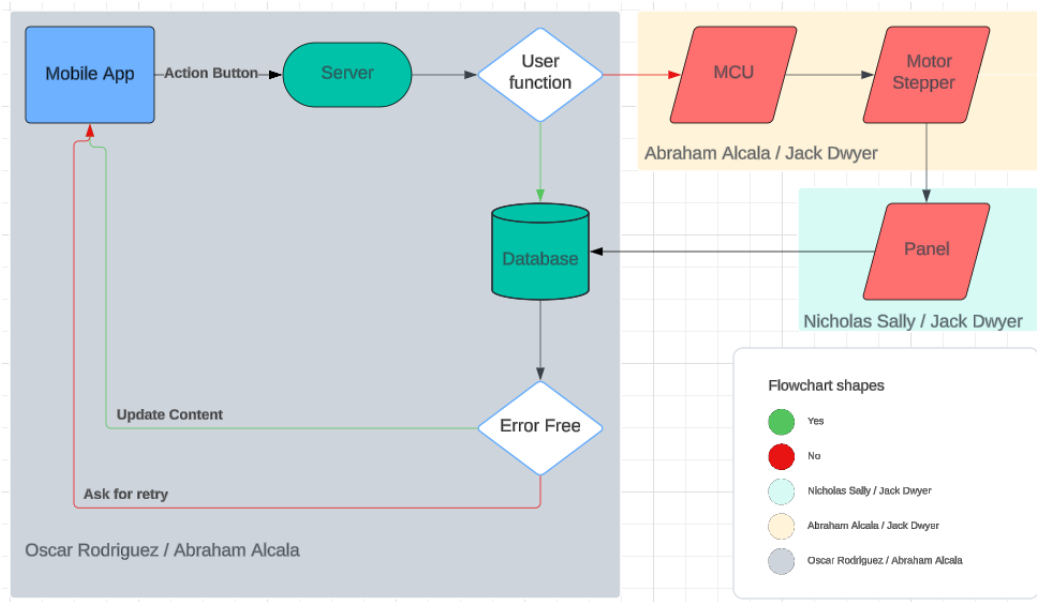


Figure 4: Mobile Software BD

## Chapter 3 – Research

This chapter delves into the technical research underpinning the project's design and development. It provides an in-depth analysis of the various components and systems necessary for building a responsive, automated electrical management system. Each section covers different aspects of the system, from the physical infrastructure required for electrical control to the choice of software and hardware that will enable smooth operation and user interaction. Each section of this chapter provides foundational research necessary for making informed design choices, ensuring that the project is both technically sound and aligned with the target requirements. The detailed comparisons and selection processes across hardware and software components serve as a guide to achieving the optimal balance between functionality, efficiency, and user experience.

### 3.1: Service Panel and Breaker

The first hardware task is choosing which service panel and circuit breaker to develop our project on. Ideally, our project would be perfectly generalizable across all makes and models of service panels and breakers; however, to simplify our initial prototype we chose the most common service panel and circuit breaker combination found in North American single-family homes per our sponsor's request.

#### 3.1.1: Service Panel Specifications

Service panels are distinguishable by a few key factors: voltage, amperage, brand, and the number of circuits supported

The voltage is the simplest of the factors since the American standard is 60Hz with 240V split between two phases. Since this is a standard, all brands will accommodate this factor

The amperage requirement for single family homes appears to be a topic of contention amongst homeowners and electricians. While we could not find a single conclusive article discussing the most popular amperage, there are several discussion boards that addressed the question. The most common conclusion was that most homes do not need above a 100A service panel, but most of them have 200A service panels installed. The reasons range from avoiding the need to rewire the house for the larger load needs of an electrified future (e.g. electric vehicles) to the trivial cost difference between the service panel sizes [1] [2]. With this information, we settled on using the 200A service panel for our project

The last major factor when determining our service panel was the brand. While service panels are a mature technology with minimal differences in function, the companies that make them have proprietary set-ups that make them incompatible with one another. Simply put, if one buys a service panel, he/she must buy the associated circuit breaker; there are no generic parts. Once again, reviewing the various discussion boards and forums shows that Square D, Siemens, and Eaton are some of the most popular brands [3] [4].

Lastly, a service panel can be defined by how many circuits it can support. The National Electric Code (NEC) dictates the minimums and maximums given the size of the home and application. To keep it as general as possible, we used the 20 to 30 circuit breaker estimates provided in a Consumer Reports article about the subject [5].

### 3.1.2: Service Panel Selection

Below is a sample of the service panels that best fit the specifications in Section 3.1.1.

*Table 2: Service Panel Selection Matrix*

|               | GE PowerMark Gold 150 Amp 16-Space 32-Circuit Indoor Main Breaker Circuit Breaker Panel | Eaton BR 100 Amp 20-Space 40-Circuit Indoor Main Breaker Load Center with Combination Cover | Square D Homeline 200 Amp 30-Space 60-Circuit Indoor Main Breaker plug-on neutral Electrical Panel |
|---------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 240V, 60Hz    | X                                                                                       | X                                                                                           | X                                                                                                  |
| Amperage      | 150                                                                                     | 100                                                                                         | 200                                                                                                |
| # of circuits | 16                                                                                      | 40                                                                                          | 60                                                                                                 |
| Cost (\$)     | 149.00                                                                                  | 108.00                                                                                      | 169.00                                                                                             |
| Notes         | Not in preferred brand                                                                  | Not immediately available                                                                   | Includes several breakers                                                                          |

The Square D Homeline 200 Amp 30-Space 60-Circuit Indoor Main Breaker plug-on neutral Electrical panel was chosen for our project. While it is the priciest of the panels, it was immediately available for purchase and included several compatible breakers. The number of supported circuits is higher than the typical number found in our research but will not dramatically alter the project in any meaningful way.

### 3.1.3: Circuit Breaker Specifications

The circuit breaker serves as the local over current protection device for entire rooms or exceptionally power-hungry devices. In the case of excess current draw, which could lead to fires or shock, the circuit breaker opens the circuit cutting off power.

Since we have chosen a Square D service panel, we need to choose the appropriate Square D circuit breakers. The service panel came as a kit, and in this kit were three HOM120 Homeline 20 Amp single-pole circuit breakers and two HOM230 Homeline 30 Amp double-pole circuit breakers. This makes sense since the most common circuit breaker size is 15A or 20A [6]. While the 30A circuit

breaker is designed for higher-power applications like water heaters, HVAC, and electrical dryers [7]. Our purchase has covered the basic breaker sizes found in a single-family home, but other breaker sizes exist. To keep the project as generalizable as possible, we must address the atypical breakers.

The two most common atypical circuit breakers we could find were single-pole tandem and high-current applications. The former consists of two switches attached to a single circuit breaker body. This option is popular when multiple rooms/spaces need to be protected but the number of spaces available on the service panel is limited. The challenge posed by this style of breaker is the switch width. A typical breaker has a switch width of approximately 1 inch. The tandem design means the switches are now one quarter of their original width (approximately 0.5 inch). This forces our actuating device to be more precise. The latter consists of two typical breakers that share a single switch. Their design is almost identical to the 30A breaker, but the biggest difference comes from the force required to turn it on and off. Larger currents always have strong switching mechanisms inside of them. To maintain the generalizability, the force provided by the actuating device must accommodate the additional force.

### 3.1.4: Service Panel Selection

Referring to our local hardware store's website for viable options, the circuit breakers below represent those closest to the specifications.

We chose the Square D Homeline 2-15 Amp Single-Pole Tandem Circuit Breaker. There was a 20A option, but the difference in the force necessary to flip the switches was negligible, so we chose the cheapest option. For the high-current breaker, we settled on the Square D Homeline 60 Amp 2-Pole Circuit Breaker(HOM260CP). Square D does make circuit breakers with high amperages including 70A, 80A, and up to 125A. However, these breakers were significantly more expensive (\$50-\$100 compared to the 60A's \$20) and not readily available. It is also unclear if a typical single-family home would be able to support such ampacity,

With the addition of the single-pole tandem and high-current circuit breakers, we are confident that most circuit breaker test cases can be represented in this project.

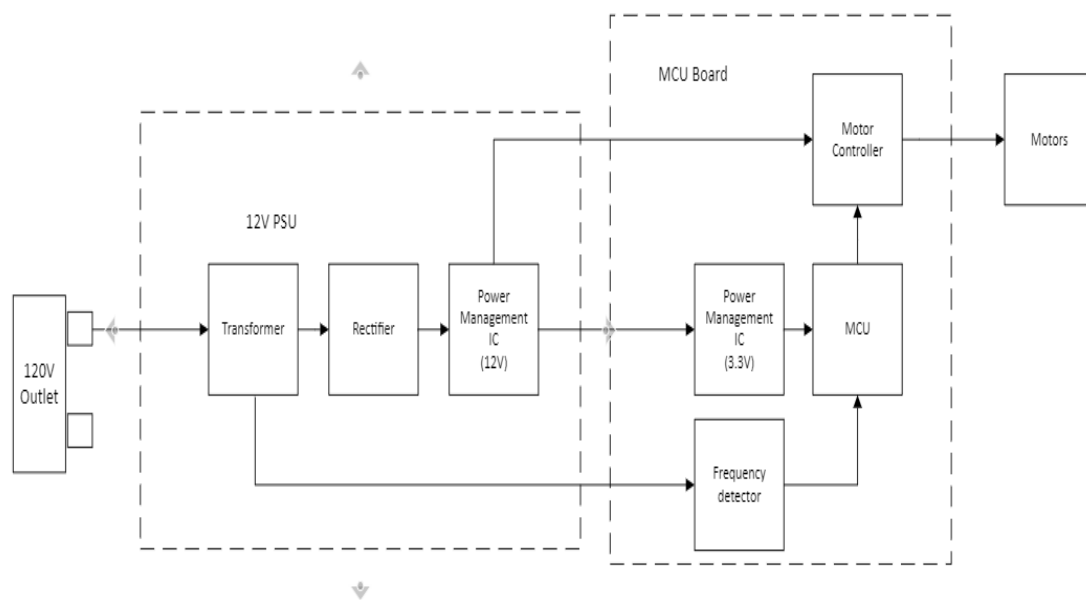
## 3.2: Power Supply

Included in the scope of this project is a power supply. This supply must provide both a 12V and a 3.3V DC output from a 120v AC input, enough amperage to sufficiently supply the motors, esp32, and other peripherals. Additionally, the supply must sufficiently precise as to not introduce undesired noise into the system.

### 3.2.1: Power Supply Overview

Figure 5 shows how we intend to power the project. A 120V wall outlet feeds a transformer. This transformer adjusts the voltage and current to acceptable levels. The signal is then rectified and fed into the 12V Power Management Integrated

Circuit (PMIC). This first PMIC acts as the source of all of the regulated power on the board, as no other part of the project will interface directly with the unregulated wall-outlet with any meaningful power draw. The first PMIC first feeds into a 3.3V PMIC. The 3.3V PMIC serves to power both the frequency detector and the MCU. The first PMIC is directly connected to the motor controller since it is expected the motors to use 12V and will need up to 4A, so any non-direct path will dramatically limit the current draw. Combined, we are expecting the project to consume a peak current of approximately 5A when the motors and the microcontroller are used together. We expect the quiescent current to fall well below since the MCU, frequency detector, and biasing power will be dramatically less than the motors. We estimate the final quiescent power at around 5W.



*Figure 5: Power Flow Block Diagram*

### 3.2.2: The Transformer

The transformer serves as the first component of the project's power system. Its role is to limit the amount of power available to the rest of the project from the wall outlet. This section covers the various transformer types, the pros and cons of each, how we properly size the transformer, and finally our choice of transformer.

#### 3.2.2.1: Transformer Overview

The transformer is the first step for both power supply units. A transformer takes a given power value and adjusts its ratio of current and voltage depending on its construction. The transformer is composed of a primary and secondary coil of wire which is usually wrapped around a ferromagnetic core. When this arrangement is exposed to alternating current the change in magnetic flux induces a higher voltage at lower current or lower voltage at higher current depending on which coil has more turns of coil [8].

While all transformers operate on a similar principle, a transformer's construction can be optimized to perform one task. Choosing the operate style of transformer is imperative to a properly function power supply.

Our transformer selection will need to accommodate a 5A peak draw and step down at or below 24V. Additionally, the transformer should minimize its distortion of the 60Hz signal coming from the outlet to ensure accurate frequency measurements.

#### *3.2.2.2 Step Up vs. Step Down vs. Isolation Transformers*

At the highest level there are only three types of transformers: step-up, step-down, and isolation. The difference between these transformers lies in the ratio of the number of turns between the primary and secondary coil. In a step-up transformer this ratio is always less than one, in a step-down transformer this ratio is always greater than one, and in an isolation transformer this ratio is always equal to 1 [9].

The name of these transformers is also indicative of their application. A step-up transformer is used when higher voltage is necessary and a step-down transformer when lower voltage is necessary. The only unique case is the isolation transformer. Since its winding ratio is always one, it does not offer any power transformation. Instead, the isolation transformer is primarily used as a safety and noise suppression device. Unlike the step-up or step-down transformers, there is no conductive path between the primary and secondary coils. Since each path is galvanically isolated, the users and electronics on the secondary side of the transformer are safe from electrical shock. The noise suppression is a quirk of inductive coupling. Since the inductors are close enough together, any noise common to both coils will be cancelled out [10].

#### *3.2.2.3: Step Up vs. Step Down vs. Isolation Transformers*

The types of transformers can be further refined by their application and construction. Some of the most common and applicable sub-categories of transformers are power, auto, and current transformers.

The power transformer is the most basic and common of all the transformer sub-categories. At their core, power transformers are designed to either step-up or step-down the voltage from one source and feed it to another with minimal frequency distortion or power loss. Since they can be used from tens of volts to thousands of volts, they can have various insulators and cooling methods [11].

The auto transformer uniquely only has one set of coils. This single coil winding acts as both the primary and secondary.

The autotransformer can be a step up, step down, or both depending on where the input and output voltage are sourced. For example, figure below, the input voltage is taken across points "A" and "B" while the output voltage is taken from "B" and "C" Since the input has more windings than the output, the output voltage is lower. The converse would also be true if the input and output were reversed [12].

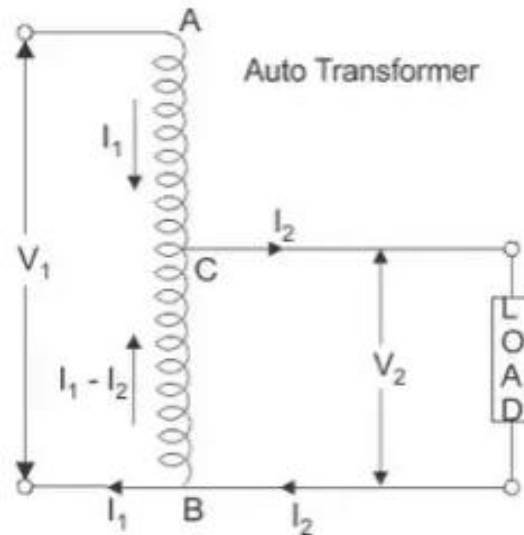


Figure 6: Example of an autotransformer set-up from Electrical.

The single coil set up has pros and cons. The size, weight, and cost associated with autotransformers is exceptionally low since less material is used in the actual construction. Furthermore, an autotransformer's output is not limited to one voltage level. It is common for autotransformers to have several taps across the coil to provide various output voltages. The autotransformer's greatest con is its lack of isolation. Since there is a single conductive path for the entire transformer, the load-side device could be exposed to full power of the source if careful circuit designed is not exercised.

The last of the common transformers is a current transformer. The current transformer takes rotating magnetic fields around a wire and transduces it into a proportional voltage via a magnetic medium [13]

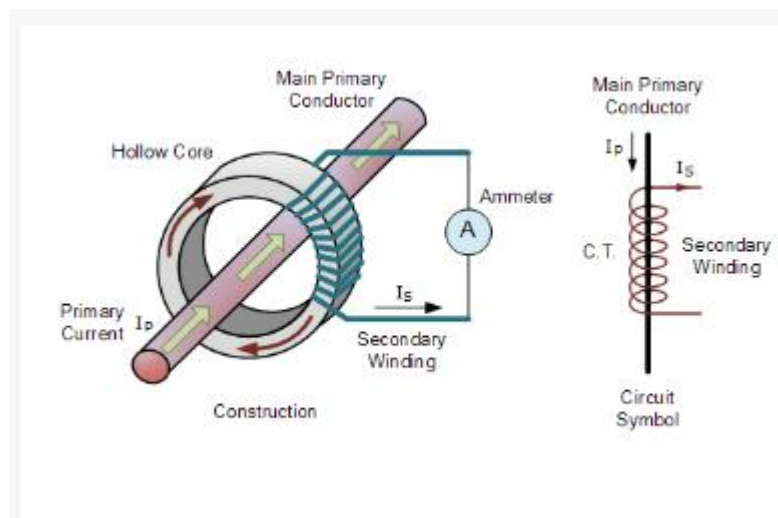


Figure 7: Current sense transformer from Electronics-Tutorial.

Since this offers the best isolation between the primary and secondary coils. Its primary use is as an intermediate step when taking current measurements, and by extension power and frequency information, from high current wires. The current transformer is a type of instrumentation transformer, which means it is designed to be as accurate as possible.

The current transformer is not designed to output a lot of power. We simply could not plug this type of transformer into a wall for power regulation. Therefore, it is immediately disqualified for use. It may be used in testing and validation, but it cannot be used in our PCBs.

#### 3.2.2.4: Transformer Comparison Summary

We have decided on a step-down power transformer for our project.

The reason behind a step-down transformer is trivial: the voltage must be reduced from 120V to approximately 12V. As for the other transformer categories, the power transformer appears to be the most well-rounded for our application.

The autotransformer would reduce some of the power management complexity by allowing us to tap along the coil to achieve 12V and 3.3V without an additional transformer or PMIC. This advantage was considered, however, since the autotransformer provides no isolation from the mains voltage there were safety concerns amongst the group. Similarly, the project is designed for the public, we want to eliminate the possibility of accidental shocks and electrocution because our product was not isolated from the mains voltage.

*Table 3: Transformer Types Comparison Matrix*

|                          | Power Transformer | Autotransformer    | Current Transformer |
|--------------------------|-------------------|--------------------|---------------------|
| Power Handling           | Very High         | 50-1000kVA or more | Low                 |
| Cost                     | \$5 - \$500       | \$50 - \$250       | \$2 - \$1500        |
| Complexity               | Low               | Moderate           | High                |
| Multiple voltage support | Yes               | Yes                | Yes                 |
| Isolation from Source    | +4000Vrms         | None               | +500Vrms            |

The current transformer was vetoed because of its power handling capability and cost. Since the current transformer is designed to provide signal information from a high-power source, it is not designed with its own power-handling abilities. We debated using the cheaper power transformer for power and the current transformer for the frequency detector but ultimately decided against it. We are expecting a 60Hz signal. At this low of frequency, it would be more economical to tap into the power transform and filter out any of the noise with a multistage filter and possibly software manipulation than it would be to have two dedicated systems.

### 3.2.2.5: Transformer Selection

The final transformer selection was based on values from Hammond Manufacturing article describing the practical values for specifying a transformer depending on load. We chose the equations that would best account for the bulk capacitance at the input of the PMIC and the rectification scheme that would minimize the voltage difference between input and output. This will help expand our options for PMIC further down the circuit chain. The equations are below [14]

$$V \text{ (Peak) D.C.} = 0.71 \times \text{Sec. V A.C.} \quad (1)$$

$$I \text{ D.C.} = 1.00 \times \text{Sec. I A.C.} \quad (2)$$

Evaluating Equation 1 for an expected peak DC voltage of 12V yields a transformer of 17Vrms. To account for diode losses and PMIC biasing, we looked for a transformer between 18-24Vrms. Since there is a one-to-one relationship between the AC and DC voltage, the transformer should be capable of outputting 4.4-5A.

We chose the 241-8-20 Transformer from Signal Transformer. We believe that it is easier to have more power and restrict it downstream instead of risking not having enough power due to unexpected losses. The reason for the 241-8-20 instead of the 187F20 was the price difference. Based on the datasheets and information compiled from Digi-Key, we could find no discernible difference between the two transformers, so we chose the cheaper option.

*Table 4:Transformer Selection Matrix*

| Transformer Name  | 166P18                                                   | 1182N9                                                      | 241-8-20                         | 187F20                           |
|-------------------|----------------------------------------------------------|-------------------------------------------------------------|----------------------------------|----------------------------------|
| Description       | Transformer, low voltage, 115V 60 Hz., 90VA, 18 C.T., 5A | Power transformer, toroid, 80 VA, with 9V @ 8.88A secondary | PWR XFMR LAMINATED 100VA CHAS MT | PWR XFMR LAMINATED 100VA CHAS MT |
| Price             | \$39.46                                                  | \$64.00                                                     | \$20.91                          | \$32.18                          |
| Secondary Voltage | 18VAC CT                                                 | 18VAC CT                                                    | 20V                              | 20V                              |
| Secondary Current | 5A                                                       | 4.44                                                        | 5A                               | 5A                               |
| Manufacturer      | Hammond Manufacturing                                    | Hammond Manufacturing                                       | Signal Transformer               | Hammond Manufacturing            |
| Stocked           | Yes                                                      | Yes                                                         | Yes                              | Yes                              |

### 3.2.3: Rectification

Rectification is the first step in voltage regulation following the transformer. It is critical to choose a rectification scheme that will provide the maximum amount of power transfer while minimizing heating, size, and loss. This section explores various rectification techniques and our final selection for the project.

### 3.2.3.1: Rectification Overview

After the power is stepped down, the second step is to rectify the AC signal. The rectifier generally consists of diodes cleverly arranged to keep a single voltage polarity (it could be positive or negative depending on the arrangement and application). The diodes are followed by some form of low-pass filter. Most literature only shows a capacitor parallel to the load. While this is the most common filtering method, the signal could be filtered with a series inductor or both components.

### 3.2.3.2: Half Wave vs. Full Wave Rectification

The first choice made when choosing a rectification topology is half wave or full wave. In a half wave rectifier, an AC signal has one polarity removed entirely. This creates a hump with a rising voltage, a peak, and then a falling voltage. During the opposite polarity's time, the signal simply drops to 0V. For a full wave rectifier, the diode arrangement causes the opposite polarity to "flip" its orientation. This functionally doubles the number of times the voltage is at the higher polarity. This results in a bumpy wave pattern with only brief drops to zero. The rectification types are best summarized in Figure 8 from Rohm Semiconductor's article.

We chose full wave rectification for two reasons: power delivery and ripple voltage. The power delivery for a half-wave rectifier is half of a full-wave rectifier. Since the transformer we have chosen outputs a maximum of 5A, we need all the transformer's power to be available. Only the full wave rectifier can provide this. The full wave rectifier also provides a smaller ripple voltage. The PMIC expects a stable input source since it is a DC/DC converter. If we used a halfwave rectifier, we would need to dramatically increase the amount and variety of capacitance/inductance available or alter our PMIC topology to accommodate the voltage swings. With the full-wave rectifier, the signal should be just DC enough to keep the PMIC and its supporting passives simple.

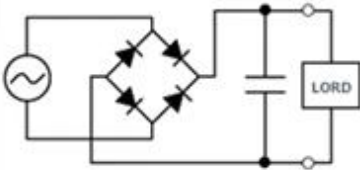
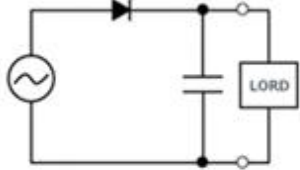
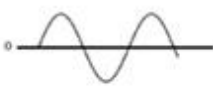
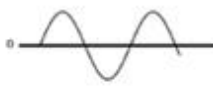
|                                                | Full-Wave Rectification                                                              | Half-Wave Rectification                                                               |
|------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Circuit Configuration                          |  |  |
| Input Voltage Waveform                         |   |  |
| Voltage Waveform After Rectification           |   |  |
| Voltage Waveform After Rectification Smoothing |   |  |

Figure 8: Rectification Comparison from Rohm Semiconductor [15]

### 3.2.4: The Power Management Integrated Circuit

We expect our project to incorporate two PMICs to accommodate the 12V required for the motors and motor controller and the 3.3V required for the MCU and other digital logic signals.

#### 3.2.4.1: *Power Management Integrated Circuit Overview*

The PMIC serves as the regulator between the rectified transformer output and the sensitive electronics downstream. The PMIC converts the rippled waveform into a near-perfect constant voltage at the desired amperage. The PMIC has similar variations to the transformers. To keep the discussion brief only the most relevant differences in PMIC will be discussed, and if a topic overlaps with the transformer section, then that section will be referenced.

#### 3.2.4.2: *Linear vs Switch Mode Power Supplies*

PMICs come in either linear or switch mode configurations. They both have different strengths and weaknesses, so choosing the correct configuration for our project will be critical for proper power delivery.

The linear configuration is the simplest. Following the power source is a linear element (usually a resistor) and a control circuit. The control circuit provides feedback from the output to maintain a constant output voltage—ideally—regardless of the output current [16].

This configuration has two primary benefits. First, the noise of this circuit is incredibly low. Since there is no switching behavior, the circuit adds very little noise to the output signal. Second, is the circuit complexity. Since the excess power is simply burned off, there is minimal external hardware necessary for the PMIC to work. The linear PMIC is not without its disadvantages. The most profound is the power loss. Instead of storing the excess energy into a capacitor or inductor, the energy is instead released as heat in a resistor. This can result in poor efficiency and lots of heat being generated if the difference between the input and the output voltage is significant, and especially if high currents are expected. Another consequence of the resistive element is that linear regulator cannot exist in the boost configuration, since resistors cannot store energy. This further reduces the number of applications the PMIC can be used [17].

The switching regulator takes advantage of how electricity interacts with inductors and capacitors. Following the source are various configurations of inductors and capacitors with some sort of feedback network pulsing the input. The pulsing action induces a back electromotive force (EMF) in the inductors which can boost or buck the input voltage. The capacitors help smooth out voltage fluctuation after boost/bucking. They also provide extra current without sacrificing voltage [18].

The pros and cons of the switching regulator are functionally the opposite of the linear regulator. For the pros, the switching regulator is dramatically more efficient. The capacitor and inductors are power storage elements, so no energy from the source is intentionally lost. Since the efficiency is higher, the amount of heat dissipated by the PMIC is also dramatically lower. For the cons, the switching

produces a lot of electric noise. This noise may affect downstream electronics directly via the connections to power, or indirectly through radiation and the harmonics from the pulses. The complexity also increases. Pulsing architecture requires additional components. In addition, most switching regulators on the market have the ability to tune the switching frequency to optimize efficiency. This is done with biasing resistors and capacitors at various pins on the IC [19].

*Table 5: PMIC Configuration Matrix*

|                     | Linear                                         | Switching                               |
|---------------------|------------------------------------------------|-----------------------------------------|
| Modes Supported     | Buck                                           | Buck, boost, flyback, buck-boost        |
| When most efficient | Low voltage different, low current consumption | Within tolerances of datasheet, tunable |
| Complexity          | Low                                            | High                                    |
| Cost                | Low                                            | High                                    |
| Noise               | Minimal Additional Noise                       | Additional noise at switching frequency |
| Heat Generated      | Depends on input/output differential           | Always low                              |

Based on the voltage difference and the current's magnitude, we needed to use a switching regulator for both the 12V and 3.3V supply. We expect 20Vrms from the transformer. This translates to approximately 29V peak voltage from the output. Even accounting for diode drops and copper resistance, the voltage difference will still be greater than 10V. This difference in conjunction with the 4-5A necessary to drive the motors means that any linear voltage regulator would overheat for our application. The same voltage difference problem is present when converting the 12V to 3.3V for the MCU, even if the current draw is less.

#### **3.2.4.3: Switching PMIC Selection**

From our research, we needed two switching voltage regulators, one that can provide 12V 5A output and one with a 3.3V 1A output. Ideally, the PMIC will have incredibly strong line regulation, which means that the output will not change if the input supply changes.

To address this design challenge, we used Texas Instruments' Webench Power Designer. Using the input voltage of 26-27V and an output voltage of 12V with an expected peak current of 5A yields circuits with the PMICs featured in Table 6 . TI Webench power Designer produced dozens of solutions. The table represents the most unique and most recommended from the software.

*Table 6: 12V PMIC Selection Matrix*

|                     | LM5148<br>RGYR | TPS56637<br>RPAR | TPS56837<br>HRPAR | LMR51450S<br>DRRR | LMR51450<br>FNDRRR |
|---------------------|----------------|------------------|-------------------|-------------------|--------------------|
| BOM Count           | 17             | 15               | 14                | 13                | 13                 |
| Estimate Cost       | \$1.93         | \$2.75           | \$3.17            | \$2.67            | \$2.67             |
| Efficiency          | 94.588%        | 96.398%          | 96.882%           | 92.963%           | 92.963%            |
| Switching Frequency | 2.075MHz       | 561.958kHz       | 480.71kHz         | 349.674kHz        | 349.674kHz         |
| IC Power Dissipated | 1.22W          | 2.24W            | 1.93W             | 4.54W             | 4.54W              |
| In Stock            | Yes            | Yes              | Yes               | Yes               | Yes                |

This exercise was repeated for the 3.3V 1A PMIC. The input parameters were 11.5-12.5V input voltage at a 12V nominal, an output of 3.3V, and a maximum output current of 1A.

*Table 7: 3.3V PMIC Selection Matrix*

|                     | TPS561201DDCR | TPS561208DDCR | TPSM861253RDX |
|---------------------|---------------|---------------|---------------|
| BOM Count           | 8             | 8             | 7             |
| Estimate Cost       | \$0.71        | \$0.71        | \$0.69        |
| Efficiency          | 92.567%       | 92.567%       | 83.124%       |
| Switching Frequency | 520.109kHz    | 520.109kHz    | 1.4 MHz       |
| IC Power Dissipated | 0.26W         | 0.26W         | 0.67W         |
| In Stock            | Yes           | Yes           | Yes           |

For the 12V PMIC, we settled on the LMR51450SDRRR. We vetoed the LM5148RGYR and TPS56637R due to their complexity. The high Bill of Materials (BOM) count means additional cost of external components and additional complexity during hardware design. The low IC power dissipation is enticing, but the difference between 1.22W and 4.54W should be trivial for our purposes. To our knowledge, we did not need to add a heat sink to account for the difference. The TPS56837HRPAR and the other two are incredibly similar. Based on the datasheets, the only discernable difference is the price. As such, we went with the cheapest option. The final selection between the LMR51450SDRRR and the LMR51450FNDRRR was completely arbitrary. In terms of function, these PMICs are identical. The only difference between the two is the package dimensions, since there are no size constraints on the project, the LMR51450SDRRR was chosen.

We chose the TPSM861253RDX for the 3.3V PMIC. The TPS561201DDCR and TPS561208DDCR are nearly identical products. The only difference the TPSM861253 is only rated to operate up to 80 C while the TPSM561201 is rated to operate up to 120 C. Our project will be operating nowhere near those ambient temperatures, so for our purposes, the circuits are identical. The main reason we chose the TPSM861253RDX is that it does not have an external inductor. Inductors require a lot of extra work and special consideration when designing Printed Circuit Boards (PCBs). We chose to minimize the complexity of the hardware design at the expense of some efficiency.

For both ICs, we did not consider the switching frequency when making our choice. Our reasoning is simple: all the PMICs operating frequencies are significantly greater than 60Hz from the outlet. The lowest switching frequency from Table 6 and Table 7 is 349.674kHz. At this frequency, even if the decoupling capacitors at the output do not eliminate the noise, a simple filter can be added further eliminate the noise with minimal modification to the circuit.

## 3.3: Frequency Detector

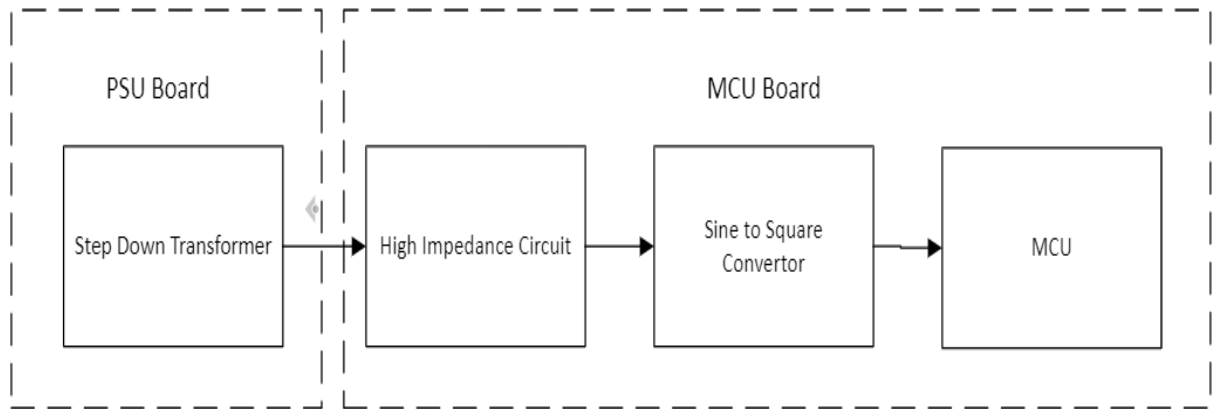
In order for the system to be able to detect and respond to power grid conditions, it is necessary to have a high accuracy frequency detector. This will be used to measure the frequency of the power grid itself. When the power grid is under load, the frequency decreases. This variance is miniscule: the U.S. power grid rarely varies by more than 0.05 Hz in either direction from its designated frequency of 60 Hz. [20] If our system can detect these small changes in frequency, it can respond appropriately, turning off breakers when a low frequency is detected, freeing up the grid. Likewise, it could respond in the inverse manner, turning on appliances when a high frequency is detected, indicating the grid currently has low load placed upon it. If the use of these devices became commonplace, managing grid load would in turn become simpler, as each household would respond dynamically to grid conditions. This could mean the end of current measures used to manage power grids near their maximum capacity, such as rolling blackouts

### 3.3.1: Frequency Detector Overview

Figure 9 shows the expected breakdown of the frequency detector from Figure 5's block diagram. This whole circuit will be housed on the MCU's PCB, but it does have a connection to the PSU board via a wire.

At a system level, the frequency detector's goal is to convert the 60Hz sine wave into 60Hz square wave to feed into the MCU. The MCU will use the time between high and low levels to extrapolate the frequency of the signal and control the motors accordingly. The frequency detector should convert the signal in a way that minimizes the loading effect, its additional noise, and any other non-ideality that could distort the input signal.

A similar frequency detector was built in a previous senior design project "Grid Stabilizing Frequency Detector." We used their project to guide our research.



*Figure 9: Frequency Detector Block Diagram*

### 3.3.2: Frequency Measuring Methods

The bulk of this research is based on a previous senior design team at the University of Central Florida. They concluded their research by choosing a cross-zero detector as their means of frequency detection. Based on the application and talks with our sponsor, we also chose a cross-zero detector; however, there are different topologies that we can choose from [21].

#### 3.3.2.1: Cross-Zero Detector Overview

At its core, a cross-zero detector is a device that responds to an input waveform crossing zero. This output is typically in the form of pulses or a square wave. The choice of output depends a lot on the input signal. If the input signal is a sawtooth or triangle wave, then the pulses may be a more accurate representation of the frequency given the shape. Since the input signal is a sine wave, we have both options at our disposal.

#### 3.3.2.2: Cross-Zero Detector Technologies

The aforementioned cross-zero detector comes in many different topologies. We explored some of the most popular including the optocoupler, NPN transistor, and comparator circuit. There will also be a discussion about using operational amplifiers instead of a comparator in case we need to make that switch given resources or complexity.

##### 3.3.2.2.1: Optocoupler

An optocoupler is a discrete IC that consists of a light source, usually a light emitting diode (LED), and a light-sensitive phototransistor [22].

To produce a Cross-zero detector with an optocoupler, the input signal is fed into the light source side of the device. Whenever the signal is sufficiently large (can be monopolar or bipolar depending on number and orientation of diodes), the light source shines bright enough to trigger the gate/base of the transistor. The transistor acts as an open circuit, which shorts the output to 0V. Once the signal drops below the forward voltage of the LED, the LED will turn off. This will also turn

off the phototransistor and create a open circuit to ground. Since the output is tied to a pull-up resistor, this causes the output to be pulled high indicating a zero-cross has occurred [23].

The most unique benefit of optocouplers is the galvanic isolation. Much like the transformers in the previous section, there is no conductive path between the LED and the phototransistor. This means that all of the downstream circuitry, the MCU in our case, would be protected from the high power of the transformer.

#### 3.3.2.2.2: NPN Transistor

Fundamentally, the NPN transistor cross-zero detector functions identically to the optocoupler but without the galvanic isolation. Instead of the input signal exciting an LED to drive the gate/base of the phototransistor, the NPN transistor's base is conductively bonded to the input signal. Besides the connections, the core behavior is identical to the optocoupler: when the input voltage is above the forward voltage of the diode, the transistor shorts out the output to zero voltage. When the input voltage drops below the forward voltage of the diode, then the output voltage swings high, producing a pulse.

#### 3.3.2.2.3: Comparator Circuit

A comparator is an amplifier with two, high-impedance input terminals that correspond to the positive and negative supply voltages of the IC. The high-impedance inputs allow for the comparator to sample the voltage with minimal distortion from the loading effect. A comparator differs from an operational amplifier due to its lack of feedback. Since there is no feedback, the amplifier always operates at its maximum open-loop gain with no means of controlling it.

A zero-cross detector with a comparator circuit is very simple to construct. One of the input terminals is tied to 0V. This acts as the reference voltage that determines the switching time. The second terminal is tied to the input, either directly or through a voltage divider. Which terminal is the reference, and which is the input depends on the desired phase difference between output and input. In the inverting set-up, the input and output polarities are flipped. For example, if the sine wave was transitioning from -1V to 1V, the output of the comparator would be the negative supply rail. By contrast, the non-inverting configuration would output the positive supply voltage. For our project, it does not matter which configuration we use [24].

#### 3.3.2.2.4: Operational Amplifier vs. Comparator

As mentioned in the previous section, the major difference between a comparator and an operation amplifier is feedback. The operational amplifier allows the output signal to be fed back into the input to control gain and be used for various calculations. By contrast, the gain of the comparator is always at its maximum and always goes rail-to-rail.

While a dedicated comparator can never be used as an operation amplifier, an operational amplifier can be used as a comparator if no feedback is provided in the circuit. This comes with a few advantages. First, operational amplifier can accept positive or negative input voltages. Voltage comparators can only accept positive

inputs, so we would have to tap the input after the transformers rectification or include a diode to remove the negative voltage entirely. With an operational amplifier, we could tap the input signal before or after rectification and still get a functioning cross-zero detector. Second, is simplicity. Most comparators require some sort of reverse polarity protection. Using an operational amplifier would eliminate the need for this additional complexity entirely.

Our research concludes that we should use a dedicated comparator instead of using an operational amplifier in an open-loop configuration. The two most relevant reasons for our project are speed and hysteresis.

Operational amplifiers are designed to be used with negative feedback, which minimizes the difference between the two inputs. In electronics design, one is introduced to the “virtual short” when doing analysis, this rule cannot be obeyed in comparator mode. If the difference between the inputs is too high, the operational amplifier will saturate. Once in saturation, the output swings of the operational amplifier dramatically slow down, often longer than the typical propagation delay presented in the datasheet. Hysteresis is best defined as the tolerance between state changes. All real circuits are noisy. This noise can temporarily push a signal above or below the switching threshold of some devices. Without hysteresis providing some tolerance, rapid switching could occur. An operational amplifier does not come with built-in hysteresis. For this reason, it is possible for the output to display the aforementioned rapid, uncontrolled switching. It is possible to introduce hysteresis to an operational amplifier but at the expense of more complicated circuitry. All dedicated comparators come with built-in hysteresis, so no additional circuitry is required. Another type of speed is the rise and fall time. When a signal transitions from high to low, there is a finite amount of time to transition between the two states. If the rise/fall time is too slow, then the software may not detect the rising or falling edge of the signal. If this happens, the MCU will not interpret the incoming signal as digital and fail to process it according.[25].

Based on the research, we used a dedicated comparator in our frequency detector. While we do not expect the non-idealities of real operational amplifiers to affect our project significantly due to the low power, low frequency, and low switching speed requirements, we also do not think the added uncertainty is worth the marginally lower price and simplicity.

*Table 8: Comparator vs. Operational Amplifier Comparison*

|                                   | Comparator     | Operational Amplifier |
|-----------------------------------|----------------|-----------------------|
| Price                             | More expensive | Less expensive        |
| Design for open-loop gain         | Yes            | No                    |
| Large input differential accepted | Yes            | No                    |
| High speed                        | Yes            | Possible              |
| Included Hysteresis               | Yes            | No                    |
| Rail-to-rail Output               | Yes            | Possible              |

### 3.3.2.3: Frequency Detector Technology Selection

For our project, we decided to use the comparator circuit as our means of frequency conversion. We believe that the comparator circuit offers the simplest, easy to use solution for our needs. We are trying to convert a very stable sine wave into a signal that can be easily read by an MCU. While pulses might provide so advantage in accuracy, assuming the pulses are narrow enough, the comparator is designed to convert an analog signal to a digital one, which is exactly what we needed. The crossing location is also significantly more tunable in the case of the comparator circuit when compared to the NPN transistor. The comparator has a minimum cross-zero location of 0V and up to the positive supply rail. By contrast, the NPN transistor's minimum cross point is the forward biasing voltage of a diode and no less. Similarly, the optocoupler's crossing point could be as high as 3V depending on the color of LED used in the process. We made the input signal as tunable as possible, but having a smaller minimum is more accommodating than a higher maximum.

Outside of technology, there is personal preference. After consulting with our group, the engineers in charge of the programming thought that having a continuous square wave input would be significantly easier to program than the pulses provided by the alternative methods. In addition to the other benefits the comparator offers, the comfort of the engineers is also important for having a successful project.

*Table 9: Cross-Zero Detector Technologies Comparison*

|                              | Optocoupler                                     | NPN Transistor                              | Comparator                    |
|------------------------------|-------------------------------------------------|---------------------------------------------|-------------------------------|
| Output Type                  | Pulses                                          | Pulses                                      | Square wave                   |
| Galvanic Isolation of Output | Yes                                             | No                                          | No                            |
| Simplicity                   | Medium                                          | Medium                                      | High                          |
| Cost                         | Most Expensive                                  | Low                                         | Medium                        |
| Crossing Location            | When less than bias voltage of the LED (varies) | When less than 0.7V (forward bias of diode) | At reference voltage of input |

### 3.3.2.4: Cross-Zero Detector Part Selection

Our cross-zero detector is based on the schematic in Texas Instruments' "Zero Crossing Detection Using Comparator Circuit" article. The only part that needs to be researched is which comparator to use since the remaining components are standard value passive components and common diodes. It is very likely that this schematic will need to modify to fit our needs. Mostly notably is that the input voltage in the figure below is 240Vrms instead of the 120Vrms and the supply voltage reaches up to 5V. Since we plan to power the comparator from the digital logic level of 3.3V, this must be modified.

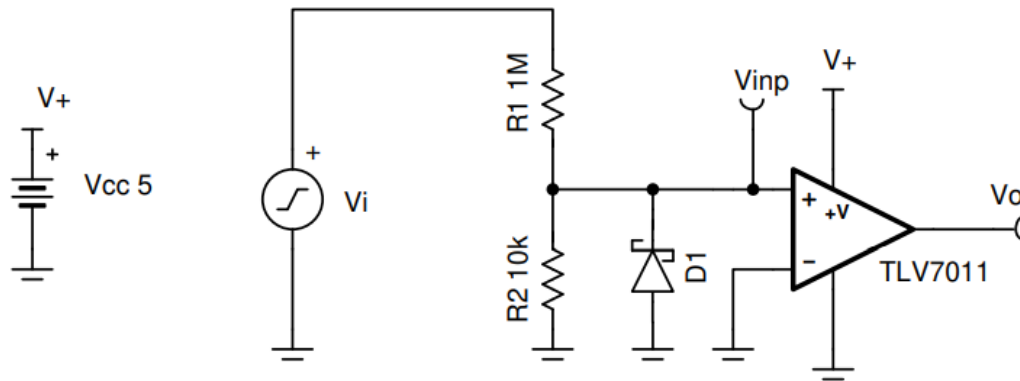


Figure 10: Cross-Zero Detector Base Courtesy of Texas Instruments

Table 10 shows a sample of the various comparators and how they compare each other.

Table 10: Comparator Comparison Matrix

|                   | TLV7011                                                                                     | TS3021                                                      | LM2903-Q1                               | TLV703x-Q1                                                                                  |
|-------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| Description       | Analog Comparators Low power, small size comparator with push-pull output 5-SC70 -40 to 125 | Analog Comparators Rail-to-rail 1.8 V high-speed comparator | Analog Comparators Dual Diff Comparator | Analog Comparators Automotive nano power comparator with push-pull output 5-SC70 -40 to 125 |
| Output Type       | Push-Pull, Open-Drain                                                                       | Push-Pull                                                   | Open-Collector outputs.                 | Push-Pull, Open-Drain                                                                       |
| Propagation Delay | <350ns                                                                                      | <100ns                                                      | <2μs                                    | <3 μs                                                                                       |
| Hysteresis        | 4.2mV                                                                                       | Unknown                                                     | Unknown                                 | 6.5mV                                                                                       |
| Price (\$)        | 0.65                                                                                        | 1.76                                                        | 0.64                                    | 0.86                                                                                        |

For our project, we opted to use the TLV7011.

First, we eliminated the LM2903-Q1. While it is the cheapest comparator of the selection, we decided against the open-collector output. Open-collector and open-drain describe transistor topologies. In either of these topologies, we would need to include a pull-up resistor, as both can only sink power. We are looking for an output that can source power, like the push-pull output.

The second comparator we eliminated was the TS3021. This comparator's greatest asset is its speed. From the datasheet, the slowest, worst-case propagation delay was under one hundred nanoseconds, which is significantly faster than any other comparators in Table 9. While a faster response time is always better, we do not think that this benefit offset the price and hysteresis issue. The price of the TS3021 is approximately double that of any other comparator we have selected. This, coupled with an unknown hysteresis value, swayed us against this comparator for our project.

The TLV7011 and the TLV703x-Q1 come from the same product line and have very similar performances. Realistically, either of these comparators would work for our application. The propagation delay of the TLV703-Q1 is three orders of magnitude greater than the TLV7011, but that still puts itself at over three orders of magnitude greater than the nominal period of the 60Hz input--16.67ms. The TLV703x-Q1 does have a slight advantage in hysteresis. The comparator has 6.5mV of hysteresis as compared to the TLV7011's 4.2mV. This means the former is more noise resistant. We settled on the TLV7011 based on its marginally lower price and existing circuit analysis. Having a step-by-step design process for a cross-zero detector will dramatically speed up the design and troubleshooting process.

## 3.4: Mechanical Systems

This section explores potential designs for the mechanical systems required. The primary objective of these systems is to be able to change the state of the breaker switches automatically and reliably.

### 3.4.1: Mechanism of Breaker Automation

As outlined earlier, the core component of making the breaker box automated or "smart" is that there must be a mechanism by which individual breakers can be turned on and off either automatically or by the end user from afar. Two methods to achieve this are:

#### 3.4.1.1: Replacing Switches with Relays

This methodology has been used successfully in the past and offers a sleek solution. The major issue with it is that it requires replacing the breaker box entirely, something that requires a trained electrician to complete and represents a large buy in for the end user. As such, we have examined methods that do not require replacing the breaker box or switches.

#### 3.4.1.2: Using Preexisting Switches

By retaining the switches in place, we are able to develop a solution that does not require disconnecting anything inside of the breaker box, thus allowing end users to install and use the device on their own, not needing special tools or training to do so. This is the method we moved forwards with.

### 3.4.2: Switching mechanism

Now, we must examine the mechanical methodology by which the breakers will be flipped. The most important requirement is that the selected system will be robust enough to produce the forces needed to flip the breakers and do so for many cycles without failure. Other important considerations are ease of installation, price, power usage, and noise level

#### 3.4.2.1: Linear Motion Device on gantry

This approach would use a two-axis gantry to position and end effector around the switches, and provide the force to flip the switches. provided the linear motion devices are powerful enough to flip the switches, this solution will be preferable as it will be simpler to mount and control, as well as being more cost effective.

#### 3.4.2.2: Individual devices (Solenoids)

This approach would pair each breaker with 2 dedicated solenoids, one to flip it on, and the other to flip it off. This would allow all breakers to be flipped simultaneously and for very fast switching. Serious drawbacks of this design are that it would be difficult to mount all of the required solenoids and may not be feasible if they are too large. Other considerations are the high cost to pair a solenoid with each switch, and the lack of expandability if new breakers are added. In addition, the control schema would be difficult to design.

#### 3.4.2.3: Selection of Switching Mechanism

We have moved forward with the linear motion device gantry design as it is simpler to implement and more cost effective. This has the downside of slower switching, as it will only have the capacity to flip one switch at a time. this must be accounted for, and it must be ensured that the speed of the gantry to move between switches is low enough to meet the requirements. The next important factor is the power of the system. We used a lead screw and guide design, to convert rotational motion from an electric motor to the linear motion that is required.

### 3.4.3: Gantry System Overview

The idea for how to physically flip the switches is to create a device similar in construction to a 3D printer. The device will have two axes, one up and down (Y), and one left to right (X). The Y axis will be used to move the end effector into place aligned with a switch, and the X axis will be used to push the end effector through the switch, turning it on or off. As each of these axes is required to deal with significant force (the Y axis in the form of gravity and the X axis in the form of the force required to flip a switch) both will be driven by lead screws similar to the z axis of a typical 3d printer. This has been chosen to avoid problems present with belt driven mechanisms, which are prone to slipping under higher loads, and are not suitable for holding against gravity as would be required for the Y axis.

The frame of the device that the motors, lead screws, and other hardware will mount to will be constructed from aluminum extrusions. This is another idea

borrowed from 3d printers. They are inexpensive, easy to mount to using T-nuts, can easily be assembled together with widely available hardware, and can be cut to size without special equipment. Additional mounting hardware will be 3d printed due to ease of production.

### 3.4.4: Guide Selection

In order to use lead screws, it is required to have a guide mechanism attached to the lead nut in order to convert the rotational motion to linear motion. The available options are to use roller wheels in the slots of the extrusions, linear guide rails, and shafts with linear bearings.

*Table 11: Guide Selection Matrix [25]*

|               | Roller wheels | Linear guide rails | rod and bearings |
|---------------|---------------|--------------------|------------------|
| Load capacity |               | x                  |                  |
| Rigidity      |               | x                  |                  |
| Accuracy      |               | x                  |                  |
| Speed         |               | x                  |                  |
| Mounting      |               |                    | x                |
| Reliability   |               |                    | x                |
| Maintenance   |               |                    | x                |
| Sizing        | x             |                    |                  |
| Cost          | x             |                    |                  |

Based on this, we have moved forward using linear guide rail. It is fairly inexpensive, easy to mount, and available in a wide range of lengths. It should further be noted that none of the authors of this paper are mechanical engineering students, and as such, this paper will not contain significant engineering analysis of the mechanical structure. Key information such as motor load will be calculated, but more advanced calculations such as stress analysis and simulation will be omitted.

### 3.4.5: Motor Load Calculations

In order to properly size the motors, their drivers, and the power supply, we must determine the amount of load that will be on the motors used to power the X and Y axes. As both axes are driven with lead screws, we must use appropriate calculations

To determine the force that the X axis must exert to successfully flip the breakers, we measured the force required with a pull scale. It was determined using this method that the largest force the X axis must be able to exert was 10 pounds.

Based on this, we use the equation provided by Haydon Kerk Pittman [26]

$$(T) = (F \times L) / (2 \times \pi \times E).$$

Required torque to exert a force is a function of the lead of the screw (a measurement of how much the thread moves up the screw in one rotation), and the efficiency of the screw

PCB linear shows that the lower end of lead screw efficiency is 30% [27]. Although our screws will likely be more efficient, we used this number as a safety factor

Using our required weight of 10 pounds (44.45 newtons), we find the following results for commonly available leads

*Table 12: Lead-Torque Matrix*

| Lead (mm) | Torque (Ncm) |
|-----------|--------------|
| 2         | 4.72         |
| 4         | 9.43         |
| 6         | 14.15        |
| 8         | 18.87        |
| 10        | 23.58        |

#### 3.4.5.1: Lead Screw Selection

As the X axis has a shorter distance to move, and requires more torque, we have used a 6mm lead. For the Y axis, less torque is required, and higher speed is beneficial, so we used an 8mm lead.

We must also consider the diameter of the lead screw. if the screw is too thin, it may face buckling due to the force and could fail. If the screw is unnecessarily thick, adding weight to the system, reducing efficiency. It would also incur cost, as larger screws have a higher price. The sweet spot appears to be at the 8mm mark for this application. This size is very commonly used with Nema 17 stepper and will be able to withstand the required load without being unnecessarily torquey.

In terms of screw length, given the area containing the switches on our selected panel, a 450 mm and 200 mm lead screw would be appropriate for the Y and X axes respectively, as this would allow for sufficient range of motion to reach all switches.

#### 3.4.6: Motor and Control Schema Comparison

In order to exert the rotational forces required to drive the lead screws, we must use an electric motor. There are a variety of options to choose from, each requiring different circuitry to drive, and each having tradeoffs.

##### 3.4.6.1: Stepper motors

This is the ideal motor to use as it keeps track of its position, only needs end sensors, and has high torque at low rpm.[28] One of the major concerns is if the motor loses steps due to a stall, it will lose its position. Fortunately, lost steps can be detected with newer stepper drivers that have current sensing capability. As we

are borrowing many design cues from 3D printers, the use of stepper motors will also simplify our design, as they are nearly universally used in such devices. This means much of the software required to drive them is commonly available, and supporting hardware is as well. An Important part of keeping track of stepper positioning is the use of end stop switches; these are used to indicate to the controller when the motor has moved to the end of its range. This can be used both to prevent machine crashes, as well as to zero-out the position of the motors upon startup.

#### 3.4.6.2: Brushless Motors with Encoders

Although brushless motors do have many benefits, they lack the low-end torque required for this application. Additionally, they are more difficult and costly to control than steppers, and require a more complex feedback loop, as for our application the precision required would need the motors to have encoders. These devices measure rotational motion, and because brushless motors have no way of knowing their position, they would be required as we must be able to move the end effector precisely. Including these would make our control system more complex, but would provide more reliable positioning than stepper motors

#### 3.4.6.3: Brushed Motors with Encoder

These motors are simpler and cheaper to drive than brushed motors or steppers but would require encoders for positioning like brushless motors. This has all of the drawbacks of brushless motors, with the addition of brushes that require maintenance.

#### 3.4.7: Comparison of Stepper Motors

We have selected the . It's high torque will allow the gantry to move quickly between switches, and flip them with ease. Due to its higher current rating, this means it would be advantageous to choose a driver with a similar current rating, so that we would be able to utilize the full power capabilities of the motor

*Table 13: Stepper motor comparison matrix*

| Supplier      | Torque (Ncm) | Amp per phase | Dimensions | Price   |
|---------------|--------------|---------------|------------|---------|
| STEPPERONLINE | 16           | 1             | 42*42*20mm | \$7.59  |
| STEPPERONLINE | 65           | 2.1           | 42*42*50mm | \$22.59 |
| STEPPERONLINE | 13           | 0.7           | 42*42*25mm | \$8.15  |
| STEPPERONLINE | 45           | 1.5           | 42*42*29mm | \$11.99 |
| Adafruit      | 20           | 0.35          | 42*42*31mm | \$14    |
| STEPPERONLINE | 59           | 2             | 42*42*48   | \$9.62  |

### 3.4.7.1: Stepper Motor Technology Explanation

There are multiple different kinds of stepper motors. The ones we have used are Bipolar. This means they have 2 phases of copper windings around the shaft. a simplified diagram of this is shown below. Each of these two phases contains multiple coils surrounding the shaft. By energizing these phases sequentially, the magnetic fields they generate are able to move the shaft very precisely. This means that if a constant voltage is applied, it will not spin but rather hold its position. This is why specialized controllers are required. [29]

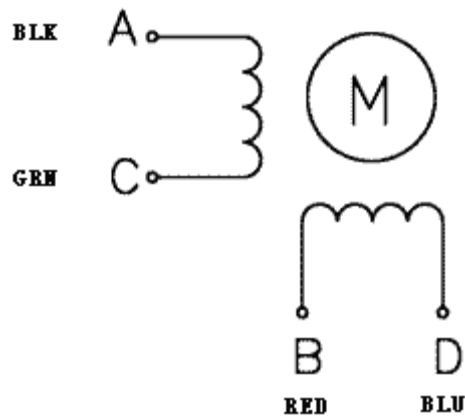


Figure 11: Bipolar stepper motor phases

### 3.4.8: Comparison of Stepper Motor Drivers

We moved forward with the TMC 5160. The cool step feature is useful to lower current while not in use, stealth chop to reduce sound, and it has sufficient current capacity. It does have a more complex control scheme than other drivers, requiring not just step and direction inputs, but SPI as well to configure its more advanced features; however, these features are worth the complexity.

Table 14: Stepper Motor Comparison Matrix

| Supplier | Model        | Continuous current (A) | Micro stepping | Special features                              |
|----------|--------------|------------------------|----------------|-----------------------------------------------|
| TI       | DRV8834      | 1.5                    | 1/32           | low voltage input                             |
| Allegro  | A4988        | 1.2                    | 1/16           |                                               |
| Allegro  | A5984        | 1.2                    | 1/32           |                                               |
| TI       | DRV8825      | 1.1                    | 1/32           |                                               |
| Toshiba  | TB67S249-FTG | 1.6                    | 1/32           | Auto Gain Control, ADMD                       |
| onsemi   | AMIS-30543   | 1.8                    | 1/128          | SPI interface, low-EMI PWM, back EMF feedback |

|                |         |     |       |                                                  |
|----------------|---------|-----|-------|--------------------------------------------------|
| Analog Devices | TMC5160 | 3   | 1/256 | CoolStep, StallGuard2, SpreadCycle, StealthChop2 |
| Analog Devices | TMC2130 | 1.2 | 1/256 | CoolStep, StallGuard, StealthChop                |
| Analog Devices | TMC2209 | 1.4 | 1/256 | SpreadCycle, Stealthchop2                        |
| Stepper Online | DM320T  | 2.2 | 12800 |                                                  |
| DFRobot        | TB6600  | 3.5 | 1/32  |                                                  |

\*note: Unlike all of the other controllers listed here, the DM320T and TB6600 are not ICs that can be mounted to either dev-boards or custom PCBs, but they are standalone devices. This has advantages and drawbacks, as they will take up more space in the design but will reduce complexity. They also include opto-isolators, which will ensure there is no back EMF. This could potentially be very useful, as it would ensure the motherboard is not damaged by the controllers.

### 3.4.9: TMC 5160 Technology Explanation

Shown below is a hardware block diagram of a typical application of the TMC 5160. Everything inside the square is integrated into the chip, and everything outside is external. This diagram shows the controller in STEP/ DIR mode, where the SPI interface is active to configure various parameters, but the signals to be sent to the motor are generated externally. One major thing to note about this motor controller is the requirement of external MOSFETs (seen on the right side). These

increase the complexity of the circuit, but allow for much larger currents to be used, and helps with heat dissipation. [30]

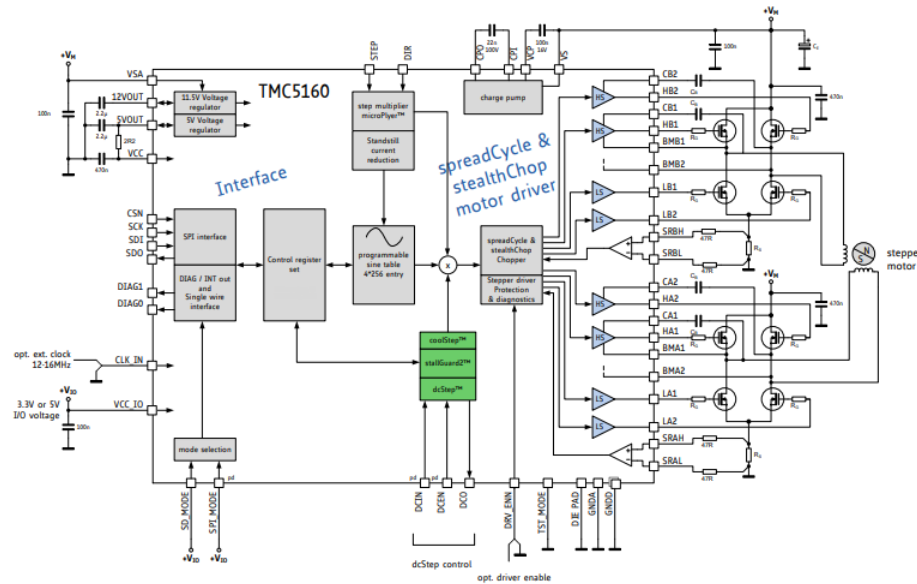


Figure 12: TMC 5160 STEP/DIR Mode BD [30]

It is also very important to note the green labeled CoolStep, this controls one of the special feature of this controller that allows it to automatically reduce the current supplied to the motor when it is not under load, greatly increasing reducing power consumption.[30]

Another important block is the one labeled SpreadCycle and StealthChop Chopper. The StealthChop feature allows the stepper motors to operate nearly silently, as opposed to most stepper drivers that cause the motor to produce a whining noise. This is a highly desirable characteristic, and a major factor in choosing this controller. It will mean the noise of operating the device will be very quiet, ensuring it does not disturb the end user.[30]

One feature that will not be used is the TMC5160s ability to home without end switches. This is accomplished using the StallGard feature, when the current reaches a certain level set by the user, it is registered as an end stop. This works in applications such as 3d printers as the motors are never required to exert high torque, so if they do it can be assumed that they have reached the end of their range of motion. However, for our application, this high torque will be necessary to flip the breaker switches. as we do not want this to register as hitting an end switch, the current threshold would have to be raised to a high level. This means that when the motor is at the end of its range of motion, it will have to exert significant force of the device itself before the end stop is registered. Due to concerns that this may damage the machine, we did not use this feature.

### 3.4.10: Physical Mounting of Gantry Structure

for the issue of mounting, two main ideas have been proposed: mounting the gantry externally, as in attached by some means to the face plate of the breaker, and mounting the gantry internally, as in discarding the face plate and building a new one. In this section we have explored both methods

#### 3.4.10.1: External mounting

This method has been proposed as it is a goal of this project for the device to be simple to mount, something that any homeowner could install quickly and without difficulty.

3m dual lock, a material similar, but much stronger than hook and loop fasteners (Velcro) could be used. In its strongest configuration it requires 60 lbs/in<sup>2</sup> to disengage.[31] This means creating a perimeter of this material around the area with the switches would be a more than sufficient mounting solution, capable of handling both the weight of the gantry, as well as the forces it will apply to the switches

One major issue with external mounting is the door that covers the switches. In order to have proper access to the switches, this door must be removed from the faceplate. On our model, this is a simple procedure, as the hinges that secure the door are just bent metal loops that are easily opened and removed with a screwdriver. However, this is not the case on all breaker boxes, some have hinges that are more difficult to remove, and it may be impossible to re-secure the door if the end user wishes to remove the product and return their breaker to its factory configuration.

Another drawback is that mounting to the faceplate would require the device to protrude further from the wall, causing a greater potential for damage, as well as being aesthetically undesirable.

#### 3.4.10.2: Internal mounting

The other idea proposed was to remove the faceplate of the breaker, and to use the mounting holes intended for the faceplate to mount the device. As all available breaker boxes use this two-part construction, this is a viable solution for any box. the user would just have to unscrew the faceplate, then screw in the device. This would end up being simpler and faster than applying a dual lock fastener and would allow the user to remove the device and replace the original faceplate very easily.

Drawbacks of this mounting scheme are that different breaker boxes have the mounting holes spaced differently. For the prototype we are creating, this is not an issue as it is only required to work on the box we have, but a theoretical market version of the product could use a schema similar to that use to mount heatsinks to motherboards; there would be multiple mounting plates included compatible with different boxes to cover the range of the market.

Due to the simplicity for the end user of internally mounting the device, we moved forward with that design.

### 3.4.11: Software Control of motors

G-code is a standard that allows for the control of devices such as CNCs and 3d printers by sending commands that include a position in space, speed, and auxiliary information as desired. As our gantry system is very similar to a CNC router, this standard will give us a simple, effective, and well documented methodology for positioning the end effector as desired [32]. The interface is simple, The G code interpreter will go line by line for the G code it receives, and it will complete each operation in order. Common operations are linear movements in 2 or 3 dimensions, circular or arc movements, end calibration procedures to correct motor positioning, and moving to a set home position. It is also possible to create custom G-code commands to perform machine specific tasks, such as tool changing.

The TMC stepper library is an open source C library designed to make interfacing with TMC stepper motor controllers. It handles the SPI interface to configure parameters such as current limiting and micro stepping. it is compatible with the driver we are using, The TMC5160[33].

GRBL is an open source C program that acts as a G-code interpreter. It is commonly used for Arduino based CNC machines and offers a simple interface to send G-code commands and control stepper motors.[34] One major limitation is that it has no SPI interface, meaning it cannot properly communicate with the motor controllers we are using

Marlin is open source 3d printer firmware written in C. it has G-code interpretation features like GRBL, but is a much more robust piece of software that includes many more customizable parameters. It also includes the SPI interface that the TMC5160 requires.[35] As this is designed for 3d printers, it has many features we do not require, but it is a robust choice nonetheless.

Fluid NC is an open source motion controller designed specifically for ESP32. It includes a G-code interpreter, is compatible with a wider range of motor controllers than GRBL; however, it does not have the SPI interface we require.[36]

Although this selection appears to indicate Marlin is the ideal choice for the motion control software, our testing found it to be too complex to successfully work. This is due to configuration using the ESP32 as well as a motor controller requiring ESP32 are uncommon and require many additional libraries that cause issues. Seeing as the TMC stepper library works very well standalone, it appears the best strategy moving forward will be to integrate it with a G-code interpreter, potentially creating a version of GRBL compatible with this library.

### 3.4.12: End effector

This will be the device that rides on the gantry and is used to apply force to the switches to flip them. In order to be able to flip the switches, it must be able to exert

44.45 newtons of force without being damaged. It is also important that the end effector does no damage the switches themselves in any way. A simple 3d printed cylinder with a rounded end will suffice, given appropriate infill. A rubberized coating of some sort could also be considered, as it may help the end effector engage the switch by providing additional grip and could reduce wear by providing a soft surface to eliminate scratching. Potential materials include spray rubberized coating, cast silicone, and vulcanized rubber such as innertube material.

## 3.5: MCU

This section delves into the considerations and decisions surrounding the selection of microcontroller units (MCUs) for our project. MCUs serve as the central processing units for embedded systems, interfacing with sensors, communication networks, and other peripherals. Given the wide range of options in the market, it is crucial to evaluate the strengths and trade-offs of potential candidates, including their suitability for our specific project requirements.

### 3.5.1: MCU vs FPGA Considerations

In developing the embedded systems for this project, we considered both microcontroller units (MCUs) and field-programmable gate arrays (FPGAs) as potential solutions. While FPGAs offer greater processing power and precision, these advantages come with certain trade-offs. FPGAs excel in applications requiring low error margins and highly specific, parallelized processing. However, these capabilities also make them a more costly option and significantly increase their power consumption, particularly compared to MCUs.

Due to their advanced processing capabilities, FPGAs are often favored for tasks demanding parallel processing. However, MCUs are specifically engineered for low-power applications and frequently offer customizable low-power modes out of the box. Moreover, FPGAs are less user-friendly, requiring a higher level of expertise to program. They are typically programmed with hardware description languages like Verilog or VHDL, which cater to niche industry applications and are less accessible to general developers.

In contrast, MCUs generally come with a suite of integrated peripherals, such as analog-to-digital converters and communication interfaces like UART and I2C, which simplifies the development process. This feature-rich design, combined with extensive documentation and an established community of support, makes MCUs far more approachable. MCUs are typically programmed in widely-used languages like C or Python and have robust support ecosystems, including integrated development environments (IDEs) and comprehensive debugging tools. Debugging on an MCU is relatively straightforward compared to the FPGA experience, which can require more advanced tools and is often limited by hardware-centric documentation that varies greatly between vendors.

Once a developer is familiar with the basics of microcontroller programming, transitioning to a new board or even a new manufacturer's MCU tends to be straightforward. By contrast, FPGAs require an entirely different design and

implementation approach, demanding specialized knowledge and a customized approach to error handling. While FPGAs offer scalability and can handle future processing demands without the limitations of an MCU, they also necessitate careful design considerations for features like error handling, which typically must be designed from scratch.

MCUs tend to offer robust environmental resistance out of the box, often featuring built-in temperature sensors and watchdog timers to enhance operational stability. FPGAs, on the other hand, may be more susceptible to environmental factors, such as temperature fluctuations or electromagnetic interference, and may require specialized enclosures to ensure reliable performance.

### 3.5.2: Selecting Microcontrollers for the Project

For our project, which involves sensor data collection and wireless communication, MCUs are the ideal choice. Given the need for consistency and ease of use across multiple deployment locations, it is beneficial to select equipment that can seamlessly interface with both sensor networks and our overarching communication network. By leveraging the flexibility of MCUs, we can utilize pre-existing open-source software and easily configure the devices for various tasks across our network.

Numerous microcontrollers fit our project's requirements, including options that support operating systems, providing even greater versatility. This flexibility allows us to choose compatible software as needed, ensuring that our chosen hardware remains adaptive to project demands while benefiting from the reliability, efficiency, and developer-friendly nature of MCUs.

#### 3.5.2.1: *Raspberry Pi Series*

The Raspberry Pi series offers intriguing options for embedded systems in this project, though each model presents unique advantages and trade-offs. Traditional Raspberry Pi boards offer a hybrid experience that blends features of both embedded systems and desktop computers. Equipped with a full Linux environment, the Pi provides extensive software compatibility, allowing us to use virtually any programming language, which can simplify debugging and facilitate software testing models,

##### 3.5.2.1.1: Traditional Pi

The Pi's hardware specifications are robust: a 1.8 GHz quad-core CPU with at least 1 GB of RAM and 32 GB of storage, along with 2.4 and 5 GHz wireless connectivity, Bluetooth, and BLE support. This capability, while impressive, can be excessive for our requirements, and the power draw reflects this. Operating at 5V with a minimum 3A supply, it demands significantly more power than traditional microcontroller boards. Additionally, while the Pi has GPIO pins capable of UART, SPI, and I2C, it lacks built-in ADC and DAC support, which limits its suitability for analog signal applications. This model is also on the higher end of the price spectrum for embedded systems, and the overhead from running a full OS can be inefficient for specialized tasks.

### 3.5.2.1.2: Raspberry Pi Zero

The Raspberry Pi Zero, a streamlined version of the traditional Pi, offers some trade-offs that may suit specific needs. It provides essential features without superfluous additions, such as unpopulated GPIO pins and reduced memory, requiring less power at 1.2A. The Zero model includes 2.4 GHz b/g/n Wi-Fi with an onboard antenna and retains Bluetooth and BLE capabilities, making it ideal for networked applications on a budget.

While the Zero has reduced processing power—a 1 GHz single-core CPU and 512 MB RAM—it maintains enough functionality for efficient development and flexibility in embedded applications. The limitations of lacking ADC/DAC and additional communication options like Zigbee remain, as does the comparatively high power requirement.

### 3.5.2.1.3: Raspberry Pi Pico

The Raspberry Pi Pico series is the Pi family's microcontroller option, specifically the RP2040 chip. It aligns more closely with traditional microcontrollers like the ESP32, STM32, and Arduino Nano. Although the Pico lacks the full Linux environment found on other Pi models, it offers significantly lower power requirements, operating between 1.8-5.5V and averaging around 38mA. Programming is limited to C/C++ and MicroPython, but the Pico has ample GPIO support, including UART, SPI, I2C, ADC, and PWM.

The Pico W model includes 2.4 GHz 802.11n wireless capabilities with WPA3 security and can function as a soft access point, enhancing its potential for IoT applications. With a dual-core 133 MHz processor, 264 KB RAM, and at least 2 MB of flash memory, the Pico series is a cost-effective solution that's capable of handling simpler tasks while maintaining network connectivity.

### 3.5.2.2: Comparison with ESP32, STM32, and Arduino Nano

The comparisons between the non-Raspberry Pi boards are listed below. The ESP32, STM32, and Arduino Nano all have a variety of different features, each with their pros and cons. The sections below go into greater detail and solidify our microcontroller selection.

#### 3.5.2.2.1: ESP32

The ESP32 stands out in the realm of embedded systems for its robust features and affordability. Designed specifically with IoT in mind, the ESP32 includes both Wi-Fi and Bluetooth (including BLE), allowing it to handle wireless communication needs without additional components. Its dual-core processor, running at up to 240 MHz, is powerful enough for demanding applications, enabling real-time data processing and simultaneous multitasking. With up to 520 KB SRAM and support for both SPI and I2C protocols, the ESP32 is versatile and able to connect multiple sensors while processing complex data streams.

The ESP32 also includes analog and digital pins, PWM, and ADC, which allow for a wide variety of sensor integrations. This versatility makes it suitable for a range of tasks, from environmental monitoring to motor control. However, it lacks the broader GPIO support of some other microcontrollers like the STM32. Additionally,

while its Wi-Fi capabilities are ideal for networked projects, they contribute to a higher power draw compared to simpler MCUs, particularly in continuous-use scenarios. Programming the ESP32 primarily uses C/C++ in the Arduino IDE or Espressif's own ESP-IDF, with substantial community support and documentation available. Overall, the ESP32 is an excellent choice for applications needing wireless connectivity and real-time processing, making it ideal for IoT projects where device-to-cloud communication is essential.

#### 3.5.2.2.2: STM32

The STM32 series by STMicroelectronics is a family of ARM Cortex-based microcontrollers known for their high performance and reliability in industrial applications. The STM32 family offers a wide range of models that cater to various application needs, from ultra-low power devices to high-performance chips with extensive memory and peripheral support. These microcontrollers are highly modular and can be configured to meet specific needs, with support for up to 512 KB of SRAM and 2 MB of flash storage in high-end models, as well as extensive GPIO options.

While STM32 lacks built-in Wi-Fi or Bluetooth, it shines in scenarios requiring precision and stability, particularly in applications where analog signal processing is crucial. It supports advanced peripherals, including DAC and ADC channels, timers, and CAN bus, making it suited for applications needing precise analog measurements or real-time motor control. The STM32 also has flexible power modes that can significantly reduce power consumption, making it suitable for low-power applications where battery life is a priority.

Programming the STM32 typically involves using C/C++ in environments like STM32CubeIDE, which provides tools for configuring the hardware and managing low-level details. The learning curve is steeper than that of the ESP32 or Arduino Nano, but this is offset by the STM32's flexibility and performance. In summary, STM32 is best for applications where robust analog control, extensive GPIO options, and reliability are needed, though it lacks built-in wireless capabilities, requiring external modules for networked applications.

#### 3.5.2.2.3: Arduino Nano

The Arduino Nano is a simple, compact microcontroller board based on the ATmega328P, commonly used for prototyping and beginner-level projects. Operating at 16 MHz with 2 KB SRAM and 32 KB of flash memory, it is less powerful than the ESP32 or STM32. However, it is highly accessible and has a low power requirement, often operating efficiently with a 5V or 3.3V power source, which makes it ideal for battery-powered applications where simplicity and cost-efficiency are key.

The Nano includes a limited number of GPIO pins, but it supports UART, I2C, and SPI communication, providing the essentials for connecting various sensors and peripherals. Despite its limitations in processing power and memory, the Nano is user-friendly, with an extensive community and a wealth of open-source libraries available in the Arduino ecosystem. This ease of use makes it a great option for

straightforward tasks such as reading sensor data or controlling basic actuators, but it falls short when it comes to more complex applications or those requiring networked communication, as it lacks built-in wireless capabilities.

Expanding the Nano's functionality requires add-on modules like Wi-Fi or Bluetooth shields, which increase the cost and complexity. Additionally, its limited processing capabilities mean it struggles with real-time applications or tasks that require substantial data processing. For basic, non-networked tasks or applications with minimal real-time requirements, the Nano is a practical choice. However, for IoT applications requiring wireless communication, the Nano's limited power and lack of built-in connectivity make it less suitable compared to the ESP32 or even the Raspberry Pi Pico.

*Table 15: MCU Decision Matrix*

| Name           | RAM    | Clock Speed | Core Count | Cost     |
|----------------|--------|-------------|------------|----------|
| Raspberry Pi 4 | 1 GB   | 1.8 GHz     | 4          | High     |
| Pi Zero        | 512 MB | 1 GHz       | 1          | Moderate |
| Pi Pico        | 264 KB | 133 MHz     | 2          | Moderate |
| ESP32          | 520 KB | 240 MHz     | 2          | Low      |
| STM32          | 512 KB | 72 MHz      | 1-2        | Low      |
| Arduino Nano   | 2 KB   | 16 MHz      | 1          | Low      |

## 3.6: Wireless Communication Protocols

Wireless communication protocols are plentiful and have a variety of uses for many different applications. This section delves into the most popular protocols and comparisons amongst them to determine which protocol is best suited for our project/implementation.

### 3.6.1: Wi-Fi

Wi-Fi is a widely used wireless communication protocol, primarily designed for high-speed data transmission over short-to-medium distances. Operating on the 2.4 GHz and 5 GHz frequency bands, Wi-Fi supports data rates that can exceed 100 Mbps, making it ideal for applications where bandwidth is critical. Its relatively high data throughput makes it suitable for applications requiring streaming, video transmission, or large data packet transfers. Additionally, Wi-Fi benefits from the availability of existing infrastructure in many locations, as routers and Wi-Fi networks are prevalent in most residential, commercial, and public areas.

However, Wi-Fi's high bandwidth and low latency come at the cost of power consumption. It requires significantly more power than protocols like Bluetooth and LoRaWAN, which limits its suitability for battery-powered devices that need to operate for extended periods without charging. Wi-Fi's range is also somewhat limited compared to other protocols, with an effective range of around 50 meters indoors, though this can vary based on environmental factors. In terms of security,

Wi-Fi supports robust encryption protocols like WPA3, which is essential for applications requiring secure data transmission.

Overall, Wi-Fi is a strong choice for applications that prioritize high-speed data transfer over power efficiency. It is particularly suitable for applications within a fixed area, such as smart home devices, video-enabled security systems, and data-intensive IoT hubs. However, its high-power requirements and limited range make it less ideal for low-power or remote applications, where protocols like Bluetooth or LoRaWAN would be more appropriate.

### 3.6.2: Bluetooth

Bluetooth is a short-range wireless communication protocol that operates primarily in the 2.4 GHz band. It is designed for low-power, close-proximity communication, with a typical range of up to 10 meters for traditional Bluetooth and up to 100 meters for Bluetooth Low Energy (BLE). BLE, in particular, is optimized for minimal power consumption, making it highly suitable for battery-powered devices that need to operate over long periods. BLE devices can run on small batteries for months or even years, making Bluetooth ideal for applications such as wearable devices, fitness trackers, and proximity-based IoT solutions.

While Bluetooth's data transfer speeds are lower than Wi-Fi's, reaching around 1 Mbps for BLE and up to 3 Mbps for traditional Bluetooth, it is sufficient for applications that only need to transmit small amounts of data, such as sensor readings or status updates. Bluetooth's ability to pair with devices quickly and securely also makes it an excellent choice for personal area networks (PANs), where it facilitates connections between devices like smartphones, headphones, and smartwatches. However, due to its short range and limited data rate, Bluetooth is not suitable for applications requiring high data throughput or long-distance communication.

Bluetooth is best suited for short-range, low-power applications where small data packets are transmitted infrequently. It is commonly found in health monitoring devices, indoor location beacons, and other low-data, close-proximity IoT applications. Although its range and data rates are limited, Bluetooth's power efficiency and rapid connection capabilities make it a versatile and widely-adopted protocol for personal and wearable devices.

### 3.6.3: LoRaWAN

LoRaWAN (Long Range Wide Area Network) is a wireless communication protocol optimized for long-range, low-power applications. Unlike Wi-Fi and Bluetooth, LoRaWAN operates on sub-gigahertz frequencies, typically around 868 MHz in Europe and 915 MHz in the United States. This allows it to achieve ranges of several kilometers, even in dense urban areas, and up to 15-20 kilometers in rural settings with minimal interference. LoRaWAN is designed for low data rates, typically less than 50 kbps, which is suitable for applications where small data packets need to be transmitted intermittently, such as environmental monitoring or asset tracking.

LoRaWAN's low power consumption is one of its biggest strengths, allowing devices to operate on small batteries for years without needing frequent recharges. This is achieved through its low data rate and duty cycle, which reduces power draw. However, the trade-off for this efficiency is limited bandwidth, meaning LoRaWAN is not suitable for applications that require high-speed data transfer or real-time communication. Additionally, LoRaWAN's network architecture involves end devices communicating with a gateway, which then relays the data to a central server. This adds a layer of complexity, as setting up a LoRaWAN network requires a gateway and backend server infrastructure, although public LoRaWAN networks are available in some regions.

LoRaWAN excels in applications requiring long-range communication, minimal data transfer, and ultra-low power consumption. It is particularly well-suited for remote monitoring and IoT applications, such as smart agriculture, environmental sensing, and utility metering, where devices must operate in dispersed or isolated locations without frequent access to power sources. Although its low data rate limits its use in high-bandwidth applications, LoRaWAN's range and efficiency make it an invaluable protocol for long-range, low-power IoT deployments.

## 3.7: Hardware Programming Language Comparisons

This section provides an in-depth analysis of programming languages commonly used in hardware development, focusing on their strengths, limitations, and suitability for various embedded systems. Each language is evaluated in terms of execution speed, resource efficiency, ease of development, and compatibility with modern microcontrollers.

The goal is to guide the selection of the most appropriate language for our project, balancing performance requirements with development complexity and long-term maintainability. Whether prioritizing rapid prototyping, real-time performance, or memory safety, the comparison highlights how each language aligns with different aspects of embedded system design.

### 3.7.1: Python

Python is a high-level, interpreted language widely known for its simplicity and readability. In hardware programming contexts, Python is particularly useful for prototyping, testing, and scripting due to its low complexity and ease of understanding. Its extensive library support is one of its greatest strengths, providing pre-built modules for a wide range of functionalities, from data handling to interfacing with hardware through GPIO and other protocols.

However, Python's interpreted nature also leads to relatively slow execution speeds compared to compiled languages like C or assembly. This can be a limiting factor in applications requiring real-time processing or high-performance computations. Additionally, Python may introduce spatial overhead due to its dynamic typing and memory management model, which can result in a larger memory footprint. While this is generally manageable on higher-capacity hardware

like Raspberry Pi boards, it could be a drawback in more constrained embedded systems.

Python is an excellent choice for rapid development and when ease of use and accessibility are prioritized over raw execution speed. It's particularly well-suited for applications on microcontrollers with sufficient processing power and memory, such as the Raspberry Pi Pico or ESP32, where prototyping speed is key, and performance limitations are secondary concerns.

### 3.7.2: C

C is a foundational language in hardware programming, known for its speed and close-to-the-metal nature, allowing direct manipulation of memory through pointers and bitwise operations. It is a compiled language, which means programs written in C generally execute much faster than those in interpreted languages like Python. This makes C highly suitable for applications requiring precise control, real-time responsiveness, and efficient use of system resources. Its syntax and structure are well understood across engineering disciplines, making it easier to find team members with C experience.

The drawbacks of C include a higher logical overhead compared to high-level languages, meaning developers must manage memory manually and take care of low-level details that would be abstracted away in other languages. This complexity can lead to more challenging code maintenance and a steeper learning curve for new developers. Furthermore, C does not natively support object-oriented programming, which may require additional effort to implement certain software design patterns. However, for applications on platforms like the STM32 or Arduino Nano, C provides the level of control and performance necessary for critical, time-sensitive tasks.

C is ideal for embedded systems where precise hardware control, performance, and efficiency are paramount. It's especially useful for applications with limited resources, where every bit of memory and processing power matters, such as real-time operating systems (RTOS) on microcontrollers.

### 3.7.3: C++

C++ builds on the foundation of C, introducing object-oriented programming (OOP) principles. This language retains much of C's low-level control and efficiency but adds the flexibility of classes, inheritance, and polymorphism, which can lead to more organized and reusable code. C++ is also a compiled language, and for most hardware applications, it performs comparably to C in terms of execution speed. Many microcontroller frameworks, including the Arduino IDE, offer support for C++, allowing developers to utilize both low-level operations and higher-level abstractions.

The primary drawback of C++ is the increased complexity, as it combines both procedural and object-oriented paradigms, which can lead to more convoluted code if not managed carefully. Moreover, C++ can introduce additional overhead due to features like dynamic memory allocation, which, if not carefully managed,

can lead to fragmentation and increased memory usage. This is especially relevant in embedded contexts with constrained resources.

C++ is a powerful choice when projects demand both the efficiency of C and the organization and modularity provided by OOP. It is often used in embedded systems with more memory and processing power, such as the ESP32, and in applications where object-oriented design can simplify complex, large-scale embedded software systems.

### 3.7.4: Rust

Rust is a modern systems programming language designed to provide the performance and control of C and C++ while incorporating memory safety features. Rust's borrow checker ensures memory safety without a garbage collector, making it a compelling option for embedded systems where safety and performance are paramount. Rust's syntax is also like C++ but provides higher-level abstractions and tools for managing concurrency and memory more safely.

Rust's main drawback is that it has a relatively small ecosystem compared to more established languages like C. Its syntax, while powerful, is more complex, and it has a steeper learning curve, which can be a barrier to adoption for developers accustomed to C or Python. Moreover, its toolchain for embedded systems, while growing, is not as mature as those for C or C++.

Rust is ideal for applications where memory safety and concurrency are critical, particularly in safety-critical systems where avoiding memory errors is essential. It is well-suited for modern, complex embedded systems and is especially applicable for security-sensitive IoT applications.

*Table 16: Embedded Programming Languages*

|        | Complexity | Speed     | Documentation | Libraries |
|--------|------------|-----------|---------------|-----------|
| Python | Low        | Low       | High          | High      |
| C      | High       | High      | High          | Moderate  |
| C++    | Moderate   | Moderate  | High          | High      |
| Rust   | Very High  | Very High | Low           | Moderate  |

## 3.8: Web Stack Comparisons

Selecting the appropriate database technology is a fundamental decision in the design of IoT and sensor-driven systems. The database must efficiently handle pollutant data, balancing the requirements for real-time access, historical data storage, scalability, and data integrity. This section provides a comparative analysis of three popular database technologies—PostgreSQL, MongoDB, and MySQL—evaluating their strengths, limitations, and applicability to the needs of this project.

The comparison will emphasize factors such as schema flexibility, ACID compliance, scalability, and performance under the constraints of an IoT application. This discussion aims to guide the selection of the database that best

aligns with the project's requirements for structured and unstructured data storage, query capabilities, and overall system reliability.

### 3.8.1: Database Comparisons

In this project, selecting the right database technology is crucial for efficient data management, particularly given the range of pollutant data and the need for real-time access and historical data storage. Here's a comparison of some widely-used database options that fit well in an IoT and sensor-driven project context:

#### 3.8.1.1: PostgreSQL

PostgreSQL is a powerful, open-source relational database that provides strong support for complex queries, data integrity, and transactional integrity. It is particularly well-suited for applications where data consistency is critical. One of PostgreSQL's key advantages is its ACID compliance, which ensures that sensor data, once recorded, is accurate and protected against data corruption. Additionally, PostgreSQL has a robust indexing system and supports a range of datatypes, making it ideal for storing mixed data types, such as integer-based measurements, time-stamped data, and geolocation data.

The downsides include potentially slower write performance for extremely high data volumes, compared to NoSQL alternatives, and its structure can sometimes be too rigid for applications requiring highly dynamic data schemas. However, for a project requiring reliability and structured queries, PostgreSQL is a strong choice.

#### 3.8.1.2: MongoDB

MongoDB is a NoSQL database that uses a flexible, document-based data model. This flexibility allows for dynamic schemas, making it ideal for applications where data structures can evolve over time. MongoDB excels at handling large volumes of unstructured data and is highly scalable, which is beneficial if the project needs to accommodate many sensor nodes, each generating data continuously. It also supports horizontal scaling, making it ideal for IoT applications where the number of data sources could grow over time.

However, MongoDB is not ACID-compliant by default for multi-document transactions, meaning it sacrifices some data consistency for speed and scalability. This can be a drawback if precise, transactional integrity is critical for the application. MongoDB is an excellent choice when flexibility, scalability, and high-speed read/writes are prioritized over data integrity.

#### 3.8.1.3: MySQL

MySQL is a widely used relational database management system known for its reliability and speed. Like PostgreSQL, MySQL is ACID-compliant, ensuring data consistency and integrity. It is also highly compatible with various programming languages and environments, making it versatile. MySQL's performance is generally very strong for read-heavy applications, which is ideal if the sensor data is accessed frequently but written less often.

One limitation is that MySQL's capabilities for handling complex queries and JSON data are not as advanced as PostgreSQL's. Additionally, for large-scale, distributed data scenarios, MySQL may require more configuration and extensions. However, for structured data where data consistency is a priority, MySQL offers a balance of performance, ease of use, and reliability.

*Table 17: Database Technology Matrix*

|            | Complexity | Speed    | Documentation |
|------------|------------|----------|---------------|
| PostgreSQL | Very Low   | Moderate | High          |
| MongoDB    | Low        | High     | High          |
| MySQL      | Moderate   | Moderate | High          |

### 3.8.2: Frontend Comparisons

A responsive, user-friendly web interface is essential for monitoring pollutant data and interacting with sensor nodes. The choice of front-end technology will greatly affect the development experience, usability, and performance of the application.

#### 3.8.2.1: React

React is a JavaScript library maintained by Facebook, known for its component-based architecture and fast rendering via a virtual DOM. This makes it particularly well-suited for applications requiring dynamic, real-time updates, such as a dashboard for monitoring sensor data. React's flexibility allows for complex, responsive user interfaces, and its ecosystem includes a wide range of libraries for state management (like Redux) and UI components.

The primary drawback of React is its relative complexity, particularly in managing component states, which can lead to boilerplate code and a steeper learning curve for newcomers. However, for a project requiring real-time data display and a responsive user experience, React's speed and reactivity make it a strong choice.

#### 3.8.2.2: Vue.js

Vue.js is a progressive JavaScript framework that is known for its gentle learning curve, making it accessible to new developers while offering powerful features for more complex applications. Like React, Vue.js supports a component-based architecture and is highly performant for building dynamic interfaces. It is particularly popular for smaller projects where quick setup and ease of use are priorities, as it provides two-way data binding and reactive UI updates out of the box.

However, Vue's ecosystem is somewhat smaller compared to React's, meaning fewer third-party libraries and community resources. For a project prioritizing simplicity and rapid development, particularly if team members are newer to frontend frameworks, Vue's flexibility and ease of use make it a practical choice.

### 3.8.2.3: Angular

Angular is a comprehensive JavaScript framework maintained by Google, designed for building large-scale applications with a well-defined structure. Angular's strengths lie in its powerful templating, built-in services, and dependency injection, which make it well-suited for applications with complex business logic. It also provides tools for testing, routing, and state management, making it a more complete solution than React or Vue.

The main drawback is Angular's size and complexity, as it requires more setup and boilerplate code, leading to a steeper learning curve. For large projects with complex requirements and a team familiar with TypeScript, Angular provides a highly scalable framework with robust features and solid community support.

### 3.8.2.4: Flutter

Flutter, developed by Google, has emerged as a powerful framework for cross-platform app development, allowing developers to create applications for mobile, web, and desktop platforms using a single codebase. One of its defining characteristics is the use of the Dart programming language, which compiles to native code. This results in impressive performance and smooth user experience. Flutter's built-in rendering engine, Skia, enables developers to implement visually striking interfaces and seamless animations, making it an excellent choice for projects that require a high level of polish and responsiveness. Additionally, Flutter comes equipped with a rich set of customizable widgets and pre-designed components that significantly speed up the development process, ultimately reducing the time to deliver a user-friendly application.

When comparing Flutter to traditional web frameworks like React, Vue.js, and Angular, its cross-platform capabilities stand out as a significant advantage. While these frameworks typically focus on web applications, Flutter's ability to maintain a consistent user experience across both mobile and web environments is particularly beneficial for projects that aim for broad accessibility. However, there are some considerations to keep in mind. The Dart programming language, although relatively easy to learn, may require developers to step outside their comfort zone if they are more familiar with JavaScript or TypeScript. Additionally, while Flutter is robust for mobile development, its web support is still maturing, which may pose challenges in terms of performance optimizations compared to more established web frameworks.

The growing popularity of Flutter is supported by a vibrant community and extensive documentation, making it an appealing choice for teams looking for a cohesive and visually engaging application. This is especially valuable in a senior design project, where leveraging pre-built components and maintaining a unified codebase can streamline the development process. The combination of native-like performance, cross-platform functionality, and rich design capabilities positions Flutter as a compelling alternative, enabling teams to deliver efficient and aesthetically pleasing applications while meeting the diverse needs of users.

*Table 18: Frontend Technology Matrix*

|         | Complexity | Speed    | Documentation |
|---------|------------|----------|---------------|
| React   | Low        | High     | High          |
| Vue.js  | Low        | High     | Moderate      |
| Angular | Moderate   | Moderate | Moderate      |
| Flutter | Moderate   | High     | Moderate      |

### 3.8.3: Backend Comparisons

The backend technology is critical for managing data processing, user authentication, and communication with sensor nodes. The following languages and frameworks offer distinct advantages for different backend needs.

#### 3.8.3.1: *Golang*

Golang, or Go, is a statically typed, compiled language developed by Google. It is known for its performance, scalability, and built-in support for concurrency, making it ideal for high-performance web servers and microservices. Go's simplicity and readability help streamline development and reduce maintenance, while its performance makes it suitable for applications that require efficient data handling and rapid response times.

The downsides of Go include a smaller ecosystem compared to more established languages like JavaScript and PHP, meaning fewer libraries for specific use cases. Additionally, Go's strict type system can sometimes make rapid prototyping slower. However, for a project requiring performance, concurrency, and scalability, especially with real-time data processing from IoT devices, Go is a strong option.

#### 3.8.3.2: *JavaScript (Node.js)*

Node.js enables server-side JavaScript, providing an event-driven, non-blocking I/O model that makes it efficient for handling large volumes of simultaneous connections. This makes Node.js well-suited for IoT applications where frequent data transmissions occur, such as real-time sensor data monitoring. Its asynchronous architecture allows Node.js to handle many tasks simultaneously, which can improve scalability and reduce response times.

The primary drawback of Node.js is that its single-threaded nature can struggle with CPU-intensive tasks. Additionally, JavaScript's dynamic typing can introduce runtime errors, making code less predictable. Despite these drawbacks, Node.js's extensive package ecosystem and fast development time make it an appealing choice for projects requiring real-time data processing and rapid prototyping.

#### 3.8.3.3: *PHP*

PHP is a widely used server-side language known for its simplicity and vast ecosystem of frameworks and libraries. It excels in applications where quick development, ease of maintenance, and integration with SQL databases are essential. PHP has great tools for routing, templating, and database management,

making it a practical option for projects that require a traditional web stack with reliable performance. However, PHP has relatively lower performance compared to Go or Node.js and lacks built-in support for concurrency, making it less suitable for real-time or highly concurrent applications. Nevertheless, PHP remains a robust choice for standard CRUD applications.

#### 3.8.3.4: Python (Django/Flask)

Python, with its frameworks Django and Flask, is another popular choice for backend development. Django provides a “batteries-included” approach with a variety of built-in features for rapid development, making it suitable for applications that require complex backend logic, user authentication, and API endpoints. Flask, on the other hand, offers more flexibility and a smaller footprint.

Python’s slower execution speed compared to Go or Node.js can be a limitation for high-performance applications. However, its readability and vast ecosystem of libraries make it an excellent choice for backend applications where rapid development and ease of use are priorities. It’s also a good option for data-driven applications, as it integrates well with data science libraries for processing and analyzing sensor data.

*Table 19: Backend Technology Matrix*

|         | Complexity | Speed    | Documentation |
|---------|------------|----------|---------------|
| Golang  | Moderate   | High     | Moderate      |
| Node.js | Low        | High     | High          |
| PHP     | Moderate   | Moderate | Moderate      |
| Python  | Low        | Low      | Moderate      |

## Chapter 4 – Standards and Design Constraints

This chapter explores the critical standards, and design constraints that guide the development and implementation of our project. These standards guarantee the system’s compliance with regulatory requirements, safety protocols, and operational expectations like form factor, environmental durability, and usability. By adhering to established industry guidelines such as the National Electric Code (NEC), National Fire Protection Association (NFPA), and OSHA safety standards, the project prioritizes safety, reliability and efficiency. Furthermore, constraints such as size, weight, energy efficiency, and environmental considerations shape the project to meet consumer expectations and integrate seamlessly into residential and utility settings. This chapter outlines how these standards and constraints influence the device’s design, ensuring it stays robust, sustainable and user-friendly.

### 4.1: Form Factor

- The weight of the device should be less than 30lb as minimizing the weight is valuable for the consumer product.

- The breaker box we are utilizing is a standard Homeline 200 amp 30-space 60-circuit indoor breaker plug-on neutral electrical panel with cover with dimensions 3.65 inches in depth, 33.78 inches in height and 14.25 inches in width.
- All components have to be insulated to prevent hazards such as fires or electric shocks to the user.
- The system should be designed to not take damage from small particles such as dirt or dust.

## 4.2: Standards

In the development of this project, adherence to industry standards is very important. These standards make sure that the device obeys national regulations as they provide safety to the user and allows the device to work in an efficient way.

## 4.3: National Electric Code (NEC)

The National Electric Code (NEC) is a regionally embraceable standard for the safe installation of electrical wiring and equipment in the United States. The NEC is published by the National Fire Protection Association (NFPA) and updated every three years to keep up with the developments in technology and evolving safety needs of electric systems. [37]

The main purpose of the National Electric Code is to protect people and their property from electrical dangers through well-defined standards for electrical design, installation and inspection. The NEC is divided into articles, each article deals with specific standards on installation, components or safety keys. Some articles that are relevant to our project are:

### 4.3.1: Article 240

Article 240 talks about overcurrent protection. This article talks about how circuit breakers must be utilized to prevent overcurrent which can lead to electrical fires or damage in the device. By following this article, our system makes sure that each circuit is protected against overloads.

### 4.3.2: Article 312

Article 312 ensures the safe installation of electrical panels as it defines specific enclosure requirements such as measurements, spacing and materials that can be used for panelboards. Our panel must fit within these measurements, proving enough insulation and protection to prevent accidental contact with live parts.

### 4.3.3: Article 408

Article 408 rules the installation of panelboards requiring that they be installed with authorization for safe operation and maintenance. The breaker panel must be dead-front, this means that no live parts can be uncover and all components must be installed in a way that guarantees accessibility for both operation and maintenance.

## 4.4: National Fire Protection Association (NFPA)

The National Fire Protection Association (NFPA) is an international non-profit organization that creates and publishes standards and codes with the purpose of reducing the risk of fires and other dangerous hazards. The NFPA is responsible for creating standards that ensure the safety and effectiveness of fire protection, electrical and life safety systems.

The main purpose of the NFPA is to develop standards that help avoid and mitigate the effects of fire and electrical hazards. These standards become laws in many regions of the country and lead engineers, electricians, contractors and manufacturers on how to install, conserve, and work electrical systems safely. The NFPA's standards go past electrical safety, encompassing a wide array of fire and life safety measures, which make sure that buildings are safe for residents under normal and emergency conditions.

### 4.4.1: NFPA 70A

The NFPA 70A standard supplies additional guidelines for the installation of electrical systems in residential buildings. We have to obey these standards when it comes to the smart panel because they make sure that it works correctly but also minimizes the chances of fire hazards caused by faulty wiring, improper connections or overheating. In compliance with NFPA 70A, we have integrated fire resistant materials for wiring and connections to make sure that if a short circuit happens, the panel will remain safe. Additionally, the system will include safety mechanisms that prevent excessive current from causing overheating. [38]

## 4.5: OSHA Electrical Safety Standards

The Occupational Health and Safety Organization (OSHA) is responsible for setting standards to address several hazards such as explosions, fires, electric shock and other dangers that workers or consumers might experience when dealing with electrical systems. [39]

The primary focus of these standards are the electrical systems parts that workers work with such as circuits, high voltage sources, motors, and switches. OSHA's electrical safety standards often align with standards from the National Fire Protection Association (NFPA) and the National Electric Code (NEC) confirming comprehensive protection for all kinds of electrical systems.

### 4.5.1: Examination

The first step listed by OSHA is examination. Before any electrical system or device is utilized, the Occupational Health and Safety Organization mandates that it be examined and determined to be free from identifiable hazards that have the risk to cause injury or even death. All devices must be labeled for its planned purpose, making sure that users and workers can identify each component easily and precisely. Devices that have the intended purpose of enclosing or encapsulating other devices should have sufficient mechanical strength and durability to supply sufficient protection. All electrical insulation must be undamaged and complete and

correctly bonded to prevent electric shocks. Long exposures to heat should not make the insulation break down and the materials utilized must be fit for the anticipated voltage and current levels. Materials such as breakers, switches and relays must be classified according to their voltage, current and planned use to make sure safe operation. Other factors that were mentioned above should also be considered to contribute to the safety of workers or consumers that may come into close contact or in proximity of the device. [39]

Showcased below is a list of relevant standards developed and published by the OSHA and we must follow when we start developing our system.

#### 4.5.2: Installation and Use

OSHA requires that all devices be installed according to their planned purpose, as it is indicated by their label and manufacturer's specification. Listed or labeled equipment shall be utilized or installed in agreement with any instructions included in the listing or labeling. [39]

#### 4.5.3: Identification of Disconnecting Mains and Circuits

Proper identification of disconnecting mains and circuits is important to make sure that the user and workers can easily recognize the correct circuit breaker or switch in case of an emergency or for maintenance.

Any devices utilized to disconnect power to appliances and motors must be clearly labeled with their purpose. These labels should be durable enough to withstand the conditions that they are involved in. These conditions could be from dust, wear and tear via touching to heat and current. Additionally, every main power source, feeder, and branch circuit must be marked with tough legible labels that do not fade or wear out over time. These labels help to make sure the safety of workers during maintenance and prevent accidents like electrocution by ensuring that only the planned circuit is disconnected. [39]

A disconnect switch refers to a switch that is utilized to disconnect the conductors of a circuit from the source of electric current. Disconnect switches are essential because they allow a circuit to be opened for maintenance as the flow of electricity is paused and workers can do their work safely.

#### 4.5.4: Guarding of Live Parts and Equipment

The purpose of this standard is to protect any person who may be in the area of electrical equipment against accidental contact. Accepted enclosures are required for electrical devices operating at 50 volts or higher to prevent people from coming in contact with them. Only qualified workers should be allowed to access areas where live parts are present. For installations where live parts are uncovered to potential physical damage such as falling objects, additional protective barriers such as screen should be utilized to minimize harm and avoid hazards. [39]

### 4.5.5: Grounding

Grounding is one of the most important safety measures in electrical systems, avoiding dangerous electrical faults and protecting both users and devices. There are two kinds of grounding: (1) Electrical System Grounding and (2) Electrical Equipment Grounding. Electrical system grounding is achieved when one conductor of the circuit is deliberately connected to the earth. The purpose of this is to protect the circuit if lightning strike or other high voltage contact occur. Equipment grounding is achieved when all metal frames of equipment and enclosures that contain electrical devices or conductors are grounded by means of a permanent and continuous connection or bond. The equipment grounding conductor supplies a path for dangerous fault current to return to the system group at the supply source of the circuit if an insulation failure takes place. [39]

## 4.6: Electromagnetic Compatibility (EMC) Standards

Since our project relies on wireless communications technologies such as Wi-Fi and Bluetooth, it is important to make sure that our system complies with the Electromagnetic Compatibility (EMC) Standards. Following the EMC standards ensures that the smart panel works efficiently without creating harmful Electromagnetic interference (EMI) which could impact other electronic devices in the area. The most relevant EMC standards for our project are IEC 61000 and FCC Part 15.

### 4.6.1: IEC 61000 and FCC Part 15

The IEC 61000 standards ensure that our smart panel does not emit excessive electromagnetic interference and that it can work properly in areas where other wireless systems such as routers, phones and smart appliances are in use. Additionally, FCC Part 15 standards in the United States further describe the permissible limits for radiated and conducted emissions from electronic systems, guaranteeing that the panel's wireless communication does not interfere other systems. [40]

Fulfilling these standards is critical for the smart panel as it guarantees efficient operation in a residential area where several other wireless and wired electrical systems coexist. It also avoids the panel from causing damaging interference that could interfere with other devices such as phones and routers. Lastly, it ensures the system's resistance to interference from other wireless systems. Failure to follow these standards can result in the device having several errors, connectivity problems and even safety risks due to electromagnetic interference.

To meet IEC61000 AND FCC Part 15 standards, our project must follow the following design implementations to control electromagnetic emissions and reduce susceptibility to external interference. Sensitive components such as Wi-Fi routers and microcontrollers are enclosed in electromagnetic shielding to avoid high-frequency electromagnetic fields created by these devices from radiating out to other devices as a result decreasing the device's overall EMI emission. Another implementation are EMI filters; these filters are installed into the power supply

circuits to reduce high-frequency noise created by switching devices such as microcontrollers and relays. Decoupling capacitors are also utilized on power lines to balance the voltage and absorb high-frequency noise. One last implementation that must be done to meet the IEC61000 and FCC Part 15 standards are antennas. Proper antenna installation guarantees optimal signal strength and reduces interference from internal sources within the smart device. Also, Antenna isolation helps decrease EMI exposure and ensures clean signal transmission. [40]

## 4.7: Communication Standards

Communication standards are essential for guaranteeing seamless data exchange between components of our project. Effective communication is important for enabling coordination between the microcontroller, sensors, and external devices, such as the mobile app interface. The project leverages established protocols such as Serial Peripheral Interface (SPI), to facilitate reliable and efficient data transfer. By adhering to these communication standards, the system guarantees precise control, minimal latency, and compatibility with several peripherals. This section highlights the importance of SPI in our project's functionality and explains its role in creating a thorough and interconnected platform for real-time data acquisition, analysis and feedback.

### 4.7.1: SPI

Serial Peripheral Interface (SPI) is a synchronous, full-duplex communication protocol that is utilized for short-distance communication between microcontrollers and other components such as sensors, displays and memory devices. SPI communication has a critical role in devices as it offers an efficient and cost-effective solution for swapping data between electrical systems. SPI works on a master-slave architecture, where the master device, commonly the microcontroller, controls the data exchange. [41]

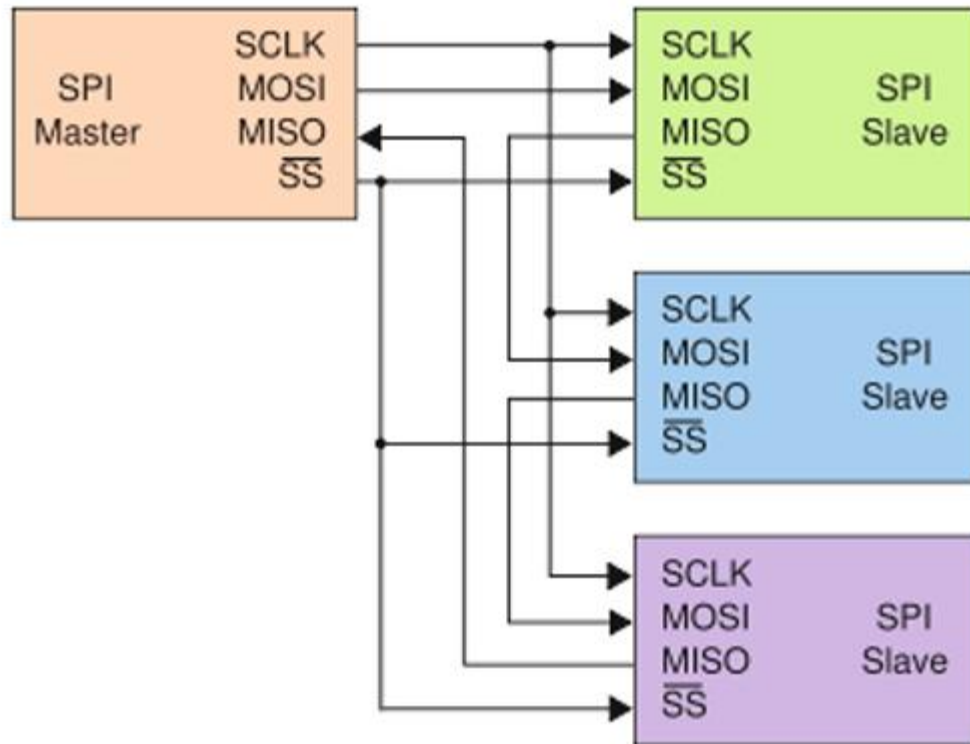
Any devices that communicate through SPI will have these four components: Serial Data Master Out Slave IN (MOSI), Serial Data Master In Slave Out (MISO), Serial Clock (SCK) and Chip Select (CS). The master device will start and mandate the data exchange, it generates the clock signal and chooses which slave device to communicate with. Then the slave devices will wait for the commands and replay through MISO. The SCK signal syncs data transmission, and the Chip Select line allows or disallows specific slave devices for communication. [41]

Serial Peripheral Interface communication follows several key electrical features. It commonly works at 3.3V or 5V, guaranteeing voltage compatibility between the master and slave devices. SPI also supports many clocking modes, set on by clock polarity (CPOL) and clock phase (CPHA), which describe the clock signal's active state and edge. There are four distinct modes determined by the CPOL and the CPHA. The CPOL bit is what determines the polarity of the clock in its idle state. The CPHA bit chooses whether the data is shifted and sampled on the rising or falling edge of the clock. The table below showcases the four different modes.

*Table 20: SPI Modes*

| SPI Mode | CPOL | CPHA | Clock Polarity at idle state | Clock Phase utilized to shift/sampled data                  |
|----------|------|------|------------------------------|-------------------------------------------------------------|
| 0        | 0    | 0    | Low                          | Data sampled on rising edge and shifted out on falling edge |
| 1        | 0    | 1    | Low                          | Data sampled on falling edge and shifted out on rising edge |
| 2        | 1    | 0    | High                         | Data sampled on falling edge and shifted out on rising edge |
| 3        | 1    | 1    | High                         | Data sampled on rising edge and shifted out on falling edge |

Data order can be pass on in either Most Significant Bit (MSB) or Least Significant Bit (LSB) order, depending on the devices features. Additionally, it is common to utilize multiple peripherals in SPI and the reason behind that is by having an individual chip select (CS) for each peripheral. When the master device desires to communicate with a particular slave, it allows that slave's chip select and can begin communicating. However, if several chip selects were enabled, the master device would be getting information on the MISO line from several slaves and would not know which data comes from which slave device. A different way of working with multiple peripherals in SPI is by daisy-chaining them together. This configuration is utilized in circumstances where the microcontroller is only outputting data and does not require to collect data, but if data had to be received, the return data will have to pass through all the peripherals before making it back to the microcontroller. The diagram below showcases what a daisy-chain configuration looks like. [41]



*Figure 13: Daisy-Chained SPI Interface*

The TMC 5160 motor controllers we are using for this project use SPI to configure control parameters. As we are using two of these controllers as slaves, with the ESP32 as the master. We have utilized a shared bus rather than a Daisy-chain architecture, meaning that The SCLK, MOSI, and MISO pins will be directly connected for the 3 devices, and each slave device will have a separate chip select pin assigned to it on the ESP.

## 4.8: User Interactivity and Data Constraints

The development of our project is driven by the knowledge that regular electrical circuit panels, even though they work, lack the ability to interact with the grid in a dynamic and automated way. With the increase of renewable energy sources such as solar and wind, which are inherently variable, the balance of the grid can be challenged during periods of high use or low supply. Even though our primary focus for the smart panel is to be controlled through an app on your phone, user interaction is also important. People need to understand the function of the device as well as having an LCD display or app interface that provides relevant data for the user's knowledge. Thus, we are implementing user constraints that will impact the design and implementation of our smart panel.

As users and homeowners, we came up with very interesting ideas of what should be done on the consumers' end and we even asked people such as our friends, family members and UCF faculty to hear some opinions that weren't ours. We

believe it was critical to get these comments as the usage of this device will be from both the utilities and consumers' point of view.

## 4.9: Consumer Constraints

For the homeowner's point of view, the main thought that we began to work on was not how this system is going to work but how is it going to benefit the user and how it will impact the user both in terms of price and interference with other electrical devices. For the utilities' point of view, the main thought was grid stabilization, cost and difficulty of installation.

The first consumer constraint that we assessed was how this device was going to benefit the consumer. We believed the best way to showcase these benefits was in the form of data, especially data such as how much money they were saving, the number of times the system worked and how much energy they were saving. The main method that we decided to take to achieve this task was a phone app. The phone app was something that would make the usage of the smart panel easier for the consumer but would add complexity to the project. The phone app not only would give the consumer data but also allows the consumer to control the smart panel directly from their phone. When the grid frequency changes, they will see it through the app and tell the smart panels what to do so the grid frequency can be balanced again. After deciding to do the phone app, we included the needed interactions with the MCU to be able to communicate with the app, which also added constraints to which MCU we could use in this project.

For the data that is going to be showcased in the app, we wanted to display general data points such as the grid frequency at the moment they log in the app, the voltage and current being utilized in the house, the amount of power that the device has saved and consumed, and maybe how much money has been saved. Lastly, the app will tell you what the device has done to restore the grid frequency if it deviates. Showing the values for voltage and current can be done utilizing analog-to-digital converters which will send the data to the MCU and then on to the app. The power consumption and cost data can be shown on the app by utilizing the voltage, current and cost-per-kilowatt-hour in software programming calculations.

### 4.9.1: Utility Constraints

For the utility constraints, we researched how the cost of the materials, the difficulty of installation and benefits to the grid frequency impact the final design and implementation. If this system is to be implemented by the utilities and installed as an add-on item, the cost would need to decrease. This impacts the material selection since a balance needs to be accomplished between the system's accuracy and the overall cost. In addition, the installation difficulty also impacts the design of the device as the device needs to be easy and simple to install for the user to engage with it. If the device cannot be installed by the consumer and needs to hire a professional, then that cost will also have an impact on the final cost of our project. Lastly, device accuracy impacts the design process as consumers would not want to buy or use a system that gives inaccurate data. Thus, material selection

and design elements for this project will center on accuracy, effectiveness and reliability.

#### 4.9.2: Homeowner Constraints

The physical size of this device needs to be big enough to encase the PCB, wiring, several small parts, power supply, etc., but small enough to be easily mountable in the wall like a traditional breaker panel. The size is considered a small constraint as the performance and precision of the device are more important. The size of our breaker panel is 3.75 inches in depth, 33.78 inches in height and 14.25 inches in width. The weight of the system should be reduced to help with installation. We are aiming for the weight of our device to be less than 30 pounds. The time constraints for this project are based on our scheduled due dates generated in Senior Design 1 and 2. We must stay on track to move toward finish the project on its corresponding due date. The cost of this device should be restricted per unit while still giving accuracy of materials and dependability of parts, designs and implementations. For our project, we were given a \$3,000 budget and we hope to stay below that line. It is vitally important that our device keeps a high level of safety across all parts, implementations and installations. Since the system will be working with high voltages and high current settings, it is important to follow all safety standards during the entire project. Lastly, given that the derivations of grid frequency are so small, it is critical that our device accurately measures the frequency to a high degree of precision, so that the device performs as supposed to and alleviates stresses on the grid.

#### 4.10: Environmental Constraints

Another big type of constraint that impacts our project and its design are environmental constraints. Since our project deals with electrical systems, smart technology and integration with the power grid, environmental constraints guarantee that our project is strong, secure and sustainable under different environmental conditions.

The first of these environmental constraints is temperature tolerance. Our smart panel needs to work dependably under different temperatures, mainly since it could be installed in areas like garages, basements or utility rooms where temperatures can vary. The device should be designed to work correctly within large temperature ranges, for example -20°C to 60°C. The materials must also be able to resist both high and low temperatures without performance failure, overheating or breaking.

Another element of the environmental constraints is humidity and moisture resistance. In areas such as basements or utility rooms, the device could be exposed to high levels of humidity and even small splashes of water. The device's panel enclosure must be assessed for moisture resistance to keep the internal parts safe from damage caused by moisture, splashes or exposure to water. Correct sealing and insulation will help maintain safety and avoid electrical shorts.

The next environmental constraint is dust and particle protection. Electrical panels in certain areas such as garages or industrial settings might be exposed to dust, dirt, sand and other small particles. The panel should be designed with a dust-resistant enclosure to avoid small particles to go into the device which could lead to malfunctions or damage the internal parts over time. Adequate filters and enclosures would be needed to meet this complaint.

Another environmental constraint of this project is energy efficiency. Decreasing energy consumption is a critical environmental concern. The smart panel itself should consume minimum power while observing and managing electrical loads. The design of our project must prioritize energy efficiency, utilizing materials that consume low power when idle. The goal of this constraint is to lower the device's overall energy footprint, which lines up with broader environmental goals of decreasing electricity usage and optimizing resource use.

An environmental constraint that also needs to be considered is the sustainability of the materials. The materials utilized in the device's construction should be sourced and created with sustainability in mind, mainly if our project is to be marketed to consumers who are environmentally conscious. The device's casing and internal parts should utilize recyclable or environmentally friendly components if possible. This could even expand to using RoHS-compliant materials which restricts the use of dangerous materials in electronics manufacturing.

Another environmental constraint to be considered when talking about constraints is End-of-Life disposal. Electronic waste management is a very important environmental consideration. Due to our project being a smart breaker panel, provisions for recycling or safe disposal at the end of the device's lifecycle should be taken into consideration. The device should be created for simple disassembly to facilitate recycling of parts such as batteries, and circuit boards.

One last environmental constraint is the environmental impact of production. The production process of the system should take into consideration its carbon footprint and the environmental impact of sourcing components. Consideration of how materials are sourced, transported and built can decrease the overall environmental impact of the product. Renewable sourcing and eco-friendly manufacturing approaches can be taken into consideration for design of the device to reduce environmental harm.

## 4.11: Remaining Constraints

Political and ethical constraints, while not directly tied to technical or environmental aspects, play a significant role in shaping the project's implementation and societal impact. Political constraints involve compliance with energy policies, cybersecurity regulations, and grid management standards that govern the integration of smart technologies into existing infrastructure. Ethical constraints guarantee that the project respects user privacy, promotes data security, and minimizes potential social disruptions, such as job displacement or misuse of consumer data. By addressing these broader constraints, the project emphasizes responsible

innovation, balancing technological advancement with societal and regulatory expectations.

#### 4.11.1: Political Constraints

A minor constraint but also significant is political constraints. Our project could potentially face political constraints depending on several elements, especially those connected to energy policy, regulations and grid management. Political constraints usually come from government laws, policies on energy consumption and the wide legal framework that mandates how new technologies are established, mainly in areas such as energy and infrastructure. Our project may be subject to national or state energy laws that regulate how smart grid panel technologies can be implemented. These policies may impact on how our smart panel is authorized to interact with the electrical grid. As our smart panel interacts with the grid and communicates through an app wirelessly, cybersecurity regulations may be implemented into the system to meet certain constraints to protect the grid and consumers from cyberattacks.

#### 4.11.2: Ethical Constraints

One final constraint that needs to be considered for our project is ethical constraints. Ethical considerations are usually important in projects that involve technology, data collection, energy management and the potential impact on both consumers and the environment. Ethical constraints guarantee that the project that the project is designed and executed in a way that respects the consumer's privacy, encourages fairness and stays away from negative social or environmental impacts. Our project may collect and transfer data about a consumer's energy consumption, load patterns and other personal information. This raises ethical concerns about data privacy and security. The device must guarantee that consumer data is protected, safely stored and only used for its designed motive. Ethically, it is important to make sure that consumers have control over their own data, know how to utilize it, and are safe from unauthorized access or data breaches. We must collaborate with data privacy policies and be clear with consumers about data collection, processing and transmitting practices. Automation and smart technologies can lead to job displacement, mainly in areas like utilities and installation services. Ethically, it is critical to consider whether the widespread usage of our device could badly affect certain jobs.

## Chapter 5 - Comparison of AI Platforms

In today's world, the growth of artificial intelligence platforms has transformed how engineering and technical teams proceed toward project development. AI platforms such as ChatGPT and Google Bard are some of the AI platforms that are utilized the most, they offer higher language processing abilities that can help in everything from technical assistance and idea generation to writing, debugging and optimizing code and even analyzing data. For our Senior Design Project, ChatGPT has become a helpful resource on overcoming challenges in our research, design and execution.

However, while AI platforms offer a large range of opportunities, they also have limitations. These platforms depend on the data that they were set on and lack the intuitive and creative problem-solving capabilities of the human brain. Additionally, the correctness and accuracy of the information generated by these AI platforms are also not 100% reliable, which can cause problems in projects like ours.

### 5.1: Technical Assistance and Idea Generation

ChatGPT is very useful in supplying technical suggestions and expanding our understanding of different ways to proceed toward implementing key features in our Smart Panel. During the first stages of our project when we were thinking about options for grid frequency detection and breaker control mechanisms, ChatGPT provided several viable ideas that helped us come up with a solution.

This insight assisted us to improve our methods toward component selection. In the same way, Google Bard gave us real-time data about the newest technologies in grid frequency detection. This comparison between platforms allowed us to choose the best components so our system could have the highest level of efficiency.

ChatGPT's ability to create and compare multiple design ideas was also very helpful in amplifying the system's performance. For example, when working on the grid fluctuation, we were trying to come up with ideas that hadn't been used before. ChatGPT proposed to use a real-time feedback loop using a phase-locked loop circuit. As we investigated the proposed idea, we realized its potential applications and how it can help maintain the stability of the system.

### 5.2: Code Refinement and Error Resolution

Another aspect where ChatGPT was very helpful and influential was in the programming section of the project. In our project, we are utilizing and programming the MSP420 microcontroller to control the breaker system based on real-time grid frequency signals. Given that all our group members don't have a strong grip on programming due to the difference in majors, ChatGPT can be a very useful tool in explaining code structures and debugging errors.

ChatGPT can be helpful when it comes to this aspect of the project as it can provide you code snippets and even debugging steps if there is something wrong with your code. Even when your code runs perfectly and there are no errors,

ChatGPT can still give some insights into how to optimize the code so it can run more efficiently. ChatGPT can also supply alternative libraries and documents that can expedite our coding process.

When coding mental blocks can happen and situations where we don't know where to begin can occur, ChatGPT can give different ideas and solutions that can provide a way forward. This resource can be very useful not only for debugging errors and explaining what is wrong but also for optimization and efficiency in the code.

### 5.3: Data Analysis Support

Another way ChatGPT and similar AI platforms can be useful in our Senior Design projects is with data analysis. As our system generates data during testing, especially variations on the grid frequency and the performance of the circuit breaker, ChatGPT can assist with arranging and explaining the data. It can also recognize trends when it comes to the panel responding to different grid conditions and giving solutions for upgrading the efficiency of the system. For example, ChatGPT could analyze the data that we collect from how the circuit panel responds to the changing grid frequency and provide us with trends, errors and sections that need improvement. It can also explain the data in different ways so we can understand it better and give us different ways to visualize it. This is very useful as we can use those data visualization techniques to showcase our data in our reports.

Another way that ChatGPT can be beneficial in data analysis is taking our data and comparing it with already existing data creating a better look into our system and if we are going in the right or wrong direction. ChatGPT can also assist us if we ever feel lost or do not know what to do with the data that you have collected, as it can supply many references to documents and explanations that can help us expand our knowledge and understand the data.

### 5.4: Limitations and Potential Pitfalls

While ChatGPT is really helpful when it comes to technical assistance and idea generation and optimizing and debugging code, there are several limitations that need us to proceed carefully. We must make sure the exactitude and accuracy of the important information being given by ChatGPT because if any of this information is inaccurate then that can cause serious damage to our project. A wrongful execution given by ChatGPT can cause the design of the circuit panel to fail, which is a serious issue due to the carefulness and precision that those implementations required so the system can do its job such as detect the change in the grid frequency and making sure to do the correct response to that grid frequency change. Depending only on AI platforms such as ChatGPT and Google Bard without using other methods of researching can have a negative impact on your project. It can push back your project timeline, as additional time will be needed to find, fix and re-test the parts that were affected by the inaccuracies.

It is not bad to use ChatGPT or other AI platforms when we feel stuck or need help, but always ensure that the information that it is being utilized is accurate and correct. Just because it is an AI that is giving the information does not mean that it is 100% correct, they make mistakes too. That's why it is important to always consult additional resources and experts and confirm that the information that was given by ChatGPT and used in the project is accurate and correct. Especially with this project as there is a possibility that it could be installed in homes, the trust and confidence of the system to work correctly could be impacted if incorrect code that was giving by an AI tool were to cause a flaw on the system or system failure. This could also affect the learning experience and reliability of the team as they did not make sure that everything on the project was 100% accurate and correct.

This is why it is very important to reserve some time to verify and validate the ideas and implementations that are given by ChatGPT before implementing them into our project. Consulting with additional resources, talking to experts in the subject and doing testing outside the project are important steps that can confirm the accuracy and correctness of the implementation in our system. Doing those steps lets us use the strengths of ChatGPT while eliminating the risks that can come with inaccurate information.

## 5.5: Creativity and Problem-Solving

Even though there are many ways to use AI platforms like ChatGPT and Google Bard to our advantage when making ideas and implementations as well as analyzing data, it is important to realize that these AI platforms lack the creativity and abstract problem-solving capabilities that the human mind contains. These AI platforms work based on patterns and information available in the data that they contain, and while they are excellent at coming up with quick ideas, creating text based on already known information, AI platforms are missing the capacity to think intuitively or in an abstract way, which is a key component when it comes to creating innovative systems such as ours. It was our team's creative problem-solving that moved the project forward when we faced issues that did not fit within pre-existing patterns.

Thinking outside the box and taking from previous experiences is something that AI technology cannot achieve when it is compared to the human mind. For example, when coming up with the mechanical components of the braker control system and how we were going to implement it, ChatGPT could not provide the same level of innovative thinking that was necessary to address the physical constraint we encountered. In cases like this one, human ingenuity is key to generating solutions that AI platforms cannot create.

## 5.6: Test Case

To achieve a better understanding on how ChatGPT can work and provide solutions with such effectiveness, we asked ChatGPT the following question: "What are the most effective methods for detecting grid frequency variations in an electrical panel system for residential use?"

### 5.6.1: Zero-Crossing Detection

- How it works: Zero-Crossing Detection identifies grid frequency by observing the zero-crossing points of an AC signal. It computes the instances when the AC waveform crosses the zero-voltage point. This method is mainly utilized for detecting frequency in real-time applications due to simpleness and cost-effectiveness.
- Pros: Easy to implement, low cost, generally utilized in AC systems, quick response time.
- Cons: Prone to make noise, not as precise when it comes to detecting minor fluctuations.

### 5.6.2: Phase-Locked Loop (PLL)

- How it works: The Phase-Locked Loop method locks onto the grid frequency and can productively track changes in frequency. The system creates an internal signal that parallels the frequency of the grid and alters accordingly to keep synchronization. This method is very popular for its accuracy and real-time tracking capabilities.
- Pros: High precision in detecting frequency, acceptable for real-time grid applications.
- Cons: Harder to implement, requires delicate and careful tuning.

### 5.6.3: Frequency Measurement Using a Microcontroller

- How it works: A microcontroller with a built-in frequency counter computes the frequency of the incoming grid signal. The microcontroller tests the signal at a high rate and calculates the grid frequency based on the time between zero-crossing points.
- Pros: It can be programmable based on the set requirements of the project; it can be integrated into an already existing control system for centralized monitoring.
- Cons: The precision relies on the sampling rate and the resolution of the microcontroller; to improve precision, you may need to install additional components.

## 5.7: Result

ChatGPT's response provided a complete outline of three common methods for detecting frequency changes. Each method has its own advantages and drawbacks, and which one is better depends on the standards that we want to set for our project such as cost, power consumption, and efficiency. We found that the Phase-Locked Loop could be the most relevant for our project as it is the most accurate and capability to handle real-time frequency variations. The zero-crossing detection method, even though it was simpler, this method was less applicable due to the noise problem and low accuracy. Lastly, the microcontroller-based approach showcased flexibility, but it becomes more complicated around the sample rate and accuracy as it would need additional components and calibration.

Overall, utilizing ChatGPT and other AI platforms to provide technical assistance and support when we are stuck is not a bad thing, on the contrary, it is very helpful. However, abusing these platforms and utilizing them without even trying to think for ourselves is when bad things tend to happen. Just because ChatGPT gives a solution, idea or implementation does not mean that is it correct or even accurate. There have been many occurrences when ChatGPT has provided a response to a question and that response when investigated and verified has been wrong. So do not let your project be ruined because you did not put in the extra time to check if the response from ChatGPT was correct and accurate. By balancing the usage of AI platforms for routine tasks and supporting them with human insight, we are able to move on in our project guaranteeing efficiency and innovation.

# Chapter 6: Hardware Design

Chapter 6 focuses on all the hardware necessary for the project to function, including but not limited to the electrical schematics and the mechanical drawings.

## 6.1: Overall electrical Schematic

The figures throughout section(s) 6.1-6.6 show all electrical connections necessary for the PCBs

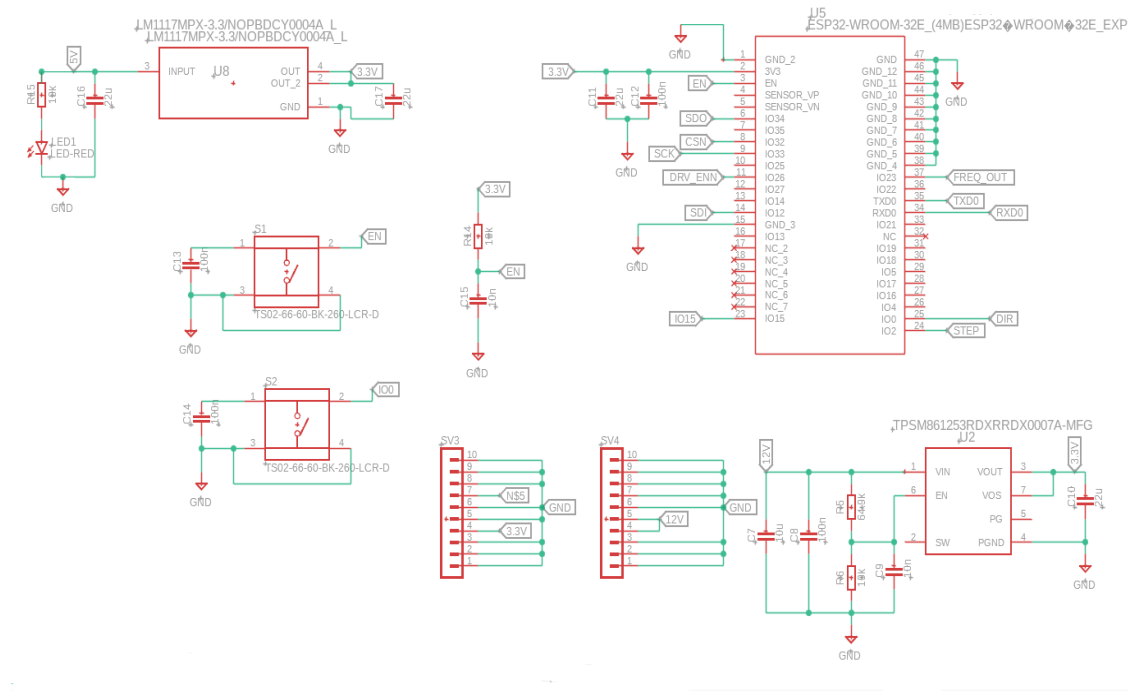


Figure 14: Overall Electrical Schematic Part 1

The overall schematic accounts for 5 different PCBs: the MCU board, the 12V PSU, the 3.3V PSU, the frequency detector, and the motor controllers. Special care was taken to assure that all net (node) names are consistent within the overall schematic. Wherever wires, represented in turquoise, have a common name, 12V for example, means that they are somehow connected to each other on the PCB.

An important note is that these subsystems will be divided into their own separate PCBs for the final projects. At Arthur Weeks's suggestion, all subsystems will be modular to help ease troubleshooting and minimize their individual impact on downstream systems. As an example, if the 12V power supply does not work, we could still hook up a benchtop power supply to run the other systems without having to redesign or destroy the PCB. Or if the ESP32 cannot be programmed successfully on the MCU board, then a development board can be used temporarily to allow for proper testing and integration of the subsystems until the schematic or PCB are corrected, ordered, and assembled.

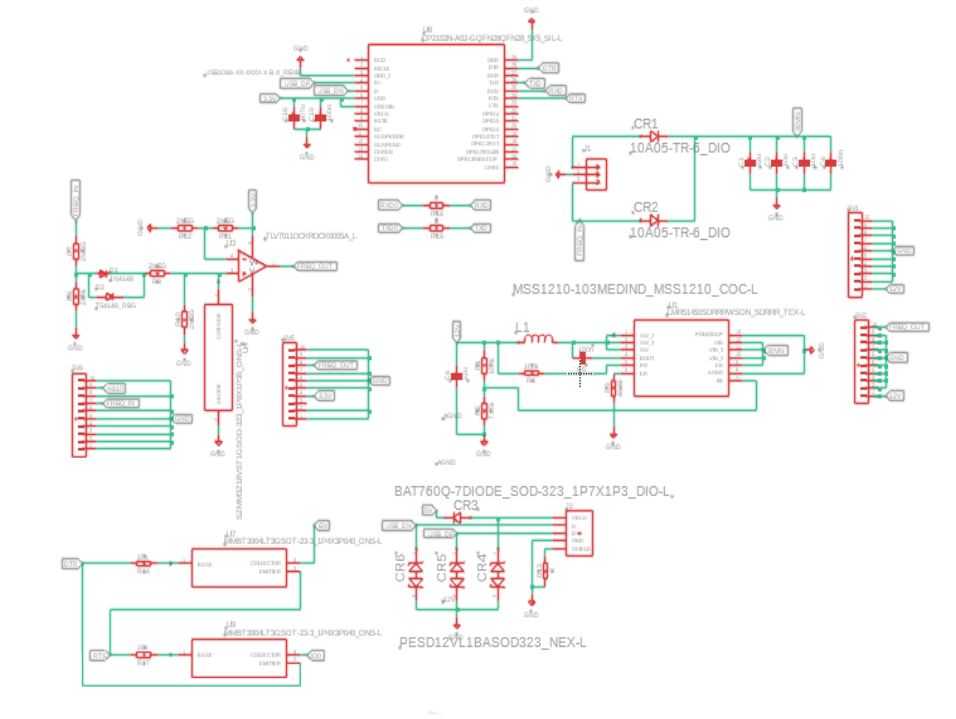


Figure 15: Overall Electrical Schematic Part 2

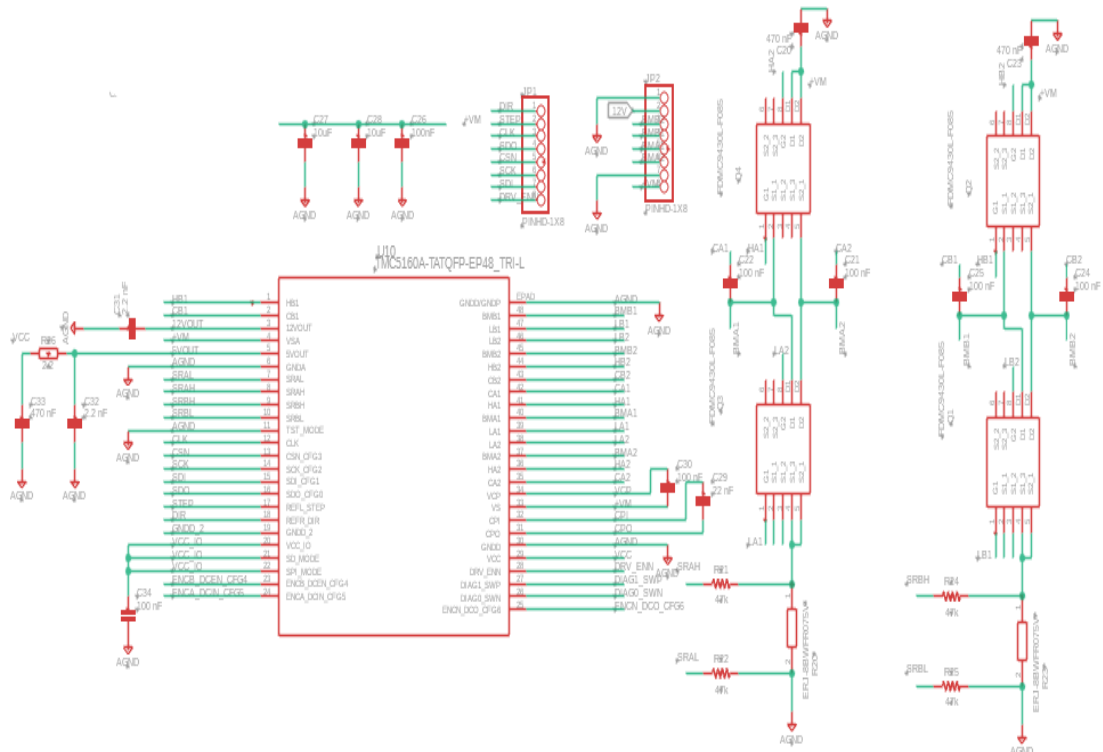


Figure 16: Overall Electrical Schematic Part 3

## 6.2: 12V PSU Electrical Schematic

The first electrical subsystem is the 12V power supply. It provides all of the regulated power for all other electrical subsystems.

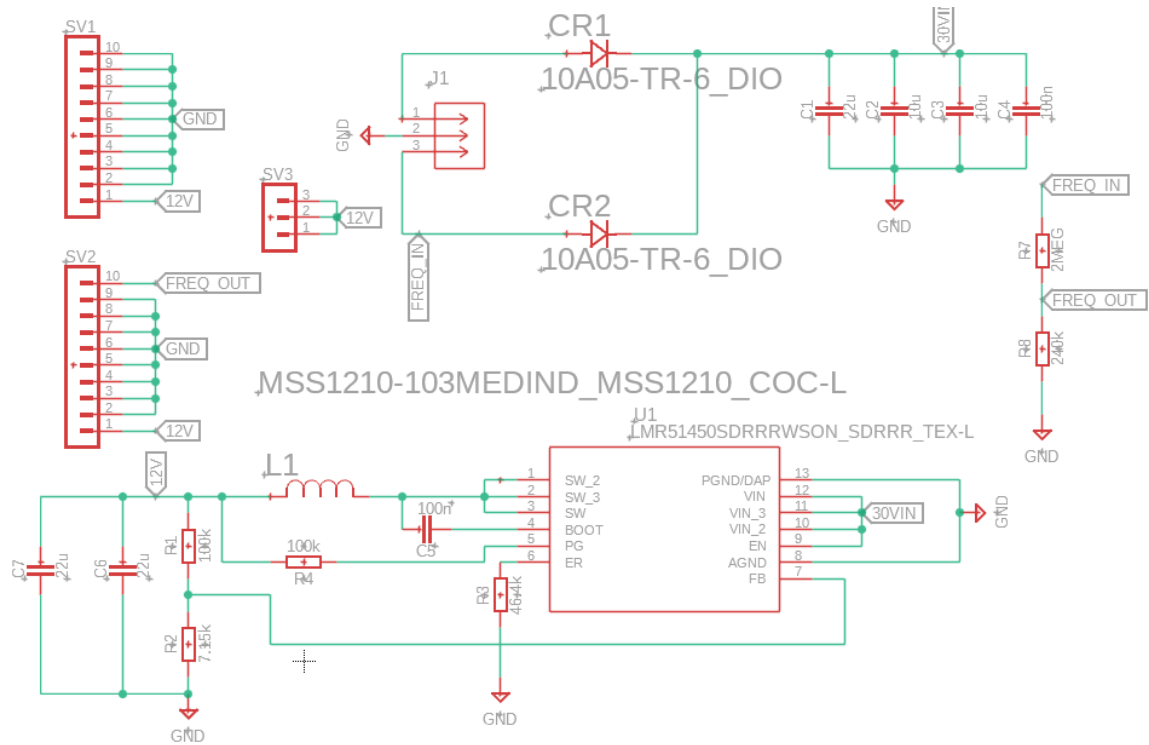


Figure 17: 12V PSU Electrical Schematic

There are four unique sections within the 12V PSU: The transformer's input from the wall, the 12V switching regulator, the voltage divider for the frequency detector circuit, and the collection of pin headers.

The first stage is the transformer's input, represented by J1, and all the circuitry connecting to it. J1 represents the terminal block used to connect the output of the step-down transformer to the PCB. Following the transformer's output is a full wave rectifier. We opted for the full wave rectifier instead of the full wave bridge rectifier due to the power savings. From the Hammond Manufacturing article mentioned in Section (XXXX), the current derating for a full wave bridge rectifier is 0.62, which would mean that the transformer must output 40% more additional current than its non-bridge rectifier counterpart. The current is the limiting factor for our project. Any additional current needs represent a more complex power management circuit and more complex thermal management. Both add cost and time to the project. Following rectification is the bulk capacitance. The output of the rectifier is a bumpy sine wave that oscillates from 0V to approximately 28V. The bulk capacitance smooths out the bumpiness into a rough approximation of a DC signal. There are currently four capacitors of various capacitances. This is to give us options when

populating the board. If some are deemed unnecessary or more is needed, we can swap out parts for more/less capacitance with minimal PCB redesign [42].

The step-down transformer is not included on this board. Instead, it is mounted to the MCU's PCB and wires will be routed to the 12V PSU board. This is to satisfy Dr. Weeks's requirement that all voltage regulators must be on separate daughter board. Since the transformer is not changing, it makes sense for it to be on the main board.

The second stage is the 12V switching regulator and all of its supporting circuitry. This circuit was not designed by us. The circuit is from Texas Instruments' Webench Power Designer tool and was validated both from the graphs provided by the Power Designer and PSpice for TI (Cadence). To obtain this specific circuit, we selected a DC/DC power converter with an expected input between 26-27V and a maximum output current of 5A. The schematic and all other relevant information from Webench will be included in the appendix and works cited [43].

This schematic also includes the first stage of the frequency detector circuit. We put the first stage as close to the input as possible for safety reasons. The frequency must be tapped as close to the source as possible to minimize distortion. Doing so means a connection to the unimpeded output of the power transformer. In case of accidental shorts or contact, we do not want to damage components or the engineer. After the voltage divider, the voltage is dropped to less than 10V and the current is expected to be on the order of microamps. This represents a far safer design than routing the unimpeded transformer signal across the MCU board where the more expensive and sensitive electronics will be held.

The last notable section is the pin headers. The pin headers serve several purposes, the most important is routing to other boards. Since the voltage regulators cannot be incorporated into the main board, the pin headers must. Since pin headers are sockets, they will also be used to mount the daughter boards to the main PCB instead of standoffs, screws, or any other mounting hardware. These boards are lightweight, and there is a minimum of 20 connection points per board, so we expect that this will be enough to hold up the daughter boards.

### 6.3: 3.3V PSU Electrical Schematic

The 3.3V PSU electrical subsystem consists solely of the 3.3V switching regulator and its supporting circuitry. Similar to the 12V switching regulator in Section 6.2, we used Texas Instruments' Webench Power Designer to get the schematic and the appropriate values. Since the input power is sourced from the 12V PSU, we did not validate the schematic in PSpice for TI or another simulation software, since the input will be incredibly stable, and the peak current requirement is only 1A.

To obtain this specific schematic, we used a 12V as the input and 3.3V and 1A as the output. The schematic and other relevant files will be attached to the appendix and works cited [44].

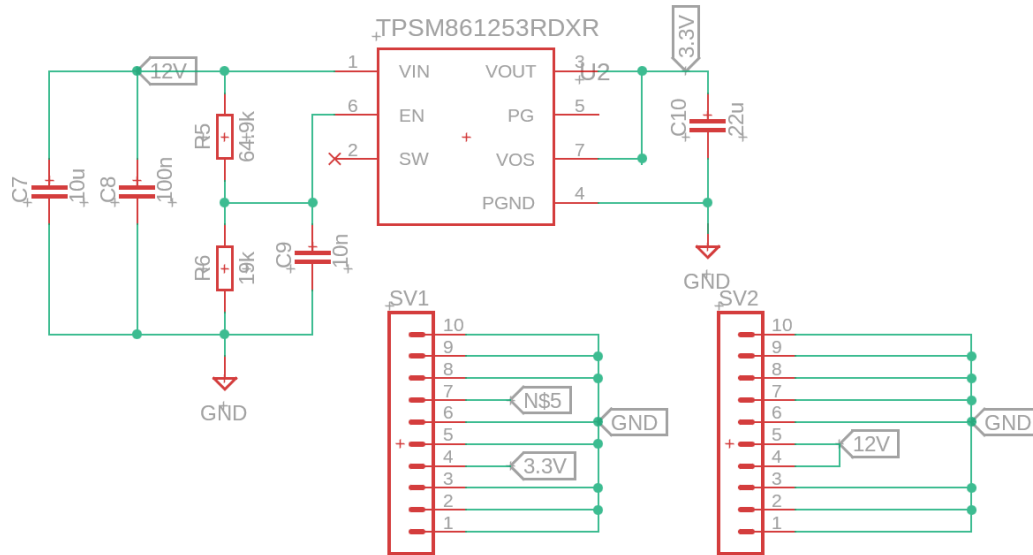


Figure 18: 3.3V PSU Electrical Schematic

## 6.4: Frequency Detector Electrical Schematic

The frequency detector shown in Figure X is our cross-zero detector designed around the TLV7011 voltage comparator.

The base of the frequency detector was previously shown in Figure 9. The base circuit consists of a voltage divider, a Zener diode, and a comparator. While this circuit should work in theory, upon simulating it with the actual, non-ideal comparator models some modifications were necessary.

The first major change is the diodes. There are three diodes in the schematic. The first two are at the input of the frequency detector and the second, represented as U4, is a Zener diode at the input of the comparator. The diodes are there to prevent reverse biasing at the comparator's input since the comparator acts erratically when the input exceeds the power rails. The diodes are there to prevent reverse biasing at the comparator's input since the comparator acts erratically when the input exceeds the power rails. The diodes are there to prevent reverse biasing at the comparator's input since the comparator acts erratically when the input exceeds the power rails. While there are three diodes, only one diode will be populated in the final design. We have all three to account for the uncertainty of the Zener diodes, as we could not find one that perfectly matched our needs, and the various packages diodes come in.

Following the diodes are another voltage divider. The differential voltage at the input is too large for the comparator to handle. The additional voltage divider helps scale down this value without distorting the frequency. It may be possible to combine this voltage divider with that on the 12V PSU, but the simulation found minimal difference in performance or a slight degradation. Having the voltage

dividers separate also allows more precise tuning of the hysteresis feedback network presented in following paragraph.

R1, R11, and R2 allow us to tune comparator's sensitivity. The voltage divider made by R11 and R12 serves as cross-zero reference point. Simply put, whenever the input voltage is above this value, the output swings to the positive supply rail, and when the input is below this value, it swings to the negative supply rail. Ideally, the non-inverting terminal would be shorted to ground to create a true 0V detection point, but due to noise inherent to the circuit, we must keep the bias point slightly above 0V or else the output will uncontrollably oscillate. Similar to R11 and R12, R1 also helps prevent unwanted oscillations. This feedback resistor allows us to provide additional hysteresis. This hysteresis makes the comparator less sensitive to noise.

All other components are either decoupling/bypass capacitors for the power supply pins or pin headers with similar functions to those mentioned in the 12V power supply section.

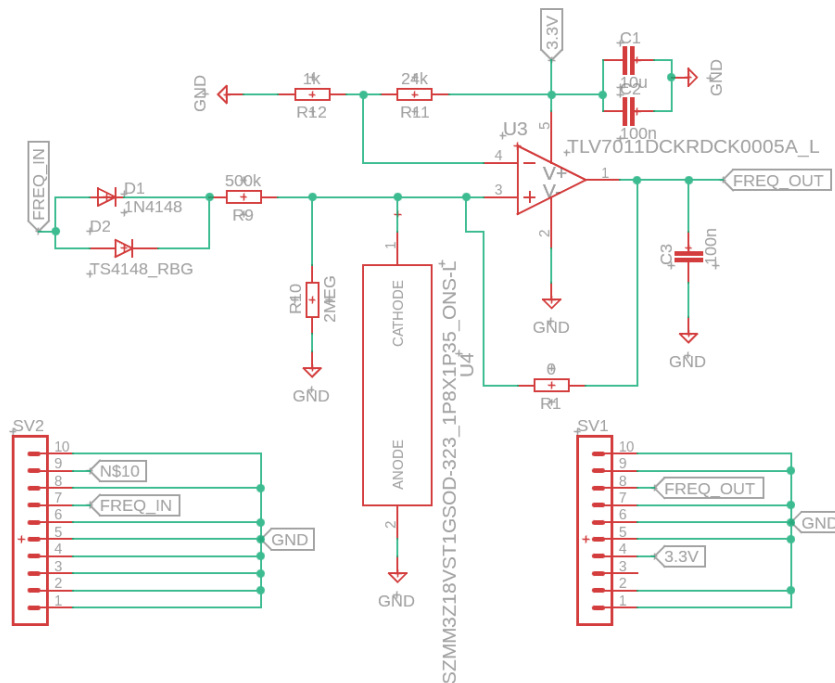
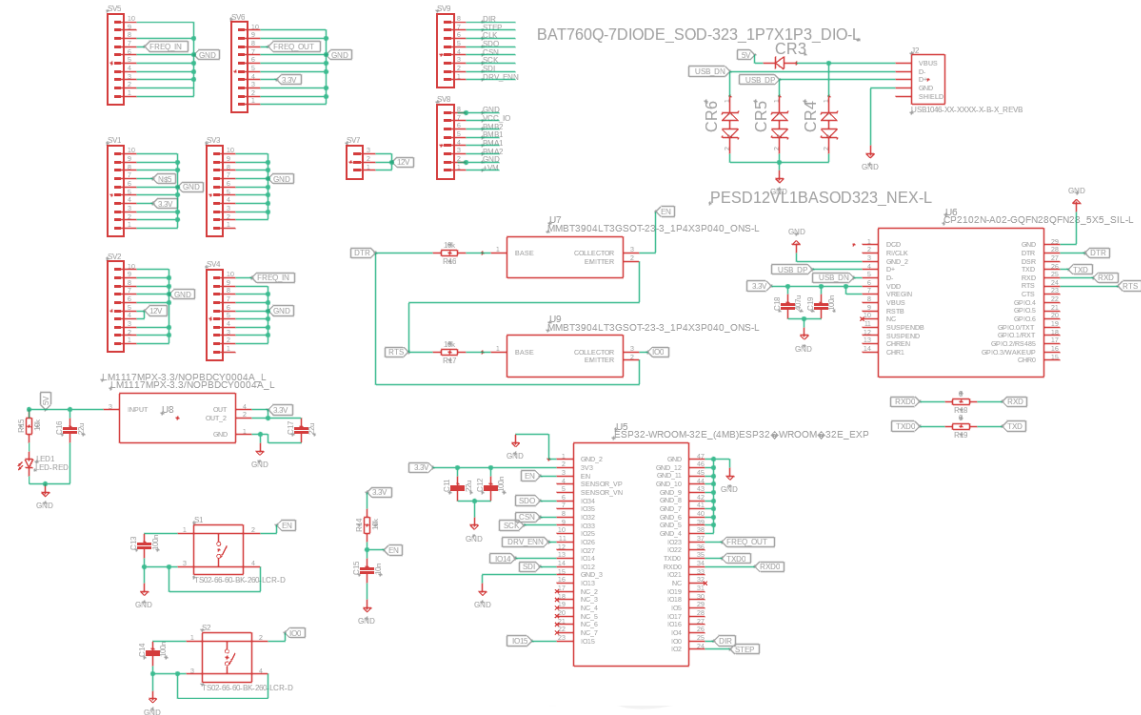


Figure 19: Frequency Detector Electrical Schematic

## 6.5: MCU Subsystem

The MCU subsystem serves as the main PCB on which the other daughter board are mounted to. Additionally, the ESP32 and all of its requisite hardware and the transformer are routed on this board. The schematic is based on the development board from Espressif Systems.

The first major component of this board is the programming hardware. The ESP32 must be programmed to perform even the most basic of tasks. We opted to use a USB-A connector for our project. The development board comes with a USB-C connector natively, but we found the connector to be too fragile and the connections too flimsy to support our needs. The USB-A connector includes electrostatic discharge (ESD) protection diodes along with a diode to protect the programming device from the power supply.



*Figure 20: MCU Electrical Schematic*

The ESP32 cannot use the USB protocol directly for programming. Instead, it uses the Universal Asynchronous Receiver/Transmitter (UART) protocol. Since most modern computers cannot directly communicate using UART, a USB-UART converter chip is necessary; this chip is represented with CP2102N.

The rest of the programming components are less critical but were added for completion. The LM1117 is a low drop-out (LDO) regulator that converts the 5V from the USB port to the 3.3V required for CP2102N and the ESP32. Technically, this is not a requirement since we could always power the ICs via the 3.3V PSU daughter board. However, we included the LDO so we could program the ESP32 independently of the other peripherals. Similarly, S1 and S2 are switches that electrically short the Enable and IO0 pins (respectively) to ground. This allows for easier programming and debugging. The last minor component of the programming hardware is the transistors U7 and U9. These are connected to the outputs of the USB/UART IC and fed into the ESP32 to actually program the MCU.

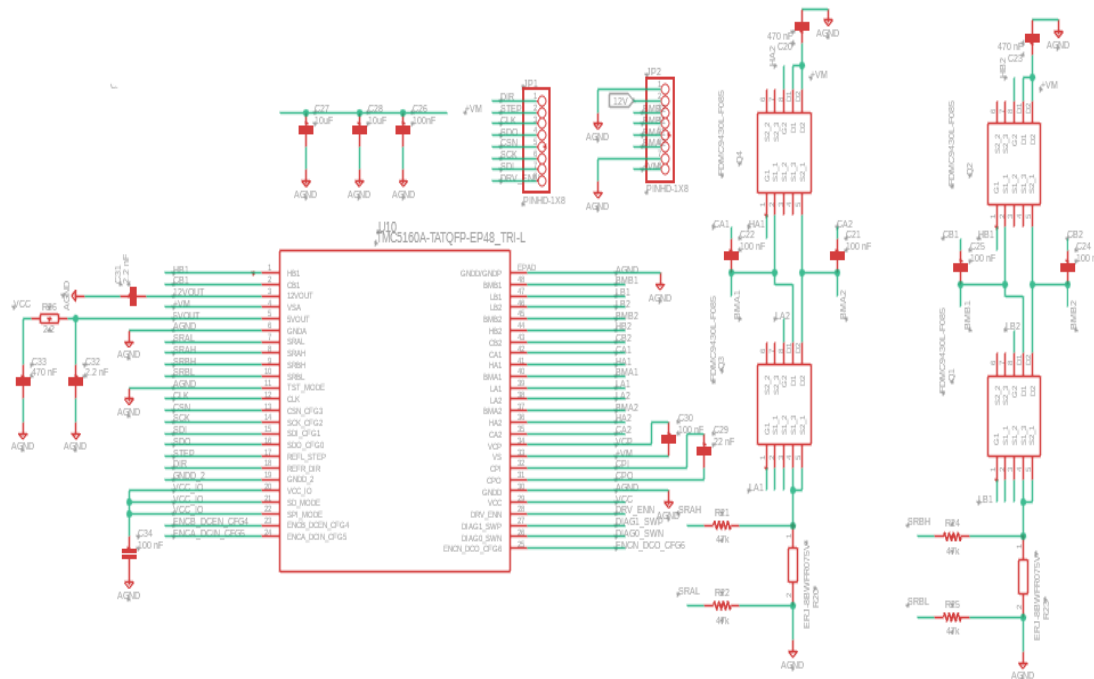
The second major component is the ESP32 itself. Most of the components attached to the ESP32 in Figure 7 are decoupling capacitors and grounds. The

remaining connections are the nets connecting to various General-Purpose Input/Output (GPIO) pins to the daughter boards. We had to be mindful of which GPIO got which connection. The motor controller connections can handle any GPIO; however, the frequency detector may need an analog to digital convertor. The ESP32 contains 2 onboard ADCs, but only one may be used with the Wi-Fi radio on. The placement for the nets were further refined based on their physical location. We tried to choose the GPIOs that were opposite of the antenna to minimize the trace length between the ESP32 and the daughter boards.

The last major component is the pin headers. The MCU PCB must accommodate all of any additional ones for the MCU's testing. An important note is that the names are inconsistent between the electrical subsystem schematics and the overall schematic. This is because the individual subsystem schematics are used for the PCB instead of only for illustrative purposes. A prime example of this is the 12V test points in Figure 2. These 12V test points are used exclusively by the 12V PSU for troubleshooting. They are labeled as 12V in the overall schematic since they do connect to that node; however, the net naming them 12V was removed from the MCU schematic to prevent having another connection corrupting the test point. This practice is continued for all test points that are exclusive to certain daughter boards.

## 6.6: Motor control

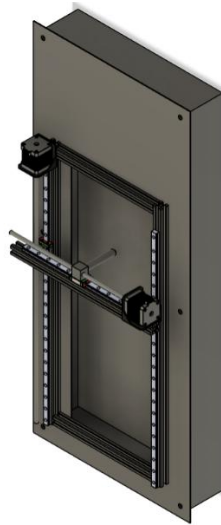
The motor control schematic consists of a few major parts: the TMC5160 itself, the 4 double MOSFETs required, some auxiliary capacitors and resistors, the precisely calibrated sense resistors, and pin headers to plug into the mother board.



The sense resistors are 0.75 ohm specially calibrated from Toshiba. They are needed to precisely measure the current being supplied to each phase of the stepper motor in order for the controllers to be able fully utilize their special features like CoolStep and StealthChop that make them so desirable. these features are explained in further depth in chapter 3. This design was based off of that of the Trimnamic SilentStepStick[45]

## 6.7: Mechanical Systems

Figure 9 shows a 3D rendering of the prototype mechanical system. It utilizes stepper motors, lead screws, aluminum extrusions, linear profile rails, and end effectors as outlined in Chapter 3. The mounting is also as outlined in chapter 3, the stock face plate has been replaced with a new plate that includes correctly spaced mounting holes so installing it is only requires unscrewing and screwing 6 screws. The plate also includes an appropriate cutout for the end effector to interface with the switches (not shown here). The mechanical system is constructed nearly entirely from off the shelf parts, the only custom parts required are mounting brackets to connect the X and Y axis gantries, and the end effector on the X axis. These parts will be 3d printed with high infill for ease of manufacturing, and to be able to withstand the required forces.



*Figure 22: Mechanical Systems CAD Model*

This model meets the size requirements we set, it does not exceed the perimeter of the breaker panel, and its total height from the panel is just under 4 inches. This is well below the 8 inch maximum we set.

Additionally, for the final mechanical design, a dust cover with a door must be included. This way the system will not be exposed, increasing the safety, and fulfilling regulatory requirements that breaker panels must have a covering. Our current plan is to mount a plastic container that can cover the entire project and any of the panel's exposed parts. Future iterations of the project may include a

well-fitted cover for the panel, but the regulations do not require such elaborate coverings to comply.

## Chapter 7 – Software Design

This chapter provides a comprehensive overview of the foundational elements of the software development process for the project. This chapter begins with a flowchart to outline the logical progression of the system's workflows, followed by an Entity Relationship Diagram (ERD) to illustrate the database structure and relationships. A Case Diagram details user interactions and functionalities, while the State Diagram describes the dynamic behavior of the system's components. The Class Diagram defines the structural blueprint of the software, highlighting the relationships between classes. Additionally, this chapter delves into API Unit Testing and Mobile Unit Testing, ensuring that backend services and mobile functionalities meet quality standards. Finally, the chapter concludes with a discussion on the User Interface, emphasizing the design principles and usability considerations that enhance user experience. Together, these sections present a holistic view of the software's architecture and design.

### 7.1: API Endpoint Flowchart

The flowchart outlines the device management system's workflow, starting with user authentication and branching into device-related actions. It visually represents processes such as account creation, login, and error handling, followed by authenticated actions like adding, updating, searching, or deleting devices. Each step includes user input validation, structured decision-making, and feedback mechanisms, ensuring clarity and efficiency. The chart also incorporates error-handling paths for scenarios like invalid logins or malformed inputs, offering users a logical and guided progression through the system.

The flowchart highlights the relationship between authentication and device management, emphasizing the reliance on a JWT to verify user resources. It demonstrates key safeguards, such as confirmation steps for device deletions and API responses that return status updates, including success or error messages. By visually detailing these interactions, the flowchart balances system complexity with user-friendly design, ensuring seamless navigation and robust error handling throughout the process.

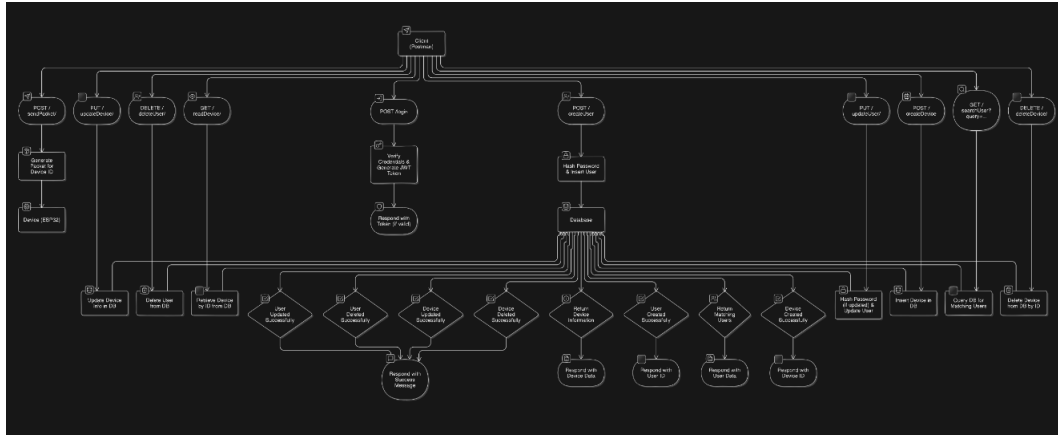


Figure 23: SW w/API endpoint Flowchart

## 7.2: Entity Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) serves as a blueprint for the database design, illustrating the relationships between various entities in the system. It provides a visual representation of the logical structure of the database, including its tables, attributes, and the connections between them. The ERD for this system consists of three primary tables: Users, Devices, and Breaker, each representing a distinct aspect of the application. These tables are interconnected to manage user accounts, devices, and breakers efficiently. Below, we have delved deeper into the purpose and structure of each table.

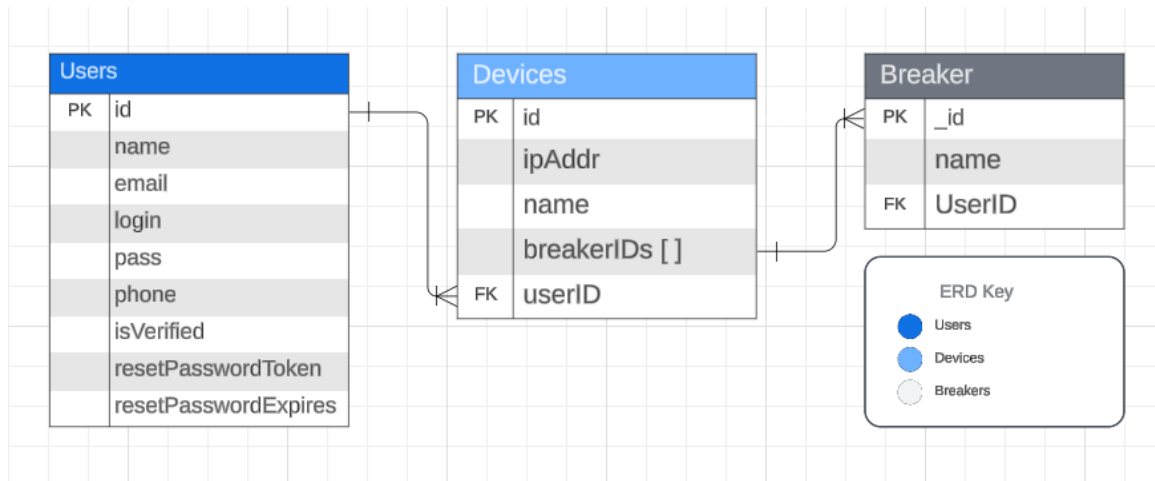


Figure 24: Entity Relationship Diagram (ERD)

### 7.2.1: Users

The Users table is a cornerstone of the database, capturing all essential information about the application's users. It includes fields such as id (a unique identifier for each user), name, email, login, and password, which are fundamental for user authentication and identification. Additionally, it includes phone for contact purposes, isVerified to indicate the account verification status, and

resetPasswordToken and resetPasswordExpires to facilitate secure password recovery processes. This table ensures robust user management while prioritizing security and user experience.

### 7.2.2: Devices

The Devices table is responsible for storing information about the devices associated with each user. Each device is uniquely identified by its id and contains attributes such as ipAddress (to locate the device on a network), name (for identification purposes), and breakerIDs[], which holds an array of associated breakers. Furthermore, the userID field establishes a direct relationship between a device and the user it belongs to, ensuring that devices are tied to the correct account.

### 7.2.3: Breakers

Lastly, the Breaker table focuses on managing the individual breakers within the system. Each breaker has a unique id, a name to distinguish it, and a userID to associate it with a specific user. This table facilitates precise control and monitoring of breakers, enabling users to manage them effectively through the application. By maintaining this association, the database ensures that breakers are securely linked to the correct user and devices.

### 7.2.4: Connections

The relationships in the database between Users, Devices, and Breakers are crucial to ensuring the scalability, flexibility, and functionality of the system. These relationships dictate how data is structured and how users interact with the control systems through the application. By exploring these one-to-many relationships, we can better understand their impact on the end user and how they influence the system's control logic.

The one-to-many relationship between Users and Devices means that a single user can own and manage multiple devices, but each device is linked to only one user. For the end user, this allows centralized control of all devices through a single account, simplifying management and enhancing usability. For example, a user with multiple devices at different locations—such as a home, office, or workshop—can easily switch between them within the application. This relationship also supports scalability, as users can add or remove devices without affecting other users, ensuring independence and streamlined management.

Similarly, the one-to-many relationship between Devices and Breakers signifies that a single device can control multiple breakers, but each breaker is associated with only one device. This setup is particularly advantageous in scenarios where a single control unit (device) is responsible for managing several breakers in a system, such as a smart power grid. For the end user, this provides granular control over individual breakers while maintaining a simplified setup, as the user only interacts with the device managing them. For instance, a user can use the application to toggle individual breakers within a circuit controlled by a single device.

From a control system perspective, these relationships allow hierarchical management and logical grouping. The User-Device relationship ensures that only authenticated and authorized users can access or modify devices, enhancing security and personalization. Meanwhile, the Device-Breaker relationship enables precise control over individual breakers, allowing the system to execute commands such as turning a specific breaker on or off or monitoring its status. This layered structure also simplifies troubleshooting and maintenance, as issues can be isolated to specific devices or breakers.

Overall, these relationships create a robust, user-friendly architecture. They allow users to easily expand their control systems, ensure security by isolating ownership, and provide a clear hierarchy for managing devices and breakers. This flexibility and logical organization directly enhance the application's functionality, making it suitable for both small-scale and large-scale implementations.

### 7.3: Case Diagram

The use case diagram outlines the flow of interactions between the user and the system, starting from account creation or login and branching into a variety of actions the user can take once authenticated. This diagram provides a clear overview of the system's functionality, user journeys, and the possible error handling paths, emphasizing both user authentication and device management.

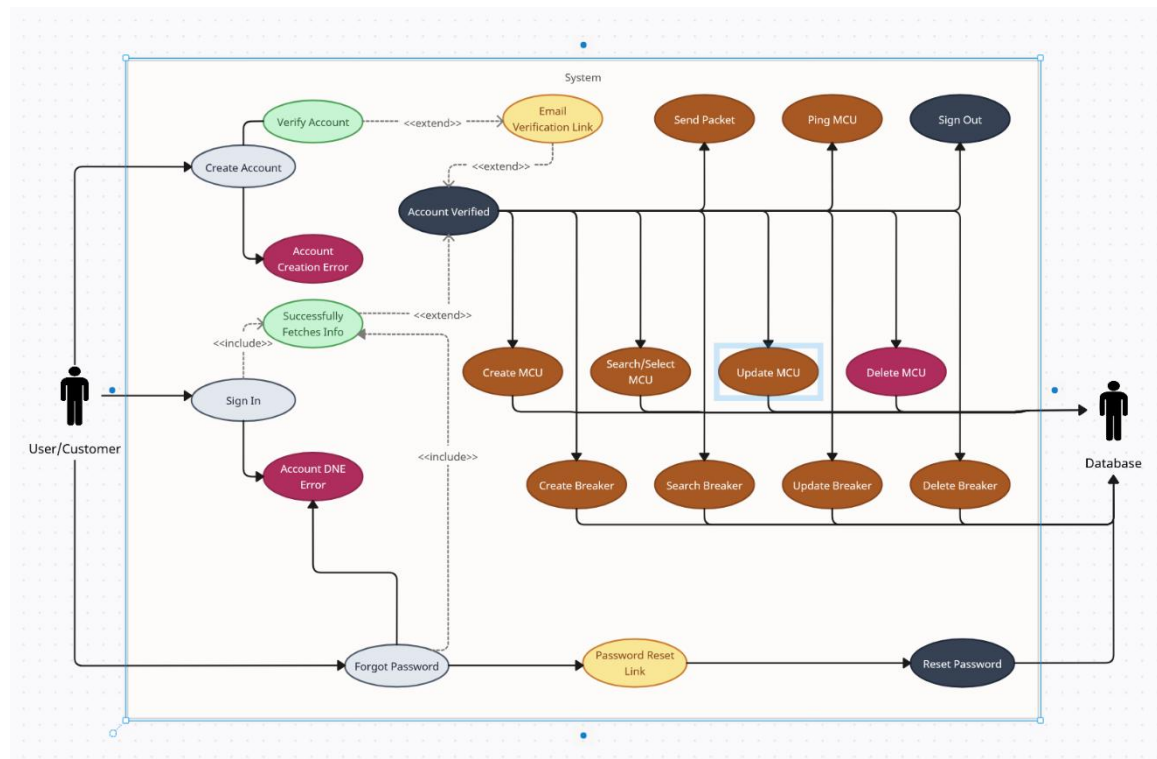


Figure 25: Use Case Diagram

### 7.3.1: Account Creation and Authentication

The process begins with the user either creating a new account or attempting to sign in. If the user chooses to create an account, they must input valid credentials, adhering to password parameters (e.g., complexity requirements). Errors in account creation—such as weak passwords or duplicate emails—will notify the user to correct the input. Once successfully created, the system sends an email verification link to the provided email address. Verification ensures the authenticity of the user and activates the account. After verifying their email, the user is redirected to the sign-in page, where they can log in with their newly created credentials.

If the user encounters an error during login, such as an "Account Does Not Exist" (DNE) error, they have the option to retry or reset their password. The password reset process involves the system emailing a reset link to the user's registered email, allowing them to create a new password. Once the user successfully logs in, the system generates a JWT (JSON Web Token), which is used to authenticate and authorize subsequent actions, ensuring that only the logged-in user can access or modify their data.

### 7.3.2: Device Management

Upon successful login, the user's JWT fetches devices associated with their account. The system enables several actions related to device management. These actions provide a flexible and user-friendly way for individuals to manage their devices in various environments, whether at home, work, or across multiple locations.

#### 7.3.2.1: *Creating a New Device*

##### **Front-End Process:**

- The user accesses the "Add Device" screen, which contains editable text fields for entering the device name and IP address. These fields are validated to ensure that the name is non-empty and the IP address conforms to a valid format (e.g., IPv4 or IPv6).
- After filling out the fields, the user clicks the "Create Device" button, which triggers a request to the back-end API.
- If the fields are invalid (e.g., empty or improperly formatted), the system displays an error message, prompting the user to fix the input before proceeding.

##### **Back-End Process:**

- The API endpoint (/createDevice) receives the request as a JSON object, which includes the name, ipAddress, and the userID (retrieved from the user's JWT token).
- The back-end validates the incoming data for completeness and correctness. For example:

- The name must not be empty.
  - The ipAddress must match a valid IP address pattern.
- If the validation passes, the system creates a new device entry in the database and associates it with the user's account.
- The API responds with a status code and a body:
  - **200**: Success. Includes a JSON response with a status message (e.g., "Device created successfully") and the id of the newly created device.
  - **400**: Bad Request. Indicates invalid input.
  - **500**: Internal Server Error. Indicates a server-side issue.
- The front-end uses the returned device ID to update the user's device list dynamically.

### *7.3.2.2: Reading/Searching an Existing Device*

#### **Front-End Process:**

- The user navigates to a device search bar on the "Device Management" screen. This bar includes a text input field where the user types the name or partial name of the device.
- As the user types, the front-end sends the search query to the back-end API in real time or on submission, depending on the implementation.
- The screen dynamically updates to display matching devices, or a "No devices found" message if no results are returned.

#### **Back-End Process:**

- The API endpoint (/searchDevices) processes the query string sent in the request. The request includes the userID (from the JWT) to ensure that only devices belonging to the authenticated user are searched.
- The back-end queries the database using the search term to find matching device names associated with the user.
- The API responds with:
  - **200**: Success. Returns a JSON array of matching devices, including their id, name, and ipAddress.
  - **404**: Not Found. If no devices match the query, the response includes a status message indicating this.
- The front-end displays the list of devices returned or the "No devices found" message based on the response.

### *7.3.2.3: Updating an Existing Device*

#### **Front-End Process:**

- The user navigates to the "Edit Device" screen, which pre-populates editable text fields with the device's current name and IP address.
- The user modifies the fields and clicks the "Save Changes" button. The system validates the new inputs to ensure compliance with format requirements.
- If the validation succeeds, the updated data is sent to the back-end API for processing.

#### **Back-End Process:**

- The API endpoint (/updateDevice/:id) receives the request as a JSON object containing the updated name, ipAddress, and the id of the device to be modified.
- The back-end validates the data, ensuring that:
  - The id corresponds to a valid device in the database.
  - The updated values meet format requirements.
- If validation passes, the system updates the database entry for the device.
- The API responds with:
  - **200**: Success. Returns a JSON object with a status message (e.g., "Device updated successfully") and the updated device details.
  - **404**: Not Found. Indicates the device ID does not exist or does not belong to the user.
  - **400**: Bad Request. Indicates validation errors for the new data.
- On success, the front-end updates the device information on the user's screen dynamically.

### *7.3.2.4: Deleting an Existing Device*

#### **Front-End Process:**

- The user selects a device from their device list and clicks the "Delete" button. A confirmation dialog appears, asking the user to confirm their intention to delete the device.
- If the user confirms, the front-end sends a request to the back-end API, including the device ID to be deleted.
- If the user cancels, no action is taken.

#### **Back-End Process:**

- The API endpoint (/deleteDevice/:id) processes the request by validating the id provided.
- The back-end ensures that:
  - The device ID exists in the database.
  - The device belongs to the authenticated user (validated via the userID from the JWT).
- If validation passes, the device is removed from the database.
- The API responds with:
  - **200:** Success. Includes a JSON response with a status message (e.g., "Device deleted successfully").
  - **404:** Not Found. Indicates the device ID does not exist or does not belong to the user.
  - **500:** Internal Server Error. Indicates issues during the deletion process.
- The front-end refreshes the device list to reflect the removal of the deleted device.

### 7.3.3: Breaker Management

Breakers, which are controlled by devices, also have their own set of user actions similarly to the ones previously mentioned.

- **Create a Breaker:** Associate a breaker with a specific device for granular control of power distribution or other functionalities.
- **Update Breaker Information:** Modify breaker details (e.g., name or identifier) to ensure accuracy in the system.
- **Delete a Breaker:** Remove a breaker from the system if it is no longer needed or has been replaced.

Breaker management allows users to exercise precise control over the system's components, offering flexibility for dynamic setups.

### 7.3.4: User Actions and Device Commands

#### User Actions and Device Commands

The system also provides additional functionality beyond device and breaker management:

- **Edit User Information:** Users can update their profile details, such as name, email, or password, ensuring their account stays accurate and secure.

- **Send Commands to a Device:** Users can interact with devices by sending specific commands (e.g., toggling LEDs or executing predefined actions).
- **Ping a Device:** This allows users to check a device's connection and status, ensuring that the system is functioning as expected.
- **Sign Out:** Users can securely log out, ensuring that their session ends and the **JWT** is invalidated.

These actions offer users comprehensive control over their account and the devices they manage, enhancing both usability and system reliability.

### 7.3.5: Error Handling and User Experience

Throughout the user journey, the system incorporates robust error-handling mechanisms to guide users through potential issues. For example:

- Account creation errors prompt users to fix invalid input.
- Login errors offer password reset options or retry opportunities.
- Deletion actions require confirmation to prevent accidental data loss.

These safeguards improve user experience, ensuring that the system remains secure, intuitive, and user-friendly. In summary, the use case diagram highlights the system's versatility and robustness in managing user accounts, devices, and breakers, while offering a seamless experience for all supported actions.

## 7.4: State Diagram

The state diagram illustrates the various states within the user and device management system, focusing on transitions triggered by endpoint interactions. Starting with user authentication, the diagram shows states like Logged Out, Logged In, and Session Expired, detailing how actions like login attempts, JWT verification, or logout influence these transitions. For device management, states include Idle, Adding Device, Editing Device, Deleting Device, and Fetching Devices, with transitions driven by API endpoints processing user inputs. Each state accounts for success and error responses, such as returning a 200 status for valid operations or a 404 for invalid resource requests, ensuring the system remains robust and responsive to user actions.

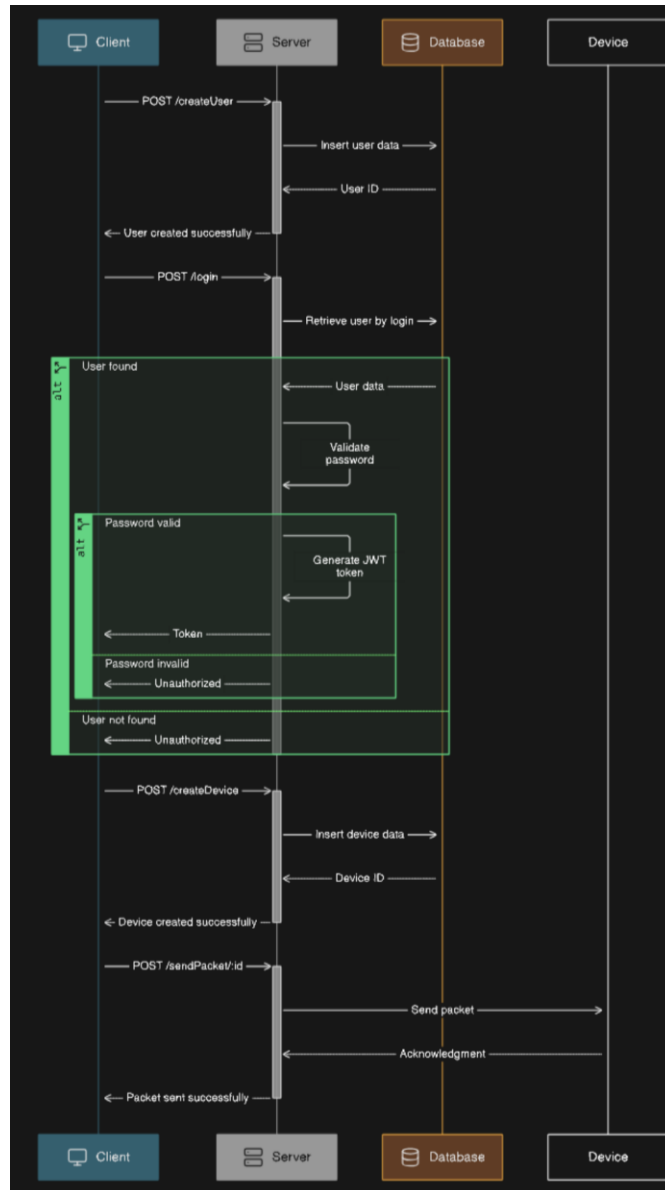


Figure 26: State Diagram

## 7.5: Class Diagram

This class diagram represents the structure of a system involving users, devices, and breakers. The User class contains information about the user, such as id, name, email, login, and other relevant details. Each user can have multiple associated Devices and Breakers. The Device class contains information about the device, including its ipAddress, name, and a list of associated breakerIDs. It also has a sendPacket() method to interact with the device. The Breaker class is used to manage breakers and is linked to a user. CRUD operations are provided for all three classes, allowing the system to create, read, update, and delete users, devices, and breakers as necessary.

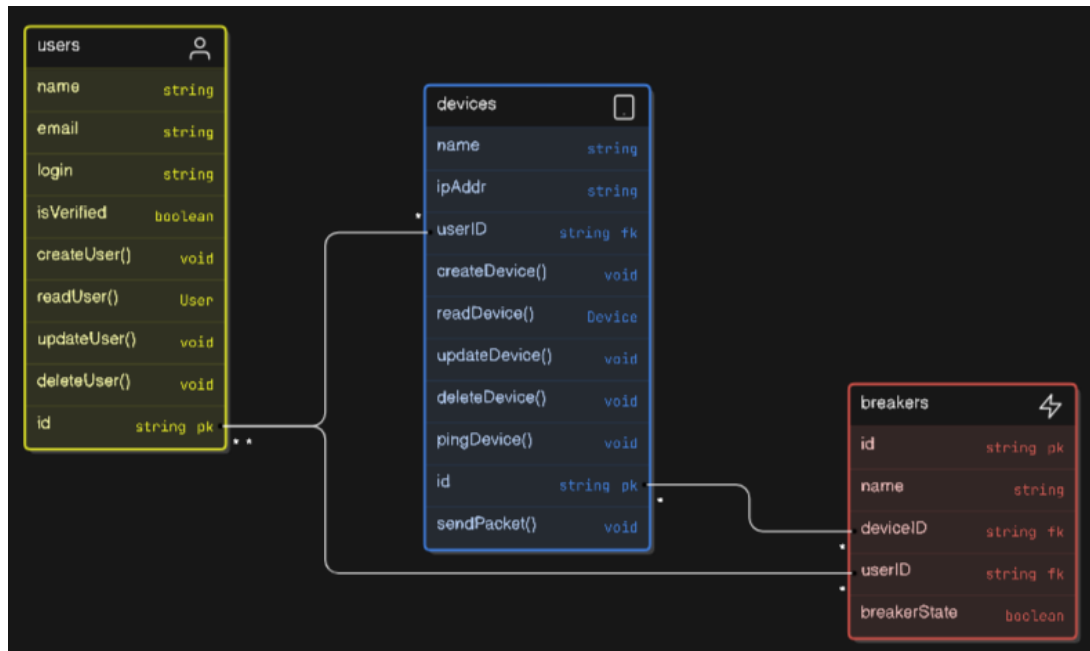


Figure 27: Class Diagram

## 7.6: API Unit Testing

The screenshot of Postman displays a series of API endpoint tests, each demonstrating different request payloads and their corresponding expected response bodies. The tests cover various operations such as creating, reading, updating, and deleting resources, and show the expected behavior of the API when handling different types of input. Each endpoint is tested with specific payloads that include parameters like user data, device configurations, and breaker information. The response bodies include status messages, data confirmation, or error messages, providing insight into the API's functionality and error handling. Overall, the screenshot shows the API's robustness and its ability to handle various scenarios for resource management.

| Smartgrid - Run results                                 |             |            |          |           |                 |
|---------------------------------------------------------|-------------|------------|----------|-----------|-----------------|
| 🔗 Ran today at 23:08:58 · <a href="#">View all runs</a> |             |            |          |           |                 |
| Source                                                  | Environment | Iterations | Duration | All tests | Avg. Resp. Time |
| Runner                                                  | none        | 1          | 9s 190ms | 34        | 18 ms           |
| RUN SUMMARY                                             |             |            |          |           |                 |
| ▶ POST createUser                                       |             |            |          | 5   0     | 1               |
| ▶ POST createDevice                                     |             |            |          | 4   0     |                 |
| ▶ GET readDevice                                        |             |            |          | 4   0     |                 |
| ▶ POST login                                            |             |            |          | 5   0     |                 |
| ▶ PUT updateUser                                        |             |            |          | 5   0     |                 |
| ▶ PUT updateDevice                                      |             |            |          | 3   0     |                 |
| ▶ DELETE deleteUser                                     |             |            |          | 4   0     |                 |
| ▶ DELETE deleteDevice                                   |             |            |          | 4   0     |                 |

Figure 28: API Unit Testing

## 7.7: Mobile Unit Testing

This section focuses on testing various services and Dart routines within the mobile application. Unit tests are designed to validate the correct execution of functions triggered by user interactions, such as button presses and data input through payloads. These tests ensure that each service, such as device management or user authentication, behaves as expected when invoked, checking for correct outputs and responses. By simulating button presses and input scenarios, the tests confirm that the app responds appropriately, triggering the expected routines, and handling edge cases or errors gracefully. This process helps ensure the reliability and smooth user experience of the mobile application by thoroughly testing its core functionalities before deployment.

## 7.8: User Interface

The User Interface (UI) follows a minimalistic approach, focusing on simplicity and clarity while offering powerful utility. Each screen is designed to be straight to the point, ensuring that users can quickly understand and navigate through the application without unnecessary distractions. The design language is clean and intuitive, with clearly labeled buttons, input fields, and actionable items that guide users through their tasks. Elements are organized logically, with ample spacing to reduce clutter, making it easy for users to focus on essential actions. This streamlined design ensures that even new users can interact with the app effectively, providing an experience that is both user-friendly and efficient.

### 7.8.1: Login Screen

The Login screen provides fields for the user to enter their email/username and password. A "Login" button is present for submitting the credentials. If the user doesn't have an account, there is a link to the Signup screen. Additionally, there is a "Forgot Password?" link that directs to the Reset Password screen for users who need to recover their account.

### 7.8.2: Sign Up Screen

The Signup screen allows new users to create an account by providing details such as name, email, password, and phone number. A "Sign Up" button submits the form. It also features a checkbox for agreeing to terms and conditions. After signing up, the user is directed to the Login screen.

### 7.8.3: Reset Password Screen

The Reset Password screen prompts the user to enter their email address. Once entered, the user receives an email with instructions for resetting their password. There is a button to submit the request, and once successful, a confirmation message appears, notifying the user that an email has been sent.

### 7.8.4: Device Management Screen

The Device Home or Management screen displays a list of the user's devices with basic details like device name and status. Users can add new devices via an "Add

Device" button or click on an existing device for more detailed information. There are options to edit or remove devices. The screen also includes a search bar for filtering devices.

### 7.8.5: Device Detail/Interaction Screen

The Device Details screen presents detailed information about the selected device, such as its IP address, device type, and operational status. It includes interactive buttons to perform actions like sending commands or performing diagnostic checks. A "Back" button allows users to return to the Device Management screen.

### 7.8.6: User Management Screen

The User Profile screen displays personal information such as the user's name, email, phone number, and account status. Users can update their profile details, change their password, or adjust notification preferences. There is also a "Delete Account" option for users who wish to remove their account. The interface includes a "Save Changes" button to apply any updates.

## 7.9: Frequency Detection Software

Modern power systems need instruments for observing and monitoring grid stability. One important part in evaluating stability is grid frequency, which can deviate due to load imbalances or power generation problems. Alongside the software design for the mobile app, the software for the detection and alert of grid frequency is one of the most important designs when it comes to the software for this project. The software that is going to be talked about was developed for the ESP32 microcontroller, exploiting its digital capabilities to measure grid frequency, detect irregularities and supply real-time feedback.

### 7.9.1: Software Overview

The software for the detection and feedback of the frequency runs on an ESP32 microcontroller interfacing with several GPIO pins such as GPIO4 which serves as the input pin to receive the external signal from the grid. The signal frequency is detected using interrupts. The GPIO2 pin monitors the onboard LED, supplying visual feedback on frequency deviations. If the LED blinks that means that the frequency went out of the pre-established range. The software consists of ISR (Interrupt Service Routine) which finds rising edges on the input signal at GPIO4 and calculates the time interval between consecutive edges to determine the signal frequency in real-time. Frequency Analysis where the measured frequency is compared to a pre-established frequency range (59-61Hz). If the frequency is out of range or if no signal is detected, the system marks this as an irregularity.

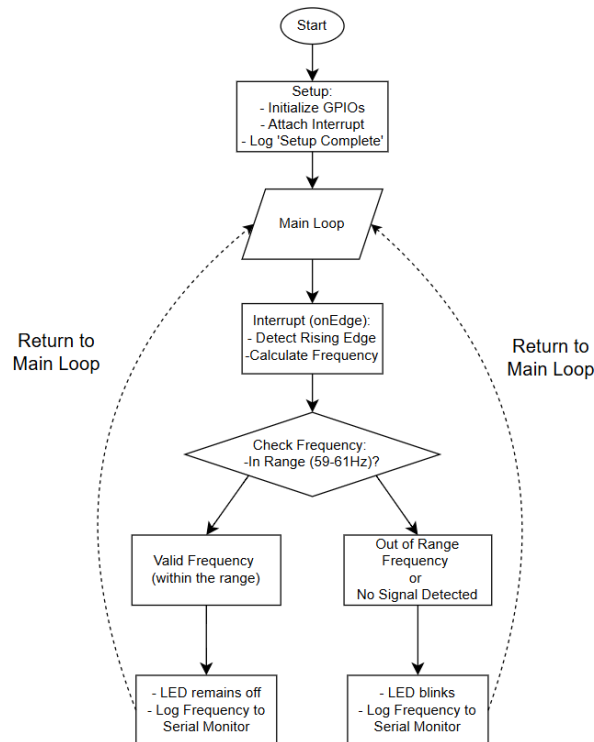
### 7.9.2: Frequency Detection and Feedback Flowchart

The flowchart outlines the ESP32's frequency detection and feedback workflow, starting with the setup step where the ESP32 is configured for its specified tasks. The GPIO pins are assigned roles, the GPIO4 pin is set as the input pin for signal detection, while GPIO2 controls the LED feedback. An interrupt is also attached to the input signal pin (GPIO4) using the `attachInterrupt()` function, allowing the

ESP32 to locate rising edges and set off the frequency calculation process. Additionally, debugging logs are printed to show that the setup is completed. The setup step gets the ESP32 ready to process signal data and perform its feedback function.

The Main Loop represents the core cycle of the software which continuously monitors the input signal, measures the frequency, and provides feedback based on the detected frequency. This loop functions indefinitely, allowing the ESP32 to dynamically respond to signal deviations. Rising edges on the input signal (GPIO4) are detected by the Interrupt (onEdge) service routine, which measures the time interval between consecutive edges. This interval is then transformed into a frequency value using the formula  $f = 1,000,000/\text{Interval } (\mu\text{s})$ . If the time interval is invalid - due to noise, loss of signal, or inconsistencies - the frequency is set to zero to specify no signal. This guarantees that the system remains robust against wrong inputs, providing reliable and accurate frequency detection under varying conditions.

Once the frequency is calculated, the software enters the Check Frequency stage, where it evaluates whether the frequency falls within the acceptable pre-established range (59-61Hz). If the measured frequency is valid and within range, the system transitions to the Valid Frequency state. In this state, the LED located in the ESP32 remains off, signaling normal operation. The measured frequency is also logged to the Serial Monitor for real-time feedback. If the frequency falls outside the acceptable range – whether below 59Hz, above 61Hz, or zero (indicating no signal) - the system transitions to the Out-of-Range Frequency state. In this state, the LED located in the ESP32 blinks to alert the user of a frequency deviation or loss of signal. This visual feedback guarantees that fluctuations are immediately noticeable to the user, supplying a critical warning mechanism. At the same time, the out-of-range frequency or “No Signal” status is logged to the Serial Monitor, giving users additional context for analysis. Both outcomes – whether the frequency is valid or out of range – lead back to the Main Loop. This cyclical structure lets the ESP32 to continuously monitor the frequency from the input signal and adapt to changes in real time.



*Figure 29: ESP32 Frequency Detection and Feedback Flowchart*

### 7.9.3: Implementation

This section focuses on how the theory of operation will be implemented into the ESP32. It will cover the code that determines the input frequency and how to verify that the code is working properly.

#### 7.9.3.1: Frequency Detection

The frequency detection software employs an interrupt-based approach to calculate the frequency of an input signal. Rising edges of an input signal are detected utilizing the ESP32's `attachInterrupt()` function. The time interval between consecutive rising edges is measured in microseconds using the ESP32's `micros()` function. This interval is then utilized to calculate the frequency using the formula:  $f = 1,000,000 / \text{Interval } (\mu\text{s})$ . If the interval is invalid due to noise or signal loss, the frequency is set to zero. This approach grants for real-time, accurate frequency detection based on external signals.

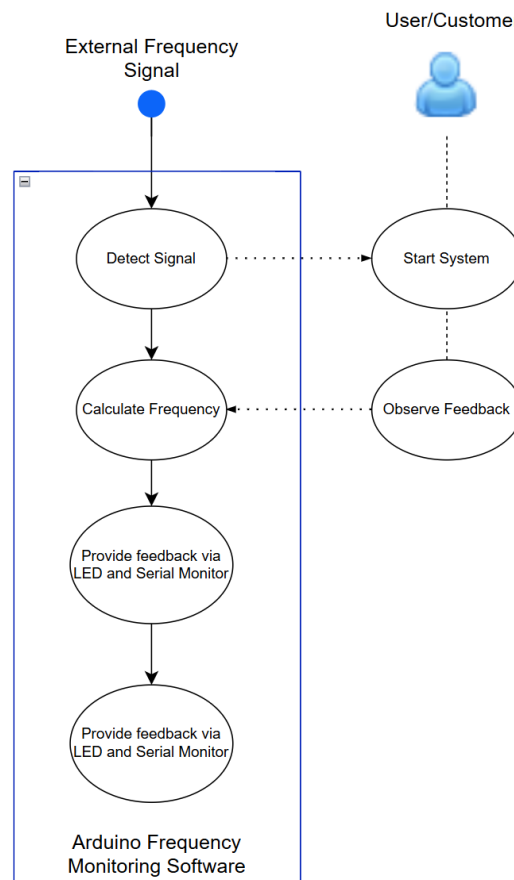
#### 7.9.3.2: Visual Feedback

An LED located in the ESP32 supplies real-time visual feedback to the user. If the LED is blinking that signifies that the frequency is out of the acceptable range (59-61Hz) or there is no signal detected. If the frequency is within the acceptable range, the LED stays off, indicating normal operation. The LED blinking implementation uses a non-blocking approach based on the `millis()` function, which makes sure that the LED feedback does not interfere with the frequency calculations or other

time-sensitive tasks. This design makes the system reliable and responsive, even under continuous operation.

### 7.10: Frequency Detection and Feedback Case Diagram

The case diagram illustrates the functionality of the Frequency Detection and Feedback software, showing its interaction with external actors and the flow of operations within the system. Two primary actors, the User/Customer and the External Frequency Signal, interact with the system. The User/Customer starts the process by starting the system and observes feedback via LED indicators and the Serial Monitor. The Outside Frequency Signal provides input to the system, which is processed through a sequence of use cases. The system first detects the signal, then calculates its frequency, and validates whether the frequency is within the acceptable pre-established range. The system supplies feedback to the user, such as blinking the LED if the frequency is out of range and displaying frequency data on the Serial Monitor. The diagram effectively separates external actors from the internal processes by using a system boundary, highlighting how the software handles frequency detection, processing, and feedback generator in an automated and logical sequence.



*Figure 30: ESP32 Frequency Detection and Feedback Case Diagram*

## 7.11: Frequency Detection and Feedback State Diagram

The state diagram illustrates the various states of the Frequency Detection and Feedback software and the transitions between them based on the input signal and calculated frequency. It clearly shows how the system processes input, validates frequency, and how it supplies appropriate feedback to the user. The system begins in the Idle State, where it waits for a signal by monitoring rising edges on the input pin (GPIO4). Once a rising edge is located, the system transitions to the Signal Detected phase, where it calculates the frequency based on the time intervals between consecutive rising edges. Depending on the calculated frequency, the system transitions into one of the two possible states: Valid Frequency or Out of Range. If the frequency falls within the acceptable range between 59-61Hz, the system enters the Valid Frequency state, where the LED remains off, and the measured frequency is displayed on the Serial Monitor, indicating normal grid conditions. If the frequency is outside the acceptable range (below 59 Hz, above 61 Hz, or 0 Hz indicating no signal), the system enters the Out-of-Range state. In this state, the LED located on the ESP32 blinks to alert the user of the frequency deviation, while the Serial Monitor logs the detected frequency. After processing the signal, the system returns to the Idle State, ready to detect the next rising edge. This state-based representation clearly showcases the real-time decision-making and feedback loop implemented in the software, guaranteeing reliable operation under varying grid conditions.

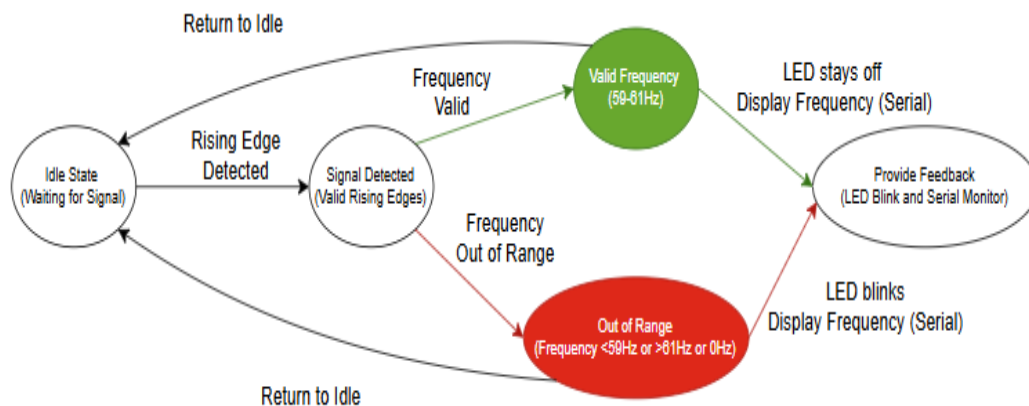


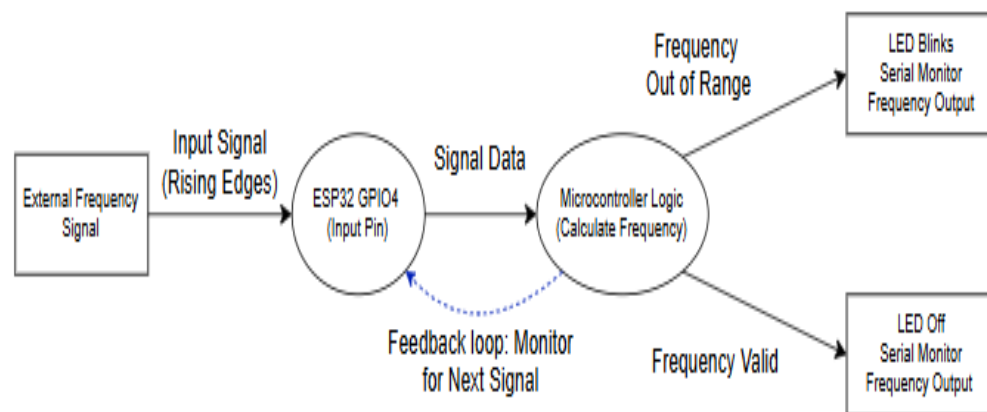
Figure 31: ESP32 Frequency Detection and Feedback State Diagram

## 7.12: Frequency Detection and Feedback Data Transfer Diagram

The data transfer diagram provides a clear representation of the flow of information within the ESP32-based frequency detection and feedback software. It outlines how data is received, processed, and used to trigger appropriate actions, ensuring the system functions effectively in real-time. The process starts with an External Frequency Signal, generated by a frequency source. This signal, which consists of rising and falling edges, is sent to the ESP32's GPIO4 Input Pin. The input signal is detected by the ESP32 as rising edges, which are critical for measuring the time interval between consecutive edges. This step marks the initial stage of data entry into the system.

The signal data from the GPIO4 is then passed to the Microcontroller Logic, where the frequency is calculated using the time interval between detected rising edges. This logic processes the data and determines whether the measured frequency is within the pre-established acceptable range (59-61Hz). If the interval is invalid due to noise or a missing signal, the frequency is set to 0 to indicate no signal. From the microcontroller Logic, the system can transition into one of two outcomes based on the calculated frequency. The first outcome is Frequency Valid, if the frequency is within the acceptable range, the system logs the frequency in the Serial Monitor and keeps the LED located on the ESP32 turned off. This state indicates that the system is operating normally without any frequency irregularities. The second outcome is Frequency Out of Range, if the frequency falls outside the acceptable range (below 59Hz, above 61Hz, or zero), the system triggers an alert. The LED blinks to provide real-time visual feedback, and the out-of-range frequency is logged to the Serial Monitor for further analysis.

The diagram also incorporates a Feedback Loop, represented by a dashed arrow from the Microcontroller Logic back to the ESP32 GPIO4 Input Pin. This loop signifies the system's continuous monitoring behavior, where it waits for the next rising edge signal after processing the current frequency. This guarantees that the system operates in a cyclical manner, always ready to detect new signals and provide feedback as necessary.



*Figure 32: ESP32 Frequency Detection and Feedback Data Transfer Diagram*

### 7.13: Mechanical Systems Software

The control loop for the mechanical systems is crucial for the successful operation of this device. The basic layout is that when a command to flip a switch is generated either by the user or automatically by the frequency detection software, these commands will be processed sequentially as g code. The sequential processing of the commands is important as the user may wish to turn on or off breakers in a specific order. Below is a block diagram describing the process in further depth.

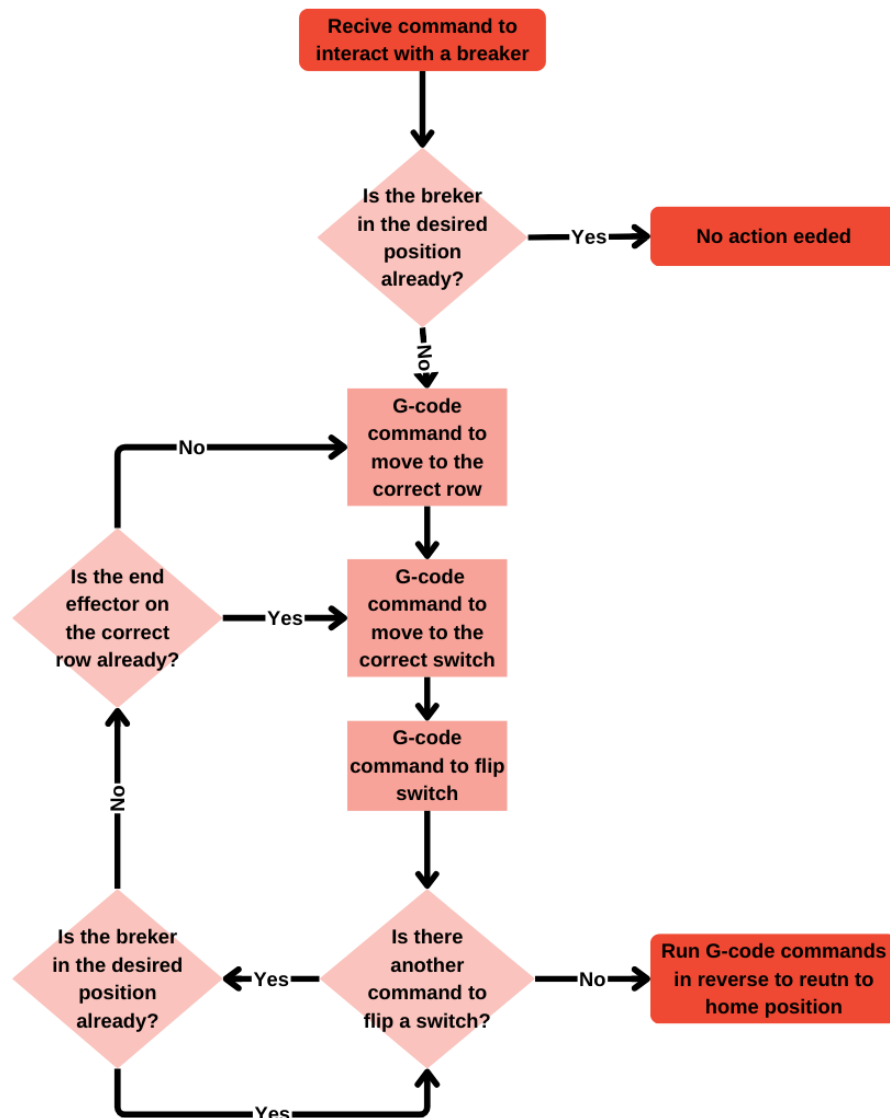


Figure 33: ESP32 Mechanical System Block Diagram

### 7.13.1: G-code

As discussed previously, the positioning of the motors will be controlled using G-code commands. The G is short for Geometric, as these commands tell the system how to move in a cartesian coordinate system.[46] The purpose of this is that by using G-code, we can control the positioning based of cartesian coordinates. If we didn't use this, we could only control the motors by telling them to spin specific amounts. Rather than telling the machine to move from it's current position to an X Y coordinate, we would have to instruct it to spin the motors however many rotations is necessary, calculating that for each movement. The following G-cod commands will be utilized in this project

*Table 21: G Code Command Matrix [47]*

| command | use                                                                                                                                                                                                                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G00     | Rapid movement: moves the end effector to desired position directly and ant maximum speed                                                                                                                                                                                                                                                           |
| G01     | Linear movement- moves end effector to desired position directly at speed set specifically in command                                                                                                                                                                                                                                               |
| G28     | Homing command- uses the end switches on each axis to obtain exact positioning of end effector. Like zeroing out a scale, by touching the end switch the software knows that the axis is at its absolute zero position. this is important upon start up as position is not stored, and may be needed in routine operation to correct for lost steps |
| G90     | Absolute positioning- this command tells the machine that any position it is told to move to is based on the zero position rather than the last position                                                                                                                                                                                            |
| G91     | Relative positioning- this command tells the machine that any position it is told to move to is based on the current position rather than the zero position                                                                                                                                                                                         |

for our purposes it's very important to break linear movement commands up into multiple pieces. This is because if we, for example, told the machine to move from the home position (top left) directly to a switch on the bottom right, the end effector would try to move directly through the other switches in the way. This would result in what is known as a crash: due to an operator error, the machine has tried to do something impossible, and now may have damaged the end effector or the breakers themselves. it could even result in a breaker being turned on or off erroneously, having potentially disastrous results for the end user (fridge is turned off and food spoils, security system goes offline, ect.) To ensure that no damage occurs, the movements will be broken into X and Y components. This way, it will be impossible to accidentally swipe through a bank of switches. Careful programing of the G-code commands will still be required to prevent damage.

## Chapter 8: System Fabrication and Construction

Chapter 8 focuses on how the research and designs from the previous chapters will be incorporated into a physical design. This section includes PCB design, mechanical construction, and any modifications to the service panel necessary to build the project.

### 8.1: PCB Design General Guidelines

While the 5 PCBs individually have their own challenges, there are a lot of similarities between them. For the sake of brevity, the rules below are common across all of the PCBs. Any further discussion about any individual PCB in Chapter 8 will focus only on the unique design criteria or challenges faced when it.

1. For any IC, the datasheet's recommended layout is followed as closely as possible. If a datasheet is not available, then follow any development board or existing circuitry online.
2. The minimum trace width is 33mil (assuming space and IC pin pitch allows) to allow for easier trouble shooting, reworking, and rerouting if necessary.
3. All IC pins supplying power must have a minimum of one 1-10 $\mu$ F bulk capacitor and one 10-100nF high frequency filter capacitor as close to the pins as possible.
4. All daughter boards must have at least 2 sets of pin headers placed on opposite sides to allow for ample ground connection and a firm mate to the MCU PCB.
5. Any pin headers not used for testing are to be grounded.
6. All boards must have at least one ground plane. If it has multiple ground planes, then stitching vias must be used throughout the board to connect the two planes, minimizing the return path impedance.
7. All high-power and/or high-current ICs must have several thermal relief vias as close as possible to the pad, ideally under the pad.
8. All configurable or optional components and all outputs, prior to thorough testing, must be open circuits are shorted with a zero-ohm resistor
9. Include as much relevant information on the silkscreen as necessary for easy assembly and troubleshooting including package layout, where the first pin is located, polarities, component type, et cetera.
10. If possible, space out digital signals as much as possible to minimize crosstalk.

Thermals are generally removed from the copper pours. Thermals refer to the non-solid pattern that a copper pour can take around components. In theory, this will help take some of the thermal stress off the component. For most 12V, 3.3V, and other power pins this feature is removed in order to maximize the amount of contact between the copper and pad thereby improving current draw. This is not a hard rule and varies dramatically across application. Practically all of the ground connections still sport the cross shape associated with the thermal since there is already plenty of contact points.

## 8.2: PCB Layout and Design

This section contains the PCB layouts for the project. It also includes any relevant discussion about nuances of the individual PCBs not covered in Section 8.1

### 8.2.1: 12V PSU PCB

There are minimal additions between our 12V PSU PCB and the recommended layout. The LMR51450 datasheet recommends plenty of contact area between the input and output pins, decoupling capacitors with stitching vias, plenty of thermal relief vias near the ground pad, and a feedback network that is isolated from the switching noise of the inductor. All but the feedback network is accounted for in the general guidelines. To shield the feedback network from the switching noise, it was routed on the bottom copper layer reasonably far away from the inductor. Ideally, the feedback network would be as close to the output as possible to better control the voltage as seen at the input of the recipient device, similar to how sense lines in multimeters or other test equipment may function, but we decided to keep the feedback on the daughter board. We chose to limit the complexity of the routing on the MCU at the expense of a few millivolts of deviation from the 12V expected by the output.

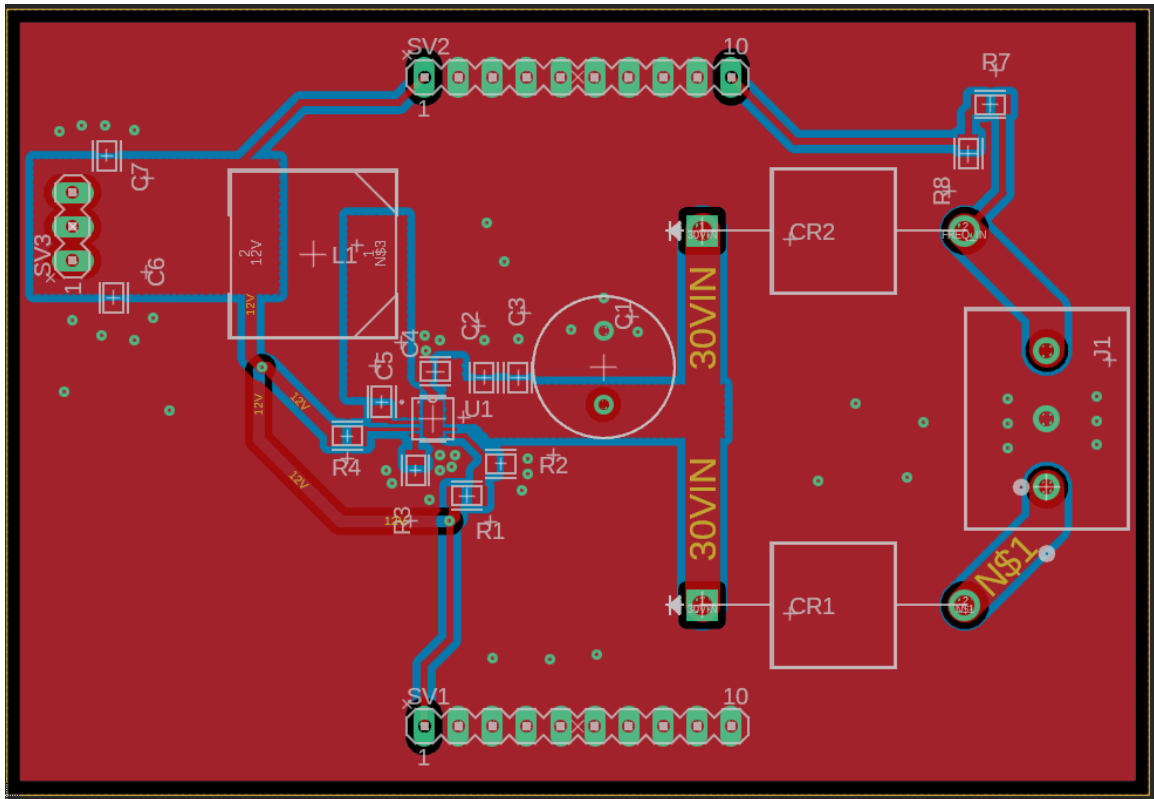


Figure 34: 12V PSU PCB Layout

The feedback network was added, but two modifications to the general guidelines were also necessary. There are additional pin headers at the 12V output and there are copper pours instead of traces at the inputs and outputs.

The additional pin headers were added to add additional current carrying capacity. Based on our research, the two most likely types of pin headers we would use are those for home appliances and industrial applications. These pin headers can support between 1-3A and 2-5A respectively. Since we planned to use the pin headers available to the senior design lab, we did not know for certain which kind we are using. To avoid fires or a possible redesign, we chose to add enough pin header to accommodate a worst-case scenario[48].

The inclusion of copper pours instead of traces is a quirk of the IC's package design. The LMR51450SDRRR is a 12 pin WSON package. Since the WSON package is small (3mm x 3mm), several pins are allocated to the input and output to accommodate large current sourcing/sinking. If we were to use traces, there would be strange overlaps and intersections that make the PCB impossible to manufacture or otherwise limit the power delivery of the board. By using a copper pour we can mold the conductive path to best fit the traces and provide the maximum power delivery while ensuring the board is as manufacturable as possible.

As for the input of the PMIC, it is simply a center-tap full wave rectification scheme. The terminal block is close to the edge for the PCB for space consideration and hosts several grounding vias. Since all of the power for the project is sourced from the transformer, all current must return to this point so minimizing the return path impedance is critical at this point.

### 8.2.2: 3.3V PSU

Similar to the 12V PSU, the 3.3V PSU makes minimal adjustments to the recommended PCB layout from its datasheet. Once again, the general guidelines account for input and output capacitors, the decoupling capacitors, and the thermal relief vias. The only explicitly mentioned addition is the feedback path needs its own unique connection to the 3.3V output instead of being attached to the copper pour and that the connection should be away from the switching pins. This is accomplished with the via seen below C10. The switching pins are on the opposite side of the output pins (near the U2 silkscreen). Separating the feedback from the switching pins are decoupling vias and the feedback path is routed on the opposite side of the board. Once again, since the surface mount component's pins are underneath the package, copper pours were used instead of traces in order to maximize the contact area for the inputs and outputs.

In Figure 9, there is an interference pattern showing bright red diagonal lines. We referenced both SnapEDA and Ultra-Librarian for the TPSM861253's PCB footprint and both had the interreference pattern. Whatever is causing the issue is inherent to the layout and not caused by a routing or clearance issue of our creation. Since it is not our fault, we should be able to override the rules to allow interference for this case. We must remain vigilant to ensure that no other design rules are broken before we allow exceptions, so that we do not accidentally allow real errors to enter the final product.

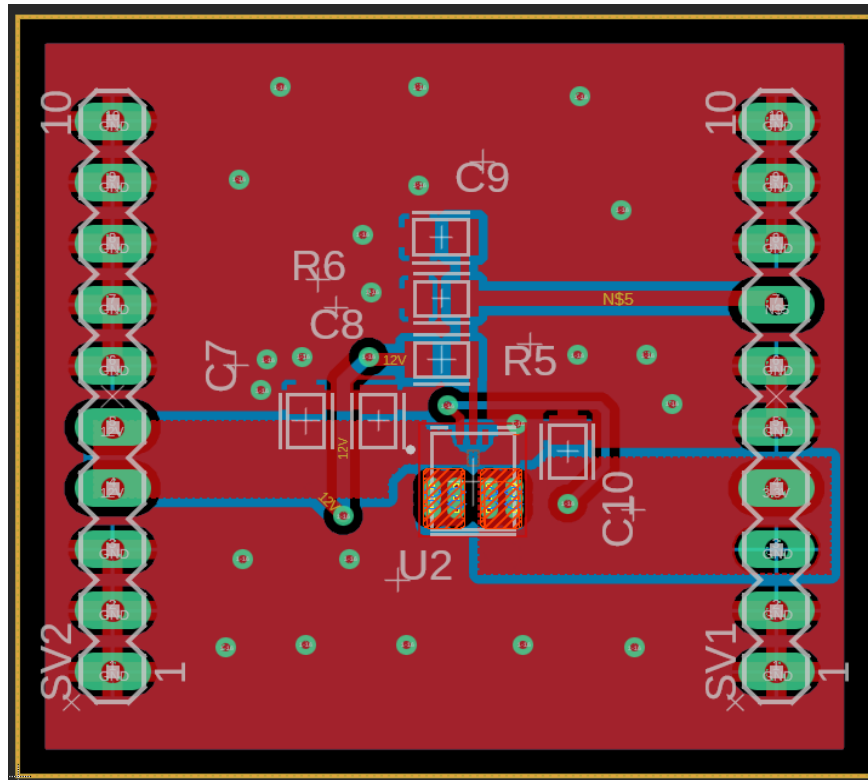


Figure 35: 3.3V PSU PCB Layout

### 8.2.3: Frequency Detector PCB

The Frequency Detector PCB is mostly designed by the team. The datasheet for the comparator has no deviations from the general guidelines.

As mentioned in Section 6.4, there are a lot of optional components that are included in the PCB. We had to choose a hierarchy of which components to place closest to the comparator. The first grouping includes the mandatory and most likely to be populated components. This grouping includes the surface mount and through hole diodes and the second voltage divider. The second grouping includes components that may be critical to the frequency detector's function as determined by testing and debugging. These include the hysteresis feedback network and the decoupling capacitor at the output. Lastly, the third group contains components that are unlikely to be populated including the Zener diode, and additional feedback components.

Since the input/output frequency is nominally 60Hz with only a few Hertz of deviation, we did not need to include any high frequency switching or Radio Frequency PCB design techniques, like impedance matching, for this PCB. Another benefit of the low frequency is that the layout may be more forgiving of noise and power reflections than if it were operating at a higher frequency.

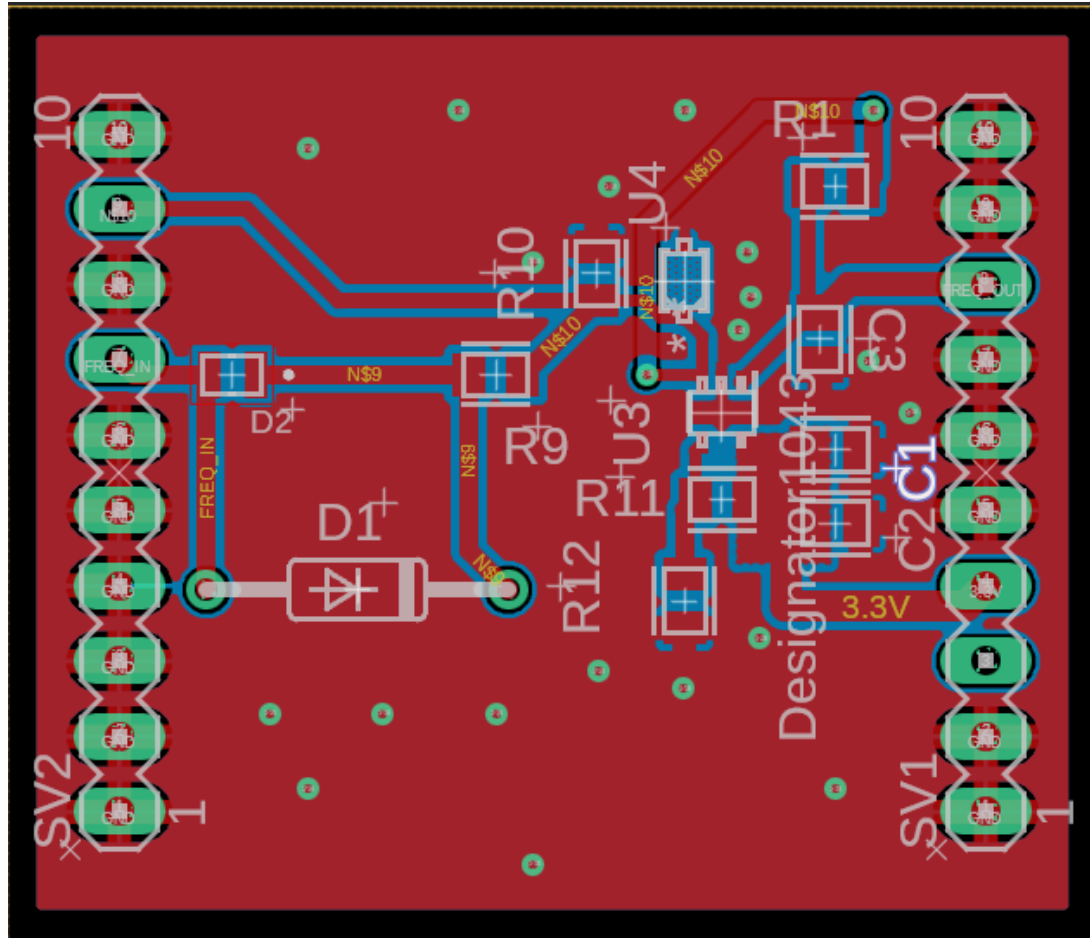


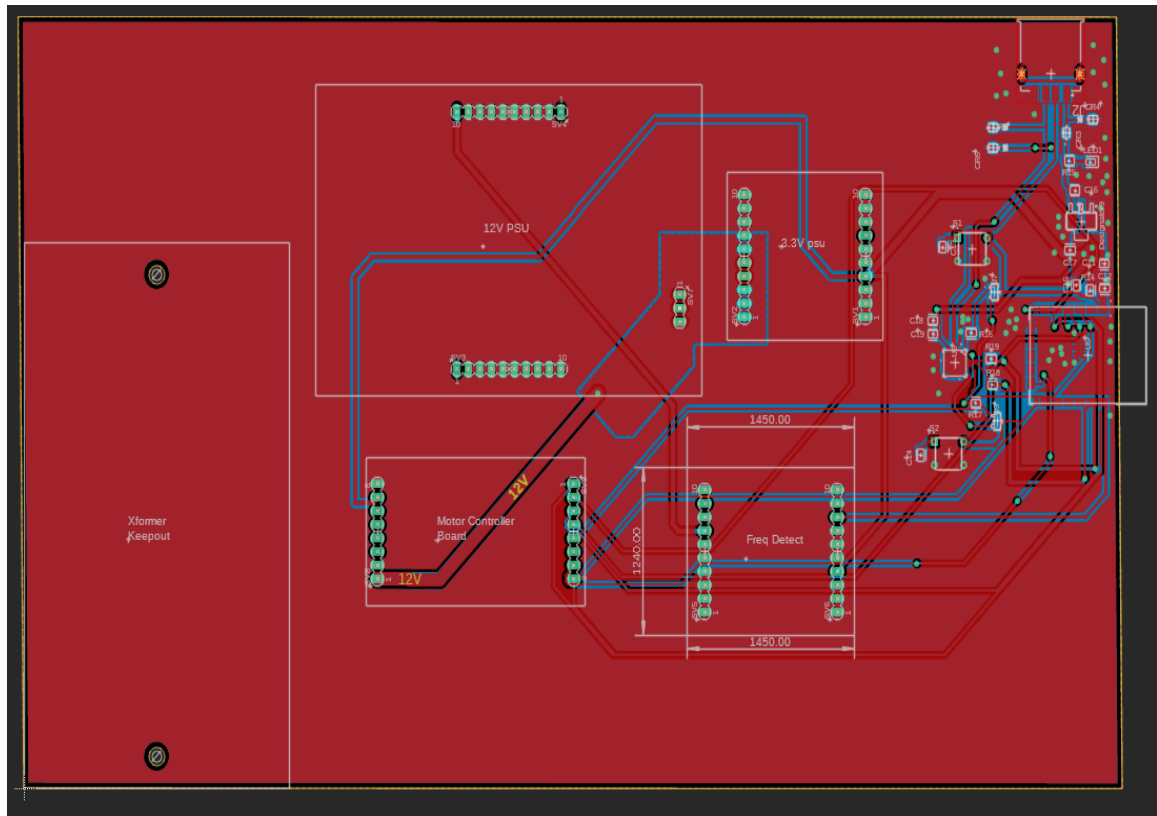
Figure 36: Frequency Detector PCB Layout

#### 8.2.4: MCU PCB

The MCU PCB is the most complex PCB. It must contain positions for all of the daughter boards as well as the ESP32's programming hardware.

The first step is to logically decide where to place the daughter boards relative to themselves and the MCU. We started by placing the transformer arbitrarily at the end of the PCB. The transformer does not have direct electrical connections (which is why it is excluded from the electrical schematic) to the MCU, only mounting holes. Since we are routing the input and outputs of the transformers with wire, its placement is less fixed than other components.

The 12V PSU's placement is the most critical of all the daughter boards. Since this board provides all the regulated power to the other peripherals, tremendous thought was put into where to place it. For this design iteration, we decided to place its output as centered on the MCU board as much as possible. This is to help minimize the trace length between the output and all the power pins. Additional trace length adds inductance and resistance which hinders the current output of the regulator.



*Figure 37: MCU PCB Layout*

The second most critical daughter board is the frequency detector. We want to place the input of the frequency detector as close to the incoming signal as possible in order to minimize the effects of frequency and phase deviation. The most causes of these distortions are EMI and noise. The best way to minimize the influence of EMI and noise is by minimizing the trace length the signal travels. Admittedly, the trace length could be significantly shorter by having the 12V PSU on the opposite (blue) side of the MCU PCB to have the input and output pins closer together. The reason we did not do this is because we are unsure where the boards will be mounted relative to the service panel. The service panel is an oddly shaped piece of metal, which may pose clearance issues and antenna placement issue (explored later in this section). Since the trace length is sub-optimal, the second-best option was to route the signal on the opposite side of the boards. We can do this without adding any additional depth to the MCU PCB since the pin headers are through hole, so no additional routing or vias are necessary to reach either side of the board.[49]

The 3.3V PSU must be centered between all the daughter boards as well as the ESP32. This subsection provides digital logic power to the comparator, ESP32, and the motor controller. While the expected current draw is significantly smaller than the 12V PSU, only about 1A maximum, we want to minimize parasitic losses as to provide the maximum chance that the boards work with minimal reworking.

The last daughter board is the motor controller board. Its placement is based on its current consumption; When on, it should draw a maximum of 4A at 12V. We decided to route the power to the MCU since there are several other signals routed through this area. We do not want to risk any unexpected signal integrity or grounding issues by presenting a large, isolated power plane in the middle of the MCU board. To ensure that enough current will flow through the traces, we consulted Digi-Key's trace width calculator for hard values. Using input parameters of 1oz/ft<sup>2</sup> for the copper thickness, 4A as the maximum current draw, and the recommended 10°C for the temperature rise, we find that we should use a minimum trace width of at least 81mils. We used two 100mil traces routed from the 12V copper pour. This will provide ample current carry capacity. The rest of the motor controller pins are simply routed to their respective GPIO. It is a similar case to the frequency detector where some adjustment to the physical board's place may provide an optimally short path, but we expect that our first iteration should work as is.

With the daughter board placement covered, the last section of PCB design is the ESP32 and its supporting hardware. While the electrical schematics were inspired by the development board, the development board provided by Espressif Systems is simply too small and too complex for us to replicate without buying it. Instead, we had to mimic the systems while adhering to our guidelines. Some of the unique aspects of this design include the antenna considerations and the USB routing.

The most obvious part of this design is that the ESP32 is hanging off the edge of the PCB. This is intentional. If an antenna, like a dipole or Wi-Fi/Bluetooth antenna included on the ESP32, is too close to conductive plane, like copper, two issues arise: image currents and unpredictable radiation pattern issues. Image currents are the complement (image) of a real current that manifests inside of conductors when moving charge is close. These currents produce their own electromagnetic waves and, if they are sufficiently close and powerful, could completely nullify the transmitting/receiving functionality of an antenna. Even if image current does not destroy the radiation pattern, having an conductor within the near-field radiation of an antenna will dramatically alter its radiation pattern in unpredictable ways. Last, if the previous two issues are somehow avoided, having a nearby conductive may act as a reflector. From a transmitting point of view, this could, once again, produce strange radiation patterns making the coverage by Wi-Fi spotty. From the transmitting side, the conductive sheet acts as a Faraday cage preventing any signal from reaching the antenna from processing information at all. To mitigate all of these possible scenarios, we simply kept any metal away from the antenna as possible, including the PCB trace, copper pour, and the service panel body.

The input of the USB connections also adds some additional challenge when placing the components on the PCB. USB-A consists of four connections: ground, positive 5V, positive differential, and negative differential. The ground and power connections are trivial to route, but the differential pair is not. A differential pair consists of two wires that carry the same information in opposite polarities. For example, if the signal on the positive terminal was high-low-high, then the negative terminal would be low-high-low. What makes routing these signals a challenge is

we are not sure how sensitive the input of the USB-UART conversion IC is to an input phase difference. Ideally, the opposite polarity signal reaches their destination at the same time. This can be done by simply making the trace length identical, even if adding additional length to one is necessary. Unfortunately, the USB-UART converter has the input pins' polarity opposite of the USB's. This means that one of the signals must travel across vias to jump the other signal. We have tried to mitigate the effects of the additional length, but on building and testing the signal integrity will verify if a redesign is necessary.

Similar to the 3.3V supply, the USB-A PCB layout has interference lines near where the connector would mate with the PCB. Once again, since these issues are inherent to the official layout, we can simply override the rule check to allow for the boards to be exported anyways.

### 8.2.5: Motor Control PCB

As outlined in Chapter 6, the circuit for the motor controller consists of the TMC5160 chip, 4 double MOS, 2 precision sense resistors, and some complimentary resistors and capacitors. The major considerations for this board were heat and power, as active cooling is recommended for both the TMC5160 and the MOSFET, and up to two amps may be traveling through the board. Currently, we plan to use a 2-layer board, similar to the other subsystems, to build this board. One side of the board will have all the active components, and the other will be strictly used for shielding, routing, and heat dissipation. In case the two-layer board is not enough to dissipate the heat, we can order a PCB with 4 layers; the additional internal layers will greatly add to the thermal mass. Additionally, external heat sinks will be mounted to the IC itself and the MOSFETs. While we expect the design to work, we can always increase the copper weight of each of the pours to provide additional thermal mass. Active cooling, like a fan, would only add unnecessary complexity and power draw from the circuit and will be avoided.

This board breaks most of the general trace size guidelines found in Section 8.1 simply due to its size. All of the traces of the motor controller are 6mils wide and that is the widest width that the IC can accommodate. This may cause issues with manufacturing and will definitely complicate the integration and troubleshooting process, so great care was taken to closely match the development board that came with motor controllers come with

*Figure 38* also contains the interference marks that the present on the other daughter boards. Once again, these are not the designers' faults and are intrinsic to the IC's layout. Once we verified that no other issues were violated, then we ignored the rule violation and continued with the manufacturing process.

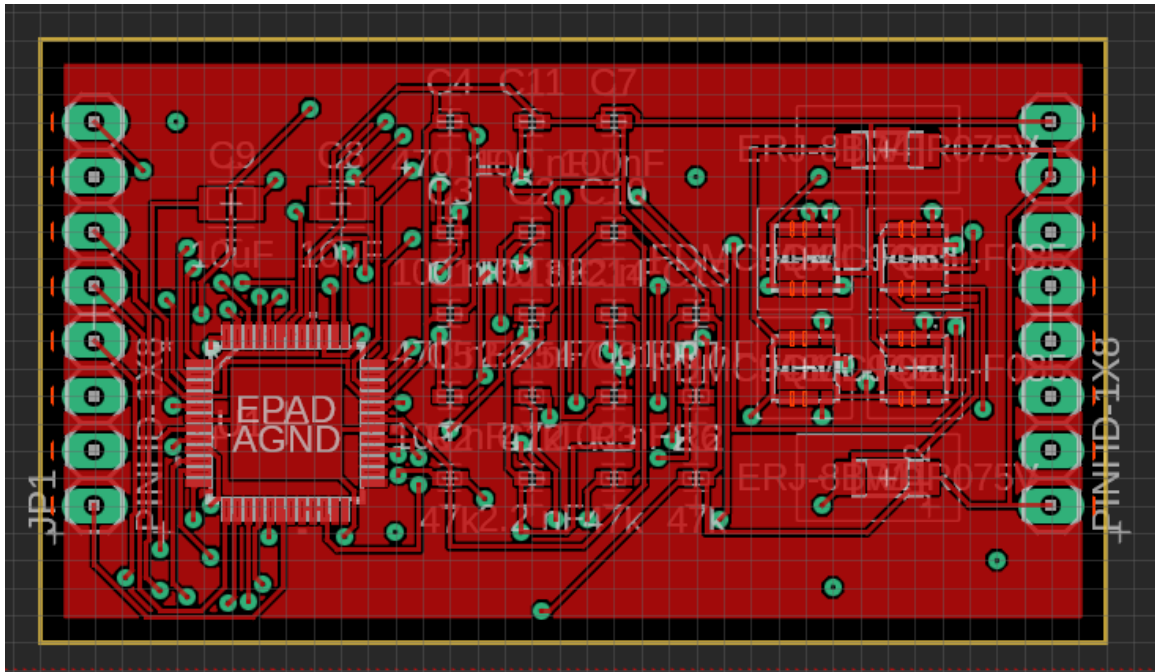


Figure 38: Motor Controller PCB Layout

### 8.3: Mechanical system fabrication and construction

As this project has a major mechanical component to it, it is very important to consider the manufacturing process for this system. Just as with PCB design, it would be easy to design a part that is beyond our manufacturing capabilities, or impossible to produce altogether. It is also very important to consider the physical properties of the materials used, as this will impact both manufacturability as well as system performance. Design choices will be informed by mechanical systems the authors are familiar with: 3d printers and CNC machines. It would be prudent to restate that there are no mechanical engineering students in this group.

### 8.3.1: Face Plate

In order for the panel to accommodate the mechanical device, one of two options must be taken. The first option is to reuse the stock faceplate; this requires removing the door by the hinges, and drilling holes into it to mount the device. The second option is to replace the face plate, creating a new one that has a cutout to allow access to the breakers; leaving the stock faceplate undamaged. This new plate could be made of either plastic or metal, with Lexan being an ideal choice, as its transparency would allow for clear viewing for demonstrative purposes.

We moved forward with the second option for the time being, as it will allow us to demonstrate the installation process more clearly, and will not incur damage to the service panel.

As Lexan does not crack or shatter like other clear plastics, it is very easy to machine. It can be cut and drilled like wood; however, precautions must be taken that it does not warm too much during tooling, as it has a low melting temperature.

To create the mounting holes a handheld drill can be used, and a router would be an ideal tool for creating the central cutout for the device to access the switches. One important consideration is that Lexan is very prone to scratches. This means the protective film must remain in place during the manufacturing and testing process, and be removed only when the device is complete, as to minimize aesthetic damage.

### 8.3.2: Frame

Although the 600mm aluminum rails will be ordered at the correct length, the shorter rails will require being cut to size. Luckily, aluminum being a soft metal, it can be cut with wood tools. To trim the extrusions to length, a standard miter saw will be used. It is important to wear breathing protection equipment as airborne Aluminum particles created by cutting or sanding can be easily inhaled and are harmful. appropriate personal protection equipment will be used. The linear profile guide rails, and lead screws will be ordered at the correct length, so no tooling is required. This is advantageous, as the hardened steel they are constructed from is much more difficult to cut than aluminum.

To mount the aluminum extrusions to the faceplate, and to mount the rails to the extrusions, T-nuts will be used. These are nuts designed to fit inside the profile of the extrusions, making it possible to attach a bolt to them at any point without the need for drilling or tapping. To attach the extrusions to themselves to form the frame of the device, T-nuts will be used alongside external brackets to increase rigidity.

### 8.3.3: Wire management

As we have a gantry-based system, one of the motors is required to move rather than stay in a fixed position. In our design, this is the X axis motor; as the Y axis moves up and down, the X axis motor moves with it. As the X axis motor requires 4 wires running to it from the motor controller, it is important that we properly manage these wires. The reasons are twofold: dangling wires would look aesthetically unappealing to Reviewers and consumers, and they have the potentiality of becoming tangled and damaged in the moving parts of the device. To solve this, we implemented cable chain.

Cable chain, also known as drag chain, is a series of interlocking plastic pieces that wires can be routed inside of.[50] The benefit they provide is that unlike loose wire, they only bend in one direction, buckling over itself to change length, while staying perfectly straight. They are used very commonly for gantry systems such as ours. By utilizing a cable chain along the Y axis, we were able to provide power to the X axis motor while keeping the wires required to do so entirely out of the way.

### 8.3.4: Linear movement considerations

For successful operation of the device, it is very important that the two linear motion devices that make up the gantry perform smoothly and reliably. To ensure this, there are some steps we can take. To make sure that the lead screws are properly aligned, they will have bearing plates put on both ends. These plates will be 3d printed and mount to the aluminum extrusions. They will house bearings that have an inner diameter that fit the lead screws precisely, meaning that the screw will not wander as it is rotated, increasing the precision of the system.[51]

Additionally, flexible couplings will be used for mounting each lead screw to the motors. These aluminum couplings are designed to accommodate the shaft of the stepper motor on one side, and the 8mm lead screw on the other. In-between, they have a long spiral cut that enables them to flex side to side by a small margin. This is very useful as it can correct for small misalignments in the system, and reduce vibration[52]. They are very commonly used in 3d printers.

To further increase precision, anti-backlash blocks will be used on the lead screws. These blocks are what thread onto the lead screw and move linearly as the screw spins. the specific blocks we used are made of Delrin, a low friction plastic .This low coefficient of friction will ensure high efficiency of the system. The important feature of these blocks is their clamping ability. by tightening the bolt, the level of grip of the lead screw can be adjusted, eliminating backlash [53]. Backlash, also called play or slop, is the amount of movement lost between the screw and the block due to imprecision in the system. [54]Because we want to be able to position our end effector precisely, it is important to reduce this error.

A design choice that requires justification is the use of a single motor and lead screw for the Y axis. It is common in 3dprinters and CNC machines to include a motor and lead screw on both sides of a large axis, the justification being that this prevents sag and increases accuracy.[55] While this is true, we have chosen not to include this feature for a few reasons. The major concern was power draw: as we are creating our own 12 volt supply, we are trying to minimize amp draw wherever possible, and another motor would majorly impact that. another concern was synching the motors, if it occurred that one of them lost steps of the G-code commands for them were written improperly, the motors could end up fighting each other and causing mechanical as well as electrical damage to the system. Instead, we have opted to include a second linear profile rail on the side without a motor. Due to these rails precision and rigidity, we believe it will be sufficient in preventing sag in the gantry system. It should also be noted that an additional motor and lead screw would require an additional control PCB, increasing the overall complexity and price of the device.

## Chapter 9 - System Testing and Evaluation

Testing and evaluation are important to ensure the reliability and efficiency of the hardware and software components in our project. This chapter explains the testing and validation process for several system components such as the power supply PCBs, frequency detector and MC board. Not only are we testing and evaluating hardware but software as well. Some of the software that we had to test and validate was the system that detects the grid frequency and gives feedback if it fluctuates. Each subsystem goes through rigorous testing to validate that it meets the required specifications and functions seamlessly within the broader system.

### 9.1: Hardware Testing and Evaluation

The hardware testing and evaluation process is critical for validating the functionality and reliability of each subsystem in the project. This section outlines the procedures utilized to test and validate the printed circuit boards (PCBs), motors and hardware structures integral to the system. Each component goes through a precise series of inspections, measurements, and operational tests to make sure it performs as intended under several conditions. Some of the testing methods include physical inspections, electrical performance verifications, and dynamical load testing. By systematically addressing potential issues at the component level, this process ensures the successful integration of all hardware into a cohesive and robust system.

#### 9.1.1: 12V PSU PCB Testing and Evaluation

The 12V power supply PCB underwent a comprehensive testing process to check its functionality. After the PCB is soldered, all solder joints are visually inspected to make sure there are no errors or weak connections. Initially, the board is powered utilizing a benchtop power supply, with no load attached to the 12V daughter board or MCU. A sine wave signal, fluctuating between 27Vp and 30Vp, is run through the terminal block.

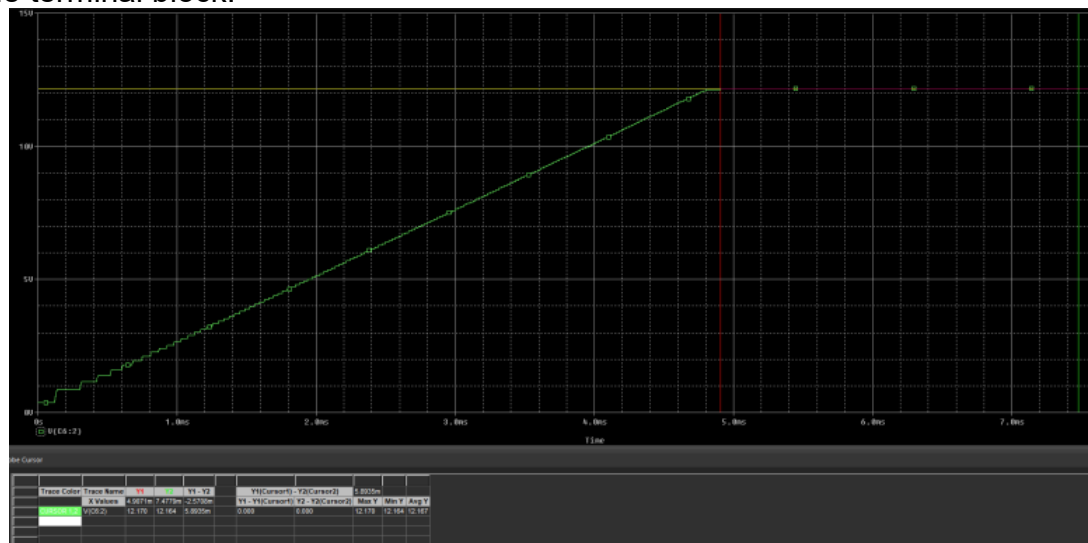


Figure 39: Simulated Output of 12V PSU

The rectifier output is probed using a digital multimeter (DMM) to ensure proper rectification. Once nominal output is reached, the board is connected to a DC load tester. Current is drawn from the PSU incrementally to prove the board's capacity to sustain 12V at a maximum current of 5A. If the board achieves these parameters, it is declared successful.

### 9.1.2: 3.3V PSU PCB Testing and Evaluation

The 3.3V PSU PCB went through similar testing to the 12V PSU PCB. After the PCB is soldered and every solder joint is inspected, the PCB is powered utilizing a 12V DC signal with a maximum current of 1A, replicating the 12V PSU. All test points on the board are probed to check the expected voltage levels. A DMM is used to make sure the output voltage is steady at 3.3V under several load conditions. Subsequently, the board is connected to a DC load tester, and current is drawn until the voltage starts to drop. If the board can provide approximately 1A of current at 3.3V without instability, the board passes the validation process.

### 9.1.3: Frequency Detector

The frequency detector PCB is tested to validate its ability to measure grid frequency accurately. After soldering, all the solder joints are inspected, and the board is powered with a 3.3V DC signal, with a current limit set to 300mA to prevent damage. Proper powering of the IC by proving the relevant pin headers. A 60Hz signal from a signal generator is applied to the input pin header, and connections are verified by probing both the IC package and pin header. The comparator's output is calculated at the output pin header to make sure the signal is a square wave matching the input frequency. If distortions are detected, further testing is conducted across the specified frequency range to recognize potential issues in the hardware or software.

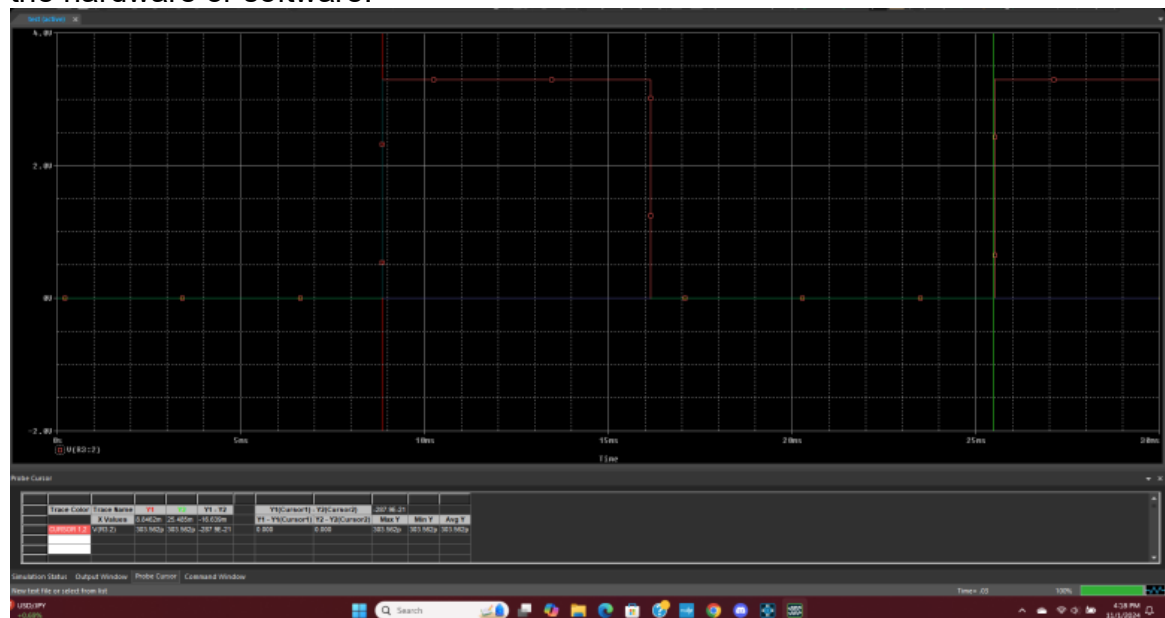


Figure 40: Simulated Output of the Frequency Detector

#### 9.1.4: Motor Controller

The evaluation for the motor control was done using commercially available development boards. These boards are designed for use in 3D printers. To validate that the controllers could meet the requirements in terms of speed and force applied through the motors, two development boards were installed on a breadboard alongside an ESP32 for control. Test code was written, and a benchtop power supply was used to provide the necessary 12volts, as well as to monitor current.

From this testing we found the TMC5160's performed very well with the selected NEMA 17 stepper motors. They displayed adequate torque to flip a breaker switch using a prototype mechanical system, sufficient speed, nearly silent operation due to the TMC's StealthChop feature, and highly reduced current draw at idle thanks to the CoolStep feature. Thanks to this validation, we moved on with the TMC5160 for further development.

#### 9.1.5: MCU Board

The MCU board goes through a series of physical and functional tests to make sure all components are properly installed and operational. After soldering and inspecting all the solder joints, a clearance and fit check is executed to verify that all daughter boards, including the transformer and antenna, fit securely and without interference. The USB-A port is checked for programming access, and the antenna placement is confirmed to avoid contact with metal surfaces. Electrical continuity tests are performed utilizing a digital multimeter (DMM) in continuity mode, verifying connections between the 12V and 3.3V PSU outputs and relevant points, GPIOs, and frequency detector input and output. The ESP32 microcontroller is tested by connecting it to a USB-A cable, detecting the grid frequency, and confirming the LED power signals when the frequency fluctuates out of the pre-established range.

### 9.2: Software Testing and Evaluation

The software testing and evaluation process is essential for guaranteeing the accuracy, reliability and performance of the system's key components. This section discusses the methods utilized to test and validate the Frequency Detection and Feedback Software on the ESP32 microcontroller and the Mobile App Software. Each software module was subjected to controlled and systematic testing to verify its functionality under different conditions. For the Frequency Detection and Feedback software, tests included frequency detection accuracy, out-of-range feedback through the LED, and signal handling using tools such as a waveform generator and an oscilloscope. The Mobile App Software underwent rigorous backend API testing using Postman, with planned frontend verification through unit testing in Visual Studio Code. These testing procedures ensure that the software meets the design requirements and functions correctly within the overall system.

### 9.2.1: Frequency Detection and Feedback Software

To test and evaluate the ESP32'S frequency detection and feedback software, a waveform generator is utilized as the external frequency source. The waveform generator is configured to output a square wave signal with a frequency of 60Hz and an amplitude of 1.650Vpp, ensuring the high and low limits aligned with the 3.3V PSU PCB that will be implemented later on. The signal wire from the waveform generator is connected to the GPIO4 pin of the ESP32, which is the designated input pin for signal detection. The ground wire is connected to one of the ESP32's GND pin to establish a shared ground reference. An oscilloscope is also connected to the signal and ground wires to visualize and verify the waveform that is being sent to the ESP32 in real-time.

The waveform generator is then utilized to replicate various frequency values to test the ESP32's detection and feedback response. The initial frequency was set to 60Hz, which lies within the acceptable pre-established range (59-61Hz). During this scenario, the LED stayed off, indicating that the software correctly recognized the frequency as valid. Next, the frequency was manually adjusted in real-time was the software was running to values outside the acceptable range, including frequencies lower than 59Hz, higher than 61Hz, 0Hz (no signal). In each of these scenarios, the LED started to blink, signaling that the frequency was out of range.

This testing confirmed the functionality of the ESP32's frequency detection and feedback software. The software was able to accurately calculate the frequency from the input signal, compare it to the acceptable range, and supply appropriate visual feedback through the LED if needed. The use of the waveform generator and oscilloscope make sure that the test conditions are precise and that the ESP32 is functioning as it is supposed to. Overall, the testing validated that the software meets its design objectives and can reliably detect and respond to frequency deviations.

### 9.2.2: Mobile App Software

To test and evaluate the software for the mobile app, a comprehensive approach is taken to assess both the backend API and the frontend functionality. For the backend API, Postman is utilized to send request to the API endpoints, both locally and via the server domain, to make sure the endpoints are working as expected. The primary criterion for success is if the response body of the API request returned a success message with a status code of 200. This shows that the request was successfully processed by the API. On the other hand, if the response body returned an error message, traceback, or a status code such as 404 (Not Found) or 500 (Internal Server Error), the request was considered a failed request, signaling a problem with the API.

For front-end verification, the unit testing suit in Visual Studio Code is planned to be used. This testing suite allows for thorough testing of the mobile app's user interface and interactions, ensuring proper functionality and alignment with the intended design. By using these testing methods, both the backend and frontend

components of the mobile app are thoroughly evaluated to guarantee efficient and reliable performance. The combination of API testing through Postman and planned unit tests for the frontend supplies confidence in the software's ability to handle real-world usage scenarios effectively.

### 9.2.2: – TMCStepper library

This is the library used to interface the TMC5160 stepper driver with the ESP 32. We found it performed well, it was able to correctly interface with the controller over SPI to configure the special features of the controllers and to set the current limit. In addition, It was able to work for the multi-device SPI configuration we used, with two motor controllers, configuring both of them correctly.

### 9.2.3: – GRBL

This is a G-code interpreter compatible with Arduino. We plan on using a version of it ported to ESP32, and adding functionality from the TMCStepper library to make it compatible with our system. When tested with an Arduino and compatible motor drivers, we found it was able to correctly move stepper motors in accordance with the G-code commands it received.

## 9.3: Overall Integration and Testing

Chapter 9.3 covers the plan to integrate the electrical, mechanical, and software components of the project into a cohesive unit. It will also discuss our current plans to test the overall project as it pertains to satisfying the system requirements as laid out in Section 2.5

### 9.3.1: Overall System Integration

The integration process for our project is structured into three primary steps to guarantee a seamless connection between hardware and software components.

Each step addresses a critical aspect of the system, from the individual functionality of components to their unified operation as a cohesive unit.

The first step in the overall integration of our project is Hardware Interface integration. This step involves connecting all the Printed Circuit Boards (PCBs) into a unified interface that can communicate effectively. This includes linking the power supply, frequency detection system, motor control circuit and additional monitoring components to the main ESP32 microcontroller. Each PCB is tested independently to ensure its functionality before being integrated into the larger system. Particular attention was paid to power distribution to avoid any interference or voltage irregularities during operation. After integration, further testing will be conducted to prove communication between the PCBs and guarantee that signals are transmitted precisely across the interface. This step establishes a stable foundation for the overall system.

The second step in the overall integration of our project is Mechanical Structure Setup. This step focuses on assembling the physical structure that houses the electrical panel. This involves making a robust and well-aligned enclosure for the

circuit breakers, wiring, and motorized switching mechanisms. Precise consideration is given to the design to guarantee that the structure could accommodate mechanical movements without compromising stability or alignment. Actuators for motorized operations will be mounted with precision to ensure reliable performance during automated tasks. Additionally, the enclosure is built to provide easy access for maintenance and upgrades while protecting internal components from external damage or environmental factors. This mechanical setup is important in guaranteeing that the system would function effectively in a real-world environment.

The last step in the overall integration of our project is Unified Software Deployment. This final step focuses on consolidating all software and ESP32-based code into a single ESP32 microcontroller. This requires merging multiple functionalities, including frequency detection, visual feedback, motor control, and communication with the mobile app. The integration process involves optimizing the software to guarantee that real-time tasks, such as frequency detection and motor control, do not interfere with each other. A lot of care is taken to keep low latency and reliable performance across all functions. Careful testing scenarios are implemented, including simulated grid frequency variations and automated circuit breaker operations, to confirm that the software can handle real-world conditions. This step makes sure that all system functionalities are centrally managed, reducing abnormalities and improving accuracy.

By following this structure approach, our team guarantees that all components – both hardware and software – will be seamlessly integrated. The process addresses potential problems, such as power distribution conflicts, mechanical misalignments, and software interference, to create a unified system that meets the project's objective for efficiency, accuracy, and scalability. This holistic integration guarantees that our project is ready to operate effectively in diverse real-world scenarios.

### 9.3.2: Overall System Testing

The last step, once all of the major subsystems are integrated together, is to test the unit as a whole against the list of requirements we set during our Divide and Conquer meeting.

Some requirements can be tested and verified before the overall system is integrated. These requirements include system security and mobile latency, since attaching the MCU to a service panel will not affect needing a password or how quickly the mobile application communicates, it is unnecessary to retest these requirements after establishing a pass.

The remaining requirements—size, weight, frequency detection, installation time, quiescent power, relay switching time, and volume—will all need to be tested/retested once the project is completed.

The size and weight requirements are the easiest to verify. We simply used a ruler and measure from the service panels face to see if our project does not extend past the 8" limit. For the weight, a simple bathroom scale will provide

enough resolution. This weight will exclude the service panel. We may include the service panel's face and the dust cover depending on our sponsor's needs.

The installation time shall cover the time it takes to modify the existing service panel, assembling the project, and mounting the project.

The relay switching time can easily be measured with a stopwatch. We make two time measurements. The first one will start once the user switches the breaker off from the mobile application and stop once the breaker is successfully switched off. The second one will start once an external signal generator trips outside of the acceptable frequency. Both of these values should remain under the 10s time frame.

The trickiest part of the overall project to accurately verify is the quiescent current. The ideal method would be to add an ammeter in series and a voltmeter in parallel at the input of the transformer (outlet side). This would provide the most accurate data, as any downstream losses (ohmic, heat, switching, et cetera) would be reflected by measurements close to the source. While ideal, there are a few issues to consider. First, we must be very careful wiring the ammeter and voltmeter. The wall outlet can provide thousands of watts which can easily damage or destroy benchtop equipment. If we were to use handheld equipment, we must ensure that the resolution and testing procedure is acceptable. In case we cannot safely or accurately measure the quiescent power, we can do so analytically. We did so by taking the current and voltage measurements at the output of the 12V PSU. Once measured, we can add back the nominal values of the upstream losses by using typical values for the regulator quiescent power, diode drops, et cetera. As long as the calculated power is more than 1W less than the worst-case calculation, we can comfortably call the requirement satisfied.

The final requirement to verify is the noise requirement. Since decibels are always a relative value, the measurement procedure must be well-defined and easily replicable for consistent results across all environments. Using any smartphone application that can measure volume in dB, place the smartphone 10 feet away from the project. Ensure that the smartphone application is using the bottom microphone (the one used for phone calls) and that the dust cover for the project is securely fastened. Once everything is in place, have the project flip the furthest breaker from its resting place. Repeat this several times and note the final, loudest volume. The project will have met the volume requirement if the volume when running does not exceed 70dB. In the case the ambient noise exceeds the maximum threshold, this test cannot be performed, and the project should be moved to another, quieter location for proper testing and validation.

## 9.4: Plan for SD2

Our plan for Senior Design 2 supplies a structured roadmap for advancing and completing our project. With clear milestones and a focus on iterative development, the team aims to integrate all components of the system into a

unified, functional design by the end of the semester. The plan emphasizes testing, refinement and collaborative efforts to address problems effectively.

Our first step in our plan for Senior Design 2 is to purchase the first iteration of PCBs during the winter break and thoroughly test them. This testing phase is very important for identifying design flaws, performance issues, or compatibility issues with other components. By testing the PCBs early, we had sufficient time to evaluate the functionality of the PCBs and determine necessary design modifications. This allows us to start Senior Design 2 with a good understanding of the PCBs' behavior and potential area for improvement. At the start of Senior Design 2, we focused on redesigning the PCBs if needed, incorporating feedback and data collected during the initial testing stage. These redesigns aim to solve any uncovered issues and optimize the PCBs for better performance and compatibility. The updated PCBs will then be obtained in time for the next round of testing. Additionally, the rest of the parts necessary for the project will be purchased between early December and mid-January, guarantee that all components are available for integration and testing during the semester.

Within the first month of Senior Design 2, we finalized the design and figure out the mechanical structure of our project. So, then we can work all semester on assembling it. This step involves constructing the physical framework to house the electrical components, such as circuit breakers, actuators, and wiring. The structure must ensure proper alignment of moving parts, supply stability for operations, and facilitate easy access for maintenance. Developing a strong mechanical setup is important to support the system's hardware during testing and real-world conditions.

Once the redesigned PCBs arrive, they will go through careful testing to verify their functionality and compatibility with the system. This stage will validate the changes made during the first initial testing stage with the first wave of PCBs. If further corrections are necessary, they will be implemented as soon as possible. We made sure that the PCBs are rigorously evaluated to avoid potential delays in our plan.

A focal point in our plan for Senior Design 2 will be to bring all software functionalities into a singular ESP32 microcontroller. This includes frequency detection, visual feedback, motor control, and communication with the mobile app. Consolidating these components guarantees efficient system operations and simplifies the overall architecture. During this phase, the software will be tested extensively to ensure that each function works as expected without interfering with others. This step is critical for achieving a seamless integration of hardware and software.

The last 6 to 8 weeks of Senior Design 2 will be dedicated to the final integration and fine-tuning of our project. This phase involves combining the mechanical structure, PCBs, and ESP32 software into a fully functional system. In-depth testing will be conducted to identify and resolve any remaining issues, guaranteeing that the system works efficiently and meets the project's goals. We also optimized performance during this stage of our plan, focusing on efficiency,

robustness, and user-friendliness. At the same time, we started working on the final project report, recording the development process, challenges, and results in detail. To make consistent progress, we are planning on holding regular meetings every Tuesday and Thursday throughout the semester. These meetings will provide opportunities to work on the project all together, share updates, and address challenges that we may encounter through the semester. By keeping a regular schedule, we fostered effective communication and stayed on track to meet deadlines.

With this comprehensive plan, we are well-positioned to address potential issues early, refine the system through careful testing, and deliver a fully integrated and functional project by the end of Senior Design 2. The structure approach guarantees that each component is thoroughly tested, optimized and ready for real-world implementation.

## Chapter 10 – Administrative Content

The administrative content covers the engineering management portion of the project. Included in this chapter are the budget, the bill of materials, work allocation, and the ideal project milestones. Similar to the design portions of the paper, this section will change, and components are purchased for the project, systems are redesigned, and deadlines are met.

### 10.1: Budget

This project's sponsor has graciously given us a \$3,000 budget for all materials and services necessary to complete the project. Our estimated budget is distributed across four major areas: the service panel, the electrical hardware, the mechanical hardware, and the software. The service panel acts as the foundation of the project. We lumped its budget with the various circuit breakers we needed to populate it. We expect this to cost \$300 dollars at the maximum. The electrical hardware represents the costliest expense. We needed 5 PCB orders at a minimum and all the support hardware to populate them. Additionally, it is very likely that the more sensitive boards, voltage regulators and the motor drivers, will need redesign. We allocated \$600 to cover the electrical hardware. The mechanical hardware is also costly but is unlikely to change since the gantry system is inherently more customizable than the PCBs. We also allocated \$400 to the gantry system, motors, mounting, and other mechanical systems for the project. Lastly, the software is our cheapest expense. We may need to purchase a domain (at a monthly rate) to host the smartphone application, but this will still only run tens of dollars. We expect the software to cost no more than \$100.

Currently, we plan to purchase all of the components out of pocket and submit the receipts to the appropriate resource for reimbursement. We still need clarification for how monthly subscriptions would work for this system. If reimbursement does not work, we can make purchases directly through UCF purchasing and have them delivered, but this process takes significantly more time than the team purchasing the product.

*Table 22: Budget Matrix*

| Grouping            | Allocated Budget |
|---------------------|------------------|
| Service Panel       | \$300            |
| Electrical Hardware | \$600            |
| Mechanical Hardware | \$400            |
| Software            | \$100            |
| Expected Total      | \$1400           |

### 10.2: Bill of Materials

The bill of materials represents the major items that have been purchased. This list will be constantly updated as more materials, tools, and services are purchased in service of the project. The table will not include common components, like capacitors or resistors, unless they are special orders or exclusive to a single

manufacturer. All of the receipts, physical and electronic, are archived in our file-sharing program.

As a note, the part number generally represents the manufacturer's part number. This helps with consistency if the product is out of stock in our source of the part.

*Table 23: Itemized BOM*

| <b>Component</b>                  | <b>Part #</b>        | <b>Description</b>                                                                            | <b>Quantity</b> | <b>Total Price (\$)</b> |
|-----------------------------------|----------------------|-----------------------------------------------------------------------------------------------|-----------------|-------------------------|
| <b><i>Service Panel</i></b>       |                      |                                                                                               |                 |                         |
| Scale                             | 133602056            | H2OX 50 Lb Mechanical Scale With Tape                                                         | 1               | 4.99                    |
| Circuit Breaker                   | 047569062803         | Square D Homeline 60 Amp 2-Pole Circuit Breaker(HOM260CP)                                     | 1               | 15.98                   |
| Circuit Breaker                   | 047569071058         | Square D Homeline 2-15 Amp Single-Pole Tandem Circuit Breaker                                 | 1               | 13.78                   |
| Service Panel                     | 785901977827         | Square D Homeline 200 Amp 30-Space 60-Circuit Indoor Main Breaker Electrical Panel with Cover | 1               | 169.00                  |
| <b><i>Electrical Hardware</i></b> |                      |                                                                                               |                 |                         |
| Transformer                       | 530-241-8-20         | Bel Signal Transformer 50\60 Hz, Laminated                                                    | 1               | 31.73                   |
| Voltage Regulator                 | LMR51450SDRRR        | Texas Instruments 36-V, 5-A regulator                                                         | 3               | 8.10                    |
| Voltage Regulator                 | TPSM861253RDXR       | Texas Instruments 3V to 17V input volt regulator                                              | 3               | 6.63                    |
| Comparator                        | TLV7011DCKR          | Texas Instruments Low power, small size comparator                                            | 3               | 1.89                    |
| MCU                               | ESP32-WROOM-32E-N8R2 | ESP32-32E with onboard antenna                                                                | 3               | 8.64                    |
| USBA-UART                         | CP2102N-A02-GQFN28   | USB to UART convertor IC                                                                      | 3               | 10.59                   |
| Motor Controller                  | TMC5160A-TA          | Stepper Motor driver IC                                                                       | 2               | 42.59                   |

| <b><u>Mechanical Hardware</u></b> |               |                                                      |         |       |
|-----------------------------------|---------------|------------------------------------------------------|---------|-------|
| Stepper motor                     | 17HS19-2004S1 | Nema 17 Bipolar 59Ncm stepper                        | 2       | 19.24 |
| Aluminum extrusion                | 2020          | Anodized Black Aluminum Profile                      | 2400 mm | 26.39 |
| 3d printed hardware               | n/a           | Custom part                                          | 3       | 0     |
| Aluminum extrusion fasteners      | n/a           | 200pcs M5 T-Nuts Screws Kit                          | 1       | 13.99 |
| Long lead screw                   | 20-LP         | 8mm Metric Lead Screw, 540 mm                        | 1       | 21.99 |
| Short lead screw                  | 20-LP         | 8mm Metric Lead Screw, 290 mm                        | 1       | 10.99 |
| Long linear guide rail            | MGN12 550mm   | Mini Linear Rail Guide+Bearing Steel Rail Block      | 2       | 54.04 |
| Short linear guide rail           | MGN12 300mm   | Mini Linear Rail Guide+Bearing Steel Rail Block      | 1       | 29.03 |
| Linear guide carriage             | MGN12H        | Included w/ rails                                    | 3       | 0     |
| Stepper motor mount               | ST-M1         | Nema 17 Bracket for Stepper Motor                    | 2       | 3.68  |
| Lead screw nut                    | 1055-Set      | Anti-Backlash Nut Block for 8mm Metric Lead Screw    | 2       | 19.98 |
| Lead screw coupling               | 222           | 5mm * 8mm Flexible Coupling                          | 2       | 13.98 |
| Lead screw bearings               | 780           | Ball Bearing 688Z 8x16x5                             | 8       | 5.94  |
| <b><u>Software</u></b>            |               |                                                      |         |       |
| Digital Ocean Droplet             | n/a           | Virtual Machine for server code to run on constantly | 12      | 84.00 |
| Domain                            | n/a           | Domain Name Service (DNS) Reservation                | 1       | 1.00  |

### 10.3: Distribution of Work

Table 24 represents the lead engineer and point of contact for each of the project's subsystems. These roles are not rigid and collaboration between all engineers is necessary and expected for a function project.

*Table 24: Distribution of Work Matrix*

| Task                        | Responsibly Engineer (Primary) | Responsible Engineer (Secondary) |
|-----------------------------|--------------------------------|----------------------------------|
| Frequency Sensor Design     | Nicholas Sally                 | Jack Dwyer                       |
| Power Supply Unit           | Nicholas Sally                 | Jack Dwyer                       |
| Arm Switching Mechanism     | Jack Dwyer                     | Nicholas Sally                   |
| Mobile App Development      | Oscar Rodriguez                | Abraham Alcala                   |
| Microcontroller Programming | Abraham Alcala                 | Oscar Rodriguez                  |
| Administrative Task         | Nicholas Sally                 | Oscar Rodriguez                  |
| Mounting Structure          | Jack Dwyer                     | Nicholas Sally                   |

## 10.4: Project Milestones

Tables 25 and 26 describe the expected timeline for the project's completion.

*Table 25: Senior Design 1 Project Timeline*

| Project Milestone                                                   | Weeks Allocated  | Status   |
|---------------------------------------------------------------------|------------------|----------|
| Group Formation                                                     | Week 1 – Week 2  | Complete |
| Meet with sponsor for initial requirements and project scope        | Week 1 – Week 3  | Complete |
| Committee Formation                                                 | Week 1 – Week 3  | Complete |
| Divide and Conquer: Documentation Initial                           | Week 2 – Week 3  | Complete |
| Divide and Conquer: Mentor approval                                 | Week 4           | Complete |
| Divide and Conquer: Documentation Revision                          | Week 4 – Week 5  | Complete |
| Functional App and Basic Internet Functionality of Devices Complete | Week 4 – Week 9  | Complete |
| All Initial Electrical Drawings Complete                            | Week 4 – Week 9  | Complete |
| 60 Page Report Completed                                            | Week 10          | Complete |
| 60 Page Report Group Meeting                                        | Week 11          | Complete |
| All Initial PCB Designed                                            | Week 9 – Week 12 | Complete |
| Finalize Bill of Material                                           | Week 5- Week12   | Complete |

| Project Milestone                  | Weeks Allocated   | Status   |
|------------------------------------|-------------------|----------|
| Plan, Build, and Record Demo Video | Week 11 – Week 14 | Complete |
| Final Report and Demo Submission   | Week 15           | Complete |

*Table 26: Senior Design 2 Project Timeline*

| Project Milestone                                                            | Weeks Allocated   | Status      |
|------------------------------------------------------------------------------|-------------------|-------------|
| Order all parts in the Bill of Material (Latest to order initial PCB design) | Week 1            | In Progress |
| Initial Project Assembly, Integration, and Troubleshooting                   | Week 2 – Week 7   | In Progress |
| Planning for redesign/adding additional functionality                        | Week 8            | Not Started |
| Ordering all parts for redesign                                              | Week 8 – Week 9   | Not Started |
| PCB Redesign                                                                 | Week 9 – Week 10  | Not Started |
| PCB Reassembly                                                               | Week 9 – Week 10  | Not Started |
| Integration and Testing                                                      | Week 10 – Week 13 | Not Started |
| Revise existing documentation                                                | Week 12 – Week 15 | Not Started |
| Live Demo and Final Paper                                                    | Week 15           | Not Started |

## Chapter 11 - Conclusion

The Smart and Grid-Interactive Electrical Panel project represents a significant step toward modernizing home electrical systems by integrating Internet of Things functionality with renewable energy compatibility and user-centric controls. As the demand for efficient, sustainable, and reliable energy solutions increases, this project addresses critical issues in energy management, grid reliability, and home automation. Our project supplies a comprehensive solution by combining real-time frequency detection, remote circuit breaker management, and dynamic grid response mechanisms into a single, modular system. These advancements not only improve residential energy management but also contribute to broader goals of sustainability and energy efficiency.

Over the course of this project, significant progress has been made in both hardware and software development. The ESP32 microcontroller functioned as the central component for our system, facilitating real-time frequency detection and user feedback. The frequency detection software successfully calculates grid frequency deviations utilizing an interrupt-based approach, with rising edges detected on the input signal. Through thorough testing using a waveform generator and oscilloscope, the software was validated to provide accurate readings under realistic conditions. The system was configured to trigger visual feedback via an LED on the ESP32 whenever the detected frequency deviated outside the acceptable pre-established range of 59-61Hz or when no signal is detected. This functionality is important for guaranteeing homeowners are alerted to potential grid instability or operational problems, laying the groundwork for a dependable, responsive energy management system.

On the hardware side, the design and testing of the power supply PCBs, motor controllers, and frequency detection circuits have been pivotal to the project's development. The modular nature of the hardware guarantees scalability and adaptability, allowing for potential future upgrades or modifications without extensive redesigns. Each subsystem went through careful evaluation to ensure compatibility and efficiency. For instance, the power supply PCBs were tested under changing load conditions to verify consistent voltage regulation, while the motor controllers were optimized for precise and responsive breaker actuation. These efforts will lead to a hardware infrastructure capable of supporting both real-time operations and remote-control capabilities.

In addition to hardware advancements, the Smart and Grid-Interactive Electrical Panel project emphasized the development of a mobile application and backend API to supply seamless user interaction and control. The mobile app was created with an intuitive interface, allowing users to monitor breaker status, observe grid frequency data, and remotely operate breakers. Postman was utilized to thoroughly test backend API endpoints both locally and via the server domain, guaranteeing secure and reliable data transmission. Future interactions will incorporate unit testing frameworks to expand refined frontend functionality and enhance the app's user experience. By integrating mobile app functionality with the hardware system, the project aligns with modern smart home standards,

closing the gap between traditional electrical panels and cutting-edge Internet of Things technology.

While the first phase of our project established a solid foundation, a number of challenges remain to be addressed in Senior Design 2. The upcoming phase of our project will focus on three pivotal objectives: testing and validating PCBs, completing the mechanical structure and consolidating all software functionalities into a singular ESP32 microcontroller. Testing the PCBs will be critical as there can be unfound issues that lie within them and must be fixed by redesigning the PCBs. The mechanical structure must be strong, guaranteeing proper alignment for motorized breaker operations and protection for internal components. Additionally, the software consolidation process will streamline functionality, allowing the ESP32 to manage frequency detection, motor control, and mobile app communication without performance bottlenecks.

The broader significance of this project extends beyond its technical components. By empowering homeowners to participate actively in demand-side energy management, the Smart and Grid-Interactive Electrical Panel contributes to grid stability and energy sustainability. The system's ability to dynamically monitor and respond to grid conditions supports the integration of renewable energy sources such as solar and wind, which are often characterized by variability. Additionally, decreasing dependency on fossil fuels and improving energy efficiency align with global efforts to combat climate change and transition to cleaner energy solutions. The cost-effective and retrofittable design of this project guarantees that it can be adopted widely, making it accessible to a broad range of users.

The Smart and Grid-Interactive Electrical Panel project exemplifies the integration of modern technology with sustainable energy practices. It closes the gap between traditional electrical systems and the future of smart home automation, offering practical solutions to contemporary energy issues. The work completed during Senior Design 1 demonstrates the feasibility of the design and sets the stage for a comprehensive and reliable implementation in Senior Design 2. By addressing important issues in energy management, grid stability, and user accessibility, the project has the potential to make a meaningful impact on both individual households and the broader energy landscape. With continued development and refinement, the Smart and Grid-Interactive Electrical Panel will serve as a benchmark for innovation in energy technology.

## Appendix A: Reference

- [1] “Why is 200 amps standard for residential panels?” Mike Holt’s Forum. Accessed Oct. 6, 2024. [Online]. Available: <https://forums.mikeholt.com/threads/why-is-200-amps-standard-for-residential-panels.2558798/>
- [2] “Building new house, 200A or 300A service?” Reddit.com. Accessed Oct 6, 2024 [Online]. Available: [https://www.reddit.com/r/HomeMaintenance/comments/1bpvnw4/building\\_new\\_house\\_200a\\_or\\_300a\\_service/](https://www.reddit.com/r/HomeMaintenance/comments/1bpvnw4/building_new_house_200a_or_300a_service/)
- [3] “Which brands of breakers/boxes are the best?” The Garage Journal. Accessed Oct. 8, 2024. [Online]. Available: <https://www.garagejournal.com/forum/threads/which-brands-of-breakers-boxes-are-best.9523/>
- [4] “Professional electricians: what breaker panel would you choose you’re your own home?” Reddit.com. Accessed Oct 9, 2024. [Online]. Available: [https://www.reddit.com/r/electrical/comments/v4rspj/professional\\_electricians\\_what\\_breaker\\_panel/](https://www.reddit.com/r/electrical/comments/v4rspj/professional_electricians_what_breaker_panel/)
- [5] D. Focareta. “Average number of circuit breakers in a home 2024.” Consumer Affairs. Accessed Oct. 9, 2024. [Online]. Available: <https://www.consumeraffairs.com/homeowners/average-number-of-circuit-breakers-in-a-home.html>
- [6] S. Williams “How to Determine the Load Capacity of Your Circuit Breaker.” Homeelectrical.com. Accessed Oct. 9, 2024. [Online]. Available: <https://www.homeelectrical.com/understanding-capacity-your-circuit-breaker.6.html>
- [7] Circuit Breaker Wholesale. “30Amp Breaker – Everything You Need to Know.” circuitbreakerwholesale.com. Accessed Oct. 9, 2024. [Online]. Available: <https://www.circuitbreakerwholesale.com/blogs/circuit-breakers/30-amp-breaker-everything-you-need-to-know#:~:text=As%20the%20name%20implies%2C%20a,water%20heating%20systems%2C%20and%20dryers.>
- [8] A. Await. “What is a Transformer, and How Does it Work?” Digikey. Accessed Oct. 9, 2024. [Online]. Available: [https://www.digikey.com/en/blog/what-is-a-transformer-and-how-does-it-work#:~:text=The%20core%20of%20the%20transformer,via%20electromotive%20force%20\(EMF\).](https://www.digikey.com/en/blog/what-is-a-transformer-and-how-does-it-work#:~:text=The%20core%20of%20the%20transformer,via%20electromotive%20force%20(EMF).)
- [9] Electrical4U. “Auto Transformer: What is it? (Definition, Theory & Diagram).” Electrical4U.com. Accessed Oct. 10, 2024. [Online]. Available: <https://www.electrical4u.com/what-is-auto-transformer/>
- [10] Industrial Quick Search. “Power Transformer.” Iqsdirectory.com. Accessed: Oct. 10, 2024. [Online]. Available: <https://www.iqsdirectory.com/articles/electric->

transformer/power-transformers.html#:~:text=Power%20transformers%20are%20electrical%20devices,generators%20and%20primary%20distribution%20circuits.

[11] "The Current Transformer." Electronics-tutorials.ws. Accessed Oct. 11, 2024. [Online]. Available: <https://www.electronics-tutorials.ws/transformer/current-transformer.html>

[12] "Design guide for Rectifier Use." Digikey.jp. Accessed Oct. 11, 2024. [Online]. Available: <https://www.digikey.jp/htmldatasheets/production/69394/0/0/1/Design-Guide-for-Rectifier-Use.pdf>

[13] "Understanding Half Wave and Full Wave Rectifier: A Comprehensive Guide." Rohm Semiconductor. Accessed Oct. 12, 2024. [Online]. Available: <https://www.rohm.com/electronics-basics/ac-dc/rectification>

[14] "Line Regulator." Rohm Semiconductor. Accessed Oct. 12, 2024. [Online]. Available: [https://www.rohm.com/electronics-basics/dc-dc-converters/dcdc\\_what4](https://www.rohm.com/electronics-basics/dc-dc-converters/dcdc_what4)

[15] "Switching Regulator." Rohm Semiconductor. Accessed Oct. 12, 2024. [Online]. Available: [https://www.rohm.com/electronics-basics/dc-dc-converters/dcdc\\_what5](https://www.rohm.com/electronics-basics/dc-dc-converters/dcdc_what5)

[16] P. Khatri. "Zero Crossing Detector Circuit." CircuitDigest.com Accessed Oct. 12, 2024. [Online]. Available: <https://circuitdigest.com/electronic-circuits/zero-crossing-detector-circuit-diagram>

[17] "Zero Crossing Detector." Geeksforgeeks.org. Accessed Oct 14, 2024 . [Online]. Available: <https://www.geeksforgeeks.org/zero-crossing-detector/>

[18] J. Bryant. "Using Op Amps as Comparators." Analog Devices. Accessed Oct. 20, 2024. [Online]. Available: <https://www.analog.com/media/en/technical-documentation/application-notes/AN-849.pdf>

[19] J. Atwal. "Zero Crossing Detection Using Comparator Circuit." Texas Instruments. Accessed Oct 20, 2024. [Online]. Available: [https://www.ti.com/lit/ab/snoa999a/snoa999a.pdf?ts=1729596510098&ref\\_url=https%253A%252F%252Fwww.google.com%252F](https://www.ti.com/lit/ab/snoa999a/snoa999a.pdf?ts=1729596510098&ref_url=https%253A%252F%252Fwww.google.com%252F)

[20] "Op-amp Comparator." Electronics-tutorial.ws. Accessed Oct. 20, 2024. [Online] <https://www.electronics-tutorials.ws/opamp/op-amp-comparator.html>

[21] "Explaining Frequency Deviation with a Few Examples." Nhz.us. Accessed Nov. 2, 2024. [Online]. [https://nhz.us/Content/Frequency%20Deviation%20and%20Phase%20Deviation%20description%20in%20Simple%20Words%20with%20Video%20and%20Examples.html#google\\_vignette](https://nhz.us/Content/Frequency%20Deviation%20and%20Phase%20Deviation%20description%20in%20Simple%20Words%20with%20Video%20and%20Examples.html#google_vignette)

- [22] V. Singh. "Difference Between Bluetooth and Wifi." Shiksha.com. Accessed: Sep. 6, 2024. [Online]. Available: <https://www.shiksha.com/online-courses/articles/difference-between-bluetooth-and-wi-fi/>
- [23] J. Kek. "Flutter vs React Native - A guide from every experience level." justinkek.com. Accessed: Sep. 6, 2024. [Online]. Available: <https://www.justinkek.com/blog/flutter-vs-react-native/flutter-vs-react-native>
- [24] Espressif Systems. "ESP32--WROOM-32-Datasheet." Espressif.com. Accessed: Sep. 6, 2024. [Online]. Available: [https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32\\_datasheet\\_en.pdf](https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32_datasheet_en.pdf)
- [25] STMicroelectronics. "Arm®-based 32-bit MCU, 150 DMIPs, up to 1 MB Flash/128+4KB RAM, USB OTG HS/FS, Ethernet, 17 TIMs, 3 ADCs, 15 comm. interfaces and camera." st.com. Accessed: Sep. 6, 2024. [Online]. Available: <https://www.st.com/resource/en/datasheet/stm32f205rb.pdf>
- [26] "Arduino Pro Mini." arduino.cc. Accessed: Sep. 6, 2024. [Online]. Available: <https://docs.arduino.cc/retired/boards/arduino-pro-mini/>
- [27] E. M. Doolin, A. E. Cruz, A. Glidewell, B. Green. "Grid Stabilizing Frequency Detector (GSFD)." University of Central Florida, Orlando, FL., United States. Accessed: Sep. 6, 2024. [Online]. Available: <https://maverick.eecs.ucf.edu/seniordesign/fa2023sp2024/g32/>
- [28] "PCB Trace Width Conversion Calculator." Digikey.com. Accessed Nov. 15, 2024. [Online]. Available: <https://www.digikey.com/en/resources/conversion-calculators/conversion-calculator-pcb-trace-width>
- [29] "Image Current." Harvard University. Accessed Nov. 17, 2024. [Online]. Available: <https://sciencedemonstrations.fas.harvard.edu/presentations/image-current>
- [30] "TMC5160/TMC5160A DATASHEET." Analog Devices. Accessed Oct. 20, 2024. [Online] [https://www.analog.com/media/en/technical-documentation/datasheets/TMC5160A\\_datasheet\\_rev1.18.pdf](https://www.analog.com/media/en/technical-documentation/datasheets/TMC5160A_datasheet_rev1.18.pdf)
- [31] T. S. Perry, "Planned U.S. Power System Experiment Means Some Clocks Will Speed Up," *IEEE Spectrum*, Jul. 2011. Accessed Oct. 25, 2024. [Online] <https://spectrum.ieee.org/power-system-experiment-in-us-means-clocks-will-speed-up>.
- [32] "Which linear guide is most suitable for my application?," *Sinadrives.com*, 2024. Accessed: Nov. 2, 2024. [Online]. Available: <https://sinadrives.com/en/which-linear-guide-is-most-suitable-for-my-application/>.
- [33] "Lead Screws 101," *www.haydonkerkpittman.com*. Accessed: Nov. 5, 2024. [Online] Available: <https://www.haydonkerkpittman.com/learningzone/whitepapers/lead-screws-101>.

- [34] "What is Leadscrew Efficiency," *pbcllinear.com*. Accessed: Nov. 7, 2024. [Online] Available: <https://pbcllinear.com/blog/2018/february/what-is-lead-screw-efficiency-in-linear-motion>.
- [35] B. Earl, "All About Stepper Motors," *Adafruit Learning System*, May 05, 2014. Accessed: Nov. 11, 2024. [Online]. Available: <https://learn.adafruit.com/all-about-stepper-motors>.
- [36] "Stepper Motors - StepperOnline," *Omc-stepperonline.com*, 2024. Accessed: Nov. 25, 2024. [Online]. Available: <https://www.omc-stepperonline.com/stepper-motors>.
- [37] "The National Electrical Code (NEC)," *Electrical Safety Foundation*." Accessed: Oct. 24, 2024. [Online]. Available: <https://www.esfi.org/workplace-safety/industry-codes-regulations/the-national-electrical-code-nec/>
- [38] Nfpa, "NFPA 70A National Electrical Code ® Requirements for One-and Two-Family Dwellings 2005 Edition." Accessed: Oct. 24, 2024. [Online]. Available: <https://edufire.ir/storage/Library/elam/NFPA%2070A-2005.pdf>
- [39] "ELECTRICAL OSHA OFFICE OF TRAINING AND EDUCATION." Accessed: Oct. 24, 2024. [Online]. Available: [https://www.osha.gov/sites/default/files/training-library\\_electrical.pdf](https://www.osha.gov/sites/default/files/training-library_electrical.pdf)
- [40] "EMC Standards," *Academy of EMC*. Accessed Oct. 24, 2024. [Online]. Available: <https://www.academyofemc.com/emc-standards>
- [41] P. Dhaker, "Introduction to SPI Interface | Analog Devices," *Analog.com*, Sep. 2018. <https://www.analog.com/en/resources/analog-dialogue/articles/introduction-to-spi-interface.html>
- [42] "LMR514x0, 36-V, 4-A/5-A Synchronous Step-Down DC/DC Converter With Low IQ." Texas Instruments. Accessed Oct. 12, 2024. [Online]. Available: <https://www.ti.com/lit/ds/symlink/lmr51450.pdf?ts=1732568058102>
- [43] "TPSM86125x 3V to 17V Input, 1A, Synchronous Buck Module in QFN Package." Texas Instruments. Accessed Oct 12, 2024. [Online]. Available: [https://www.ti.com/lit/ds/symlink/tpsm861253.pdf?ts=1732572283035&ref\\_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FTPSM861253](https://www.ti.com/lit/ds/symlink/tpsm861253.pdf?ts=1732572283035&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FTPSM861253)
- [44] "TLV701x and TLV702x Small-Size, Low-Power, Low-Voltage Comparators." Texas Instruments. Accessed Oct 12, 2024. [Online]. Available: <https://www.ti.com/lit/ds/symlink/tlv7011.pdf?ts=1732572391529>
- [45] TRINAMIC, "TMC5160 SilentStepStick," May 2019. . Accessed: Nov. 25, 2024. [Online]. Available: [https://www.mouser.com/pdfDocs/TMC5160\\_SilentStepStick\\_datasheet\\_Rev110-2.pdf](https://www.mouser.com/pdfDocs/TMC5160_SilentStepStick_datasheet_Rev110-2.pdf)
- [46] Postman, "Postman | The Collaboration Platform for API Development," *Postman*, 2021. Accessed Oct. 12, 2024 <https://www.postman.com/>

- [47] “How to Connect to Droplets with SSH | DigitalOcean Documentation,” *DigitalOcean*. Accessed Oct 12, 2024. [Online]. Available: [2024https://docs.digitalocean.com/products/droplets/how-to/connect-with-ssh/](https://docs.digitalocean.com/products/droplets/how-to/connect-with-ssh/)
- [48] Raspberry Pi, “Raspberry Pi 4 Model B specifications,” Raspberry Pi. Accessed Oct 12, 2024. [Online]. Available: <https://www.raspberrypi.com/products/raspberry-pi-4-model-b/specifications/>
- [49] Espressif, “ESP32 Overview | Espressif Systems,” *www.espressif.com*, 2023. Accessed Oct 12, 2024. [Online]. Available: <https://www.espressif.com/en/products/socs/esp32>
- [50] “STM32 32-Bit Arm® Cortex®-M MCUs,” Mouser.com, Feb. 21, 2008. Accessed Nov. 25, 2024. [Internet]. Available: <https://www.mouser.com/new/stmicroelectronics/stm32/>.
- [51] “Features,” *createy.com*, Jul. 12, 2019. Accessed Oct 12, 2024. [Online]. Available: <https://createy.com/tour/>
- [52] Lucidchart, “Online diagram software & visual solution | Lucidchart,” *Lucidchart*, 2023. Accessed Oct 12, 2024. [Online]. Available: <https://www.lucidchart.com/pages/>
- [53] D. Tao, “MQTT on ESP32: A Beginner’s Guide,” Accessed Oct 12, 2024. [Online]. Available: *www.emqx.com*. <https://www.emqx.com/en/blog/esp32-connects-to-the-free-public-mqtt-broker#prepare-an-mqtt-broker>
- [54] “ESP32 WebSocket Server: Display Sensor Readings | Random Nerd Tutorials,” Jul. 19, 2023. Accessed Oct 12, 2024. [Online]. Available: <https://randomnerdtutorials.com/esp32-websocket-server-sensor/>
- [55] “3M™ Dual Lock™ Reclosable Fasteners,” *3M*, May 2020. Accessed: Nov. 25 2024. [Online] Available: <https://multimedia.3m.com/mws/media/801947O/duallock-psarollgoods-tds-june2012-pdf.pdf>.
- [56] teemuatlut, “TMCStepper,” *GitHub*, Apr. 06, 2023. Available: <https://github.com/teemuatlut/TMCStepper>. [Accessed: Nov. 25, 2024]
- [57] “grbl,” *GitHub*, Apr. 21, 2020. Available: <https://github.com/grbl/grbl>. [Accessed: Nov. 25, 2024]
- [58] jbrazio, “What is Marlin?,” *Marlin Firmware*, Feb. 15, 2021. Available: <https://marlinfw.org/docs/basics/introduction.html>. [Accessed: Nov. 25, 2024]
- [59] bdring, “bdring/FluidNC,” *GitHub*, May 21, 2022. Available: <https://github.com/bdring/FluidNC>. [Accessed: Nov. 25, 2024]
- [60] “Beginners Guide to 3D Printing G-Code Commands | Simplify3D,” *SIMPLIFY3D*. Available: <https://www.simplify3d.com/resources/articles/3d-printing-gcode-tutorial/>

- [61] aerosusa, "The Ultimate Guide to Cable Drag Chains | AerosUSA," *AerosUSA*, Aug. 18, 2021. Available: [https://aerosusa.com/the-ultimate-guide-to-cable-drag-chains/#What\\_Are\\_Cable\\_Drag\\_Chains](https://aerosusa.com/the-ultimate-guide-to-cable-drag-chains/#What_Are_Cable_Drag_Chains). [Accessed: Nov. 25, 2024]
- [62] "Ball Bearing 688Z 8x16x5," *OpenBuilds*, 2024. Available: <https://us.openbuilds.com/ball-bearing-688z-8x16x5/>. [Accessed: Nov. 25, 2024]
- [63] "5mm \* 8mm Flexible Coupling," *OpenBuilds*, 2024. Available: <https://us.openbuilds.com/5mm-8mm-flexible-coupling/>. [Accessed: Nov. 25, 2024]
- [64] Protolabs, "Delrin Plastic Material Properties," *www.protolabs.com*. Available: <https://www.protolabs.com/en-gb/resources/blog/delrin-offers-strong-versatile-plastic-alternative-to-metal/>
- [65] "Anti-Backlash Nut Block for 8mm Metric Lead Screw," *OpenBuilds*, 2024. Available: <https://us.openbuilds.com/anti-backlash-nut-block-for-8mm-metric-lead-screw/>. [Accessed: Nov. 25, 2024]
- [66] "Thomson Industries," *Thomsonlinear.com*, 2024. Available: <https://www.thomsonlinear.com/en/support/tips/what-is-ball-nut-backlash-play-and-deflection>. [Accessed: Nov. 25, 2024]
- [67] Q. L. Wu, "What is dual Z Axis and why do I need it for my 3D Printer?," *DREMC STORE*, Jan. 16, 2023. Available: <https://store.dremc.com.au/blogs/news/what-is-dual-z-axis-and-why-do-i-need-it>. [Accessed: Nov. 25, 2024]

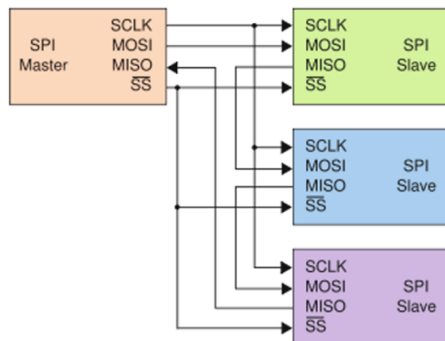
## Appendix B: Image Permissions



Good Afternoon,

As part of our Senior Design Project Report for the University of Central Florida, we are talking about communication standards and how SPI impacts the development and implementation of our project. We found your paperwork, "FR, MB91460, SPI - Daisy Chain Communication" and liked the graphic.

We were hoping to receive permission to utilize this image from your paper within the report with the appropriate citation.



from [FR, MB91460, SPI - Daisy chain Communication](#)

Thank you so much for your time,

- Abraham

Figure 41: Daisy-Chained SPI Interface Permission (Requested)

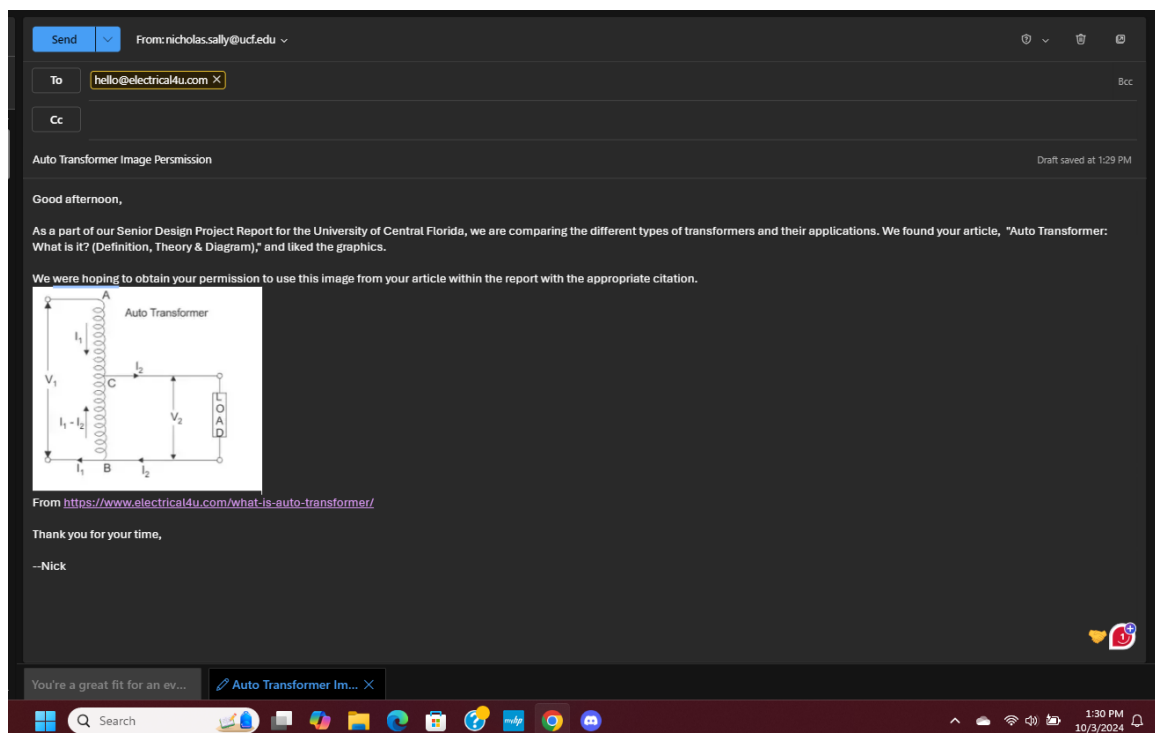
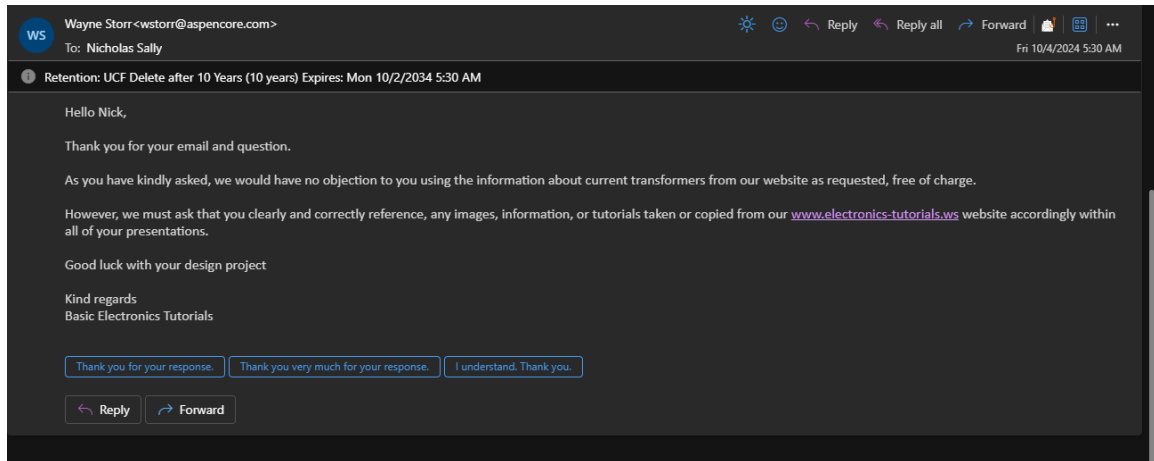
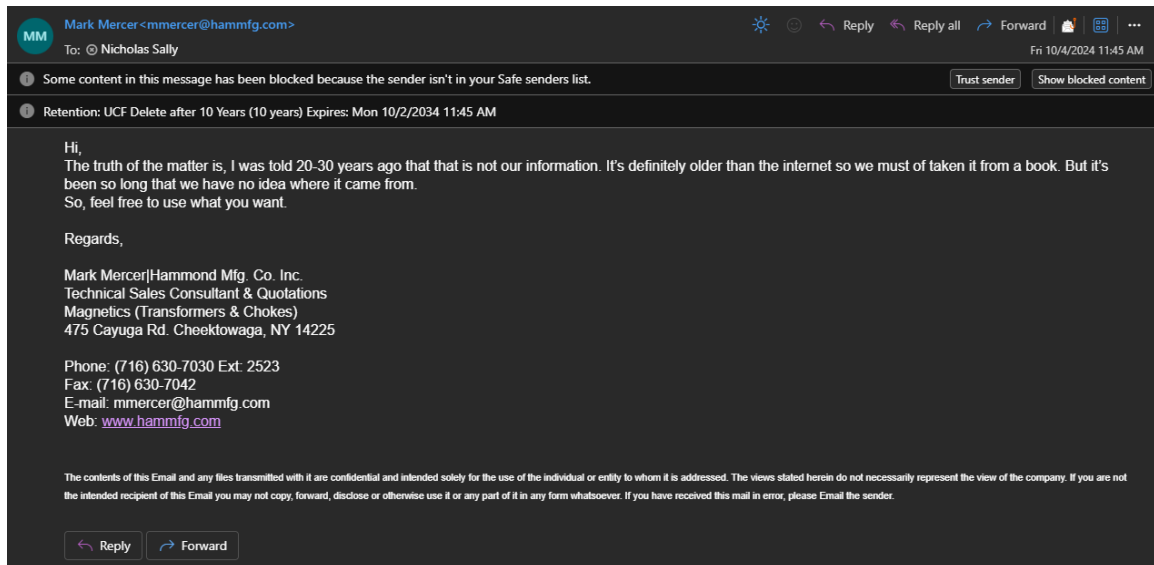


Figure 42: Autotransformer Permission (Request)



*Figure 43: Current Transformer Permission (Granted)*



*Figure 44: Transformer Specifications and Diagram Permission (Granted)*

**Send us an Email**

Name \*

Company

Email \*

Phone

For immediate assistance, please provide your phone number.

Subject \*

Message \* 

part of our project's research, we are selecting transformers and found your company's article (<https://www.digikey.jp/htmldatasheets/production/69394/0/0/1/Design-Guide-for-Rectifier-Use.pdf>) describing how to choose a transformer correctly.

My team and I are seeking your permission to use the "FULL WAVE Capacitor Input Load" picture as a part of our research. Naturally, we will provide proper citations and links to your article with your permission.

Fields marked with \* are required.

☒ I wish to receive communications from Hammond Mfg.

Fax: (519) 822-0715

**Canadian Electrical Product Sales**

Phone: (519) 822-2960  
Fax: (519) 822-0715

**Canadian Datacom Product Sales**

Phone: (877) 535-3282

**Canadian Electronic Product Sales**

Phone: (519) 886-7170  
Fax: (519) 886-4174

**Canadian Transformer Product Sales**

Phone: (519) 886-6181  
Fax: (519) 886-9540

**Main Distribution Center**

7 Nicholas Beaver Road  
Aberfoyle, Ontario, N0B 2J0  
Canada

**Quebec Office & Sales (All Products)**

Les Fabrications Hammond (Quebec) Inc.  
985 Rue Bergar  
Laval, Quebec, H7L 4Z6  
Canada  
Phone: (450) 975-1884  
Fax: (450) 975-2098

**US Office & Sales (All Products)**

Hammond Manufacturing Company Inc.  
475 Cayuga Rd  
Cheektowaga, New York, 14225  
USA  
Phone: (716) 630-7030  
Fax: (716) 630-7042

**European Office & Sales (All Products)**

Figure 45: Full Wave Rectifier Image Permission (Requested)

Send From: nicholas.sally@ucf.edu

To: copyrightcounsel@list.ti.com

Cc:

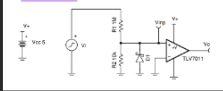
Image Permission Request for Analog Engineer's Circuit Zero Crossing Detection Using Comparator Circuit

Draft saved at 1:48 AM

Good morning,

As a part of our Senior Design project for the University of Central Florida, we needed to research cross-zero detectors for our frequency measurer. We found TI's article [Circuit Zero Crossing Detection Using Comparator Circuit](https://www.ti.com/lit/ab/sno999a/sno999a.pdf) (<https://www.ti.com/lit/ab/sno999a/sno999a.pdf>) and wanted to include parts of the article as a part of our research.

We are seeking your permission to include the image below.



We plan to use this image as an example of a comparator-based zero-cross detector. Additionally, we will compare the TLV7011 to other comparable IC's and explain their pros and cons in the circuit, ultimately choosing one for our project.

Thank you for your time,

--Nick

Select an item to read Image Permission Re...

Figure 46: Cross-Zero Detector Image Permission (Requested)

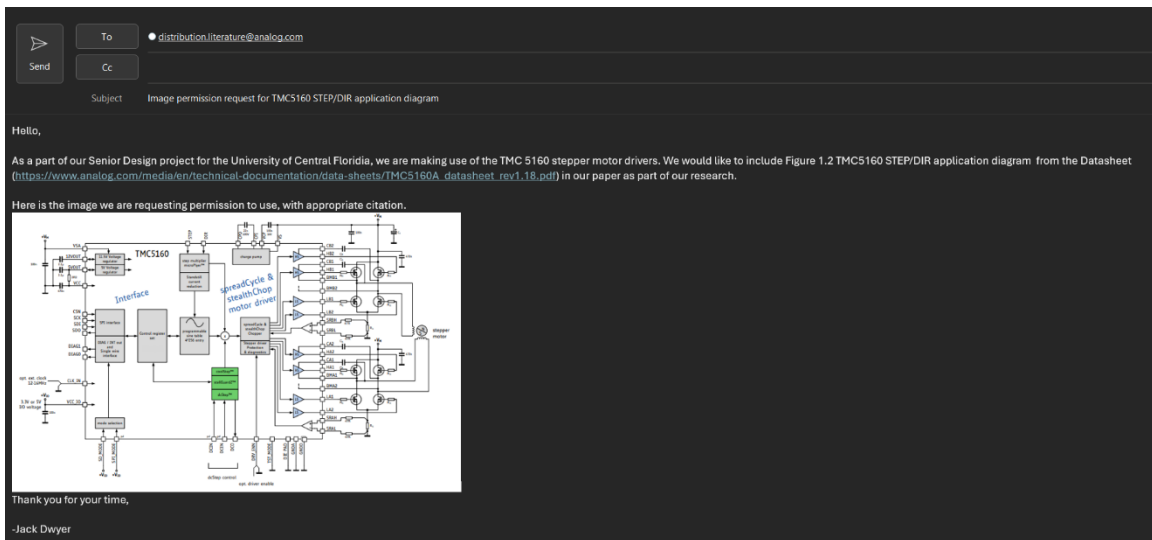


Figure 47: TMC 5160 block diagram permission (Requested)

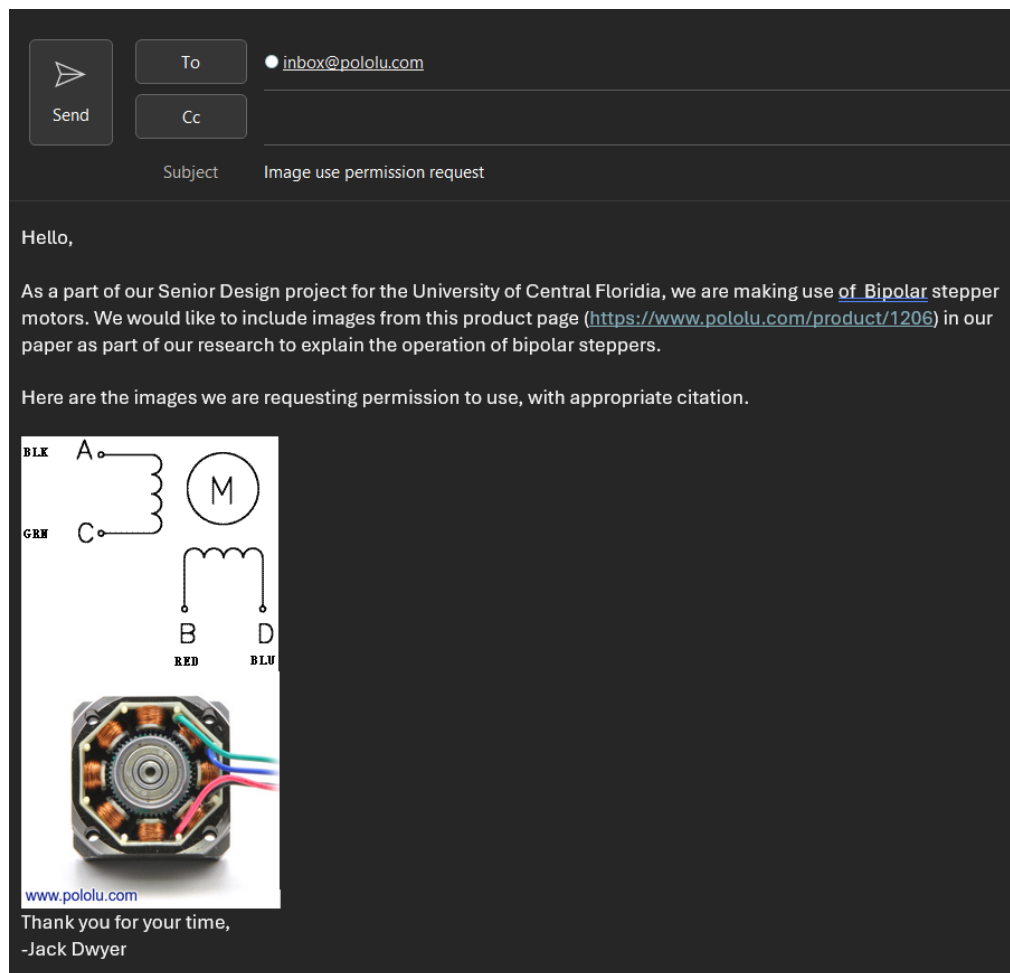


Figure 48: Stepper Motor Diagram Permission (Requested)



# Appendix C: Texas Instruments Webench Power Designer Schematics



## WEBENCH® Design Report

Design : 27 LMR51450SDRRR  
LMR51450SDRRR 26V-27V to 12.00V @ 5A

VinMin = 26.0V  
VinMax = 27.0V  
Vout = 12.0V  
Iout = 5.0A

Device = LMR51450SDRRR  
Topology = Buck  
Created = 2024-10-09 10:49:45.131  
BOM Cost = \$2.67  
BOM Count = 13  
Total Pd = 4.54W

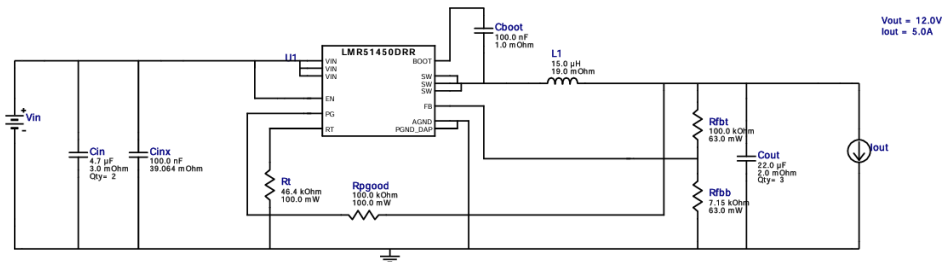


Figure 49: 12V PSU Webench Schematic Courtesy of Texas Instruments



## WEBENCH® Design Report

Design : 31 TPSM861253RDX  
TPSM861253RDX 11.5V-12.5V to 3.30V @ 1A

VinMin = 11.5V  
VinMax = 12.5V  
Vout = 3.3V  
Iout = 1.0A

Device = TPSM861253RDX  
Topology = Buck  
Created = 2024-10-09 11:41:25.685  
BOM Cost = \$0.69  
BOM Count = 7  
Total Pd = 0.67W

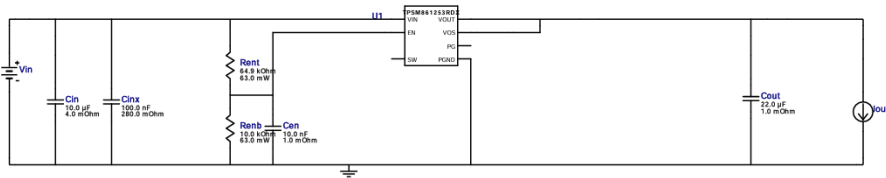


Figure 50: 3.3V PSU Webench Schematic Courtesy of Texas Instruments